

ECSE 4560/ 6560

Modern Communication Systems

Lecture 2: Wireless Systems Overview

Part 1

Ish Jain, Fall 2025



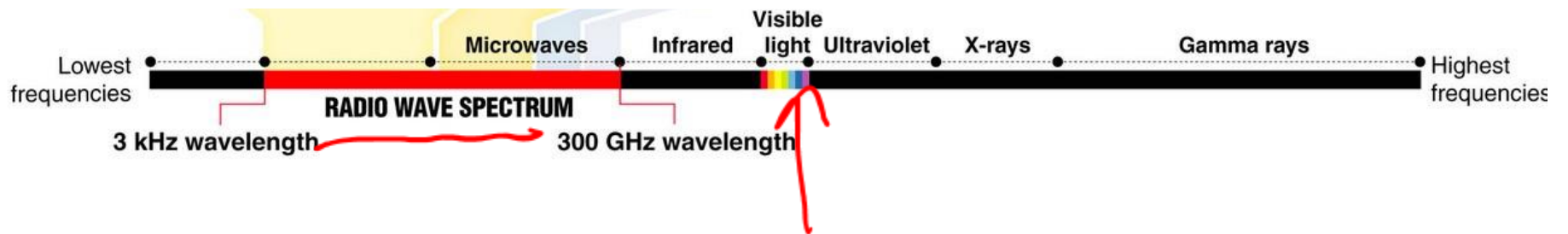
Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- GradeScope for assignment submission (You will be added)
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
 - Single attempt; Don't forget to submit, o/w not graded.
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Wireless**
 - 80% attendance mandatory, In-class participation 5%
- Projects
 - A list of papers will be posted soon
 - You may choose a different paper different (requires approval)
 - Choose early and get it approved (office hours, email, Webex)
 - No two groups should choose the same papers/project
 - Group formation due 9/12

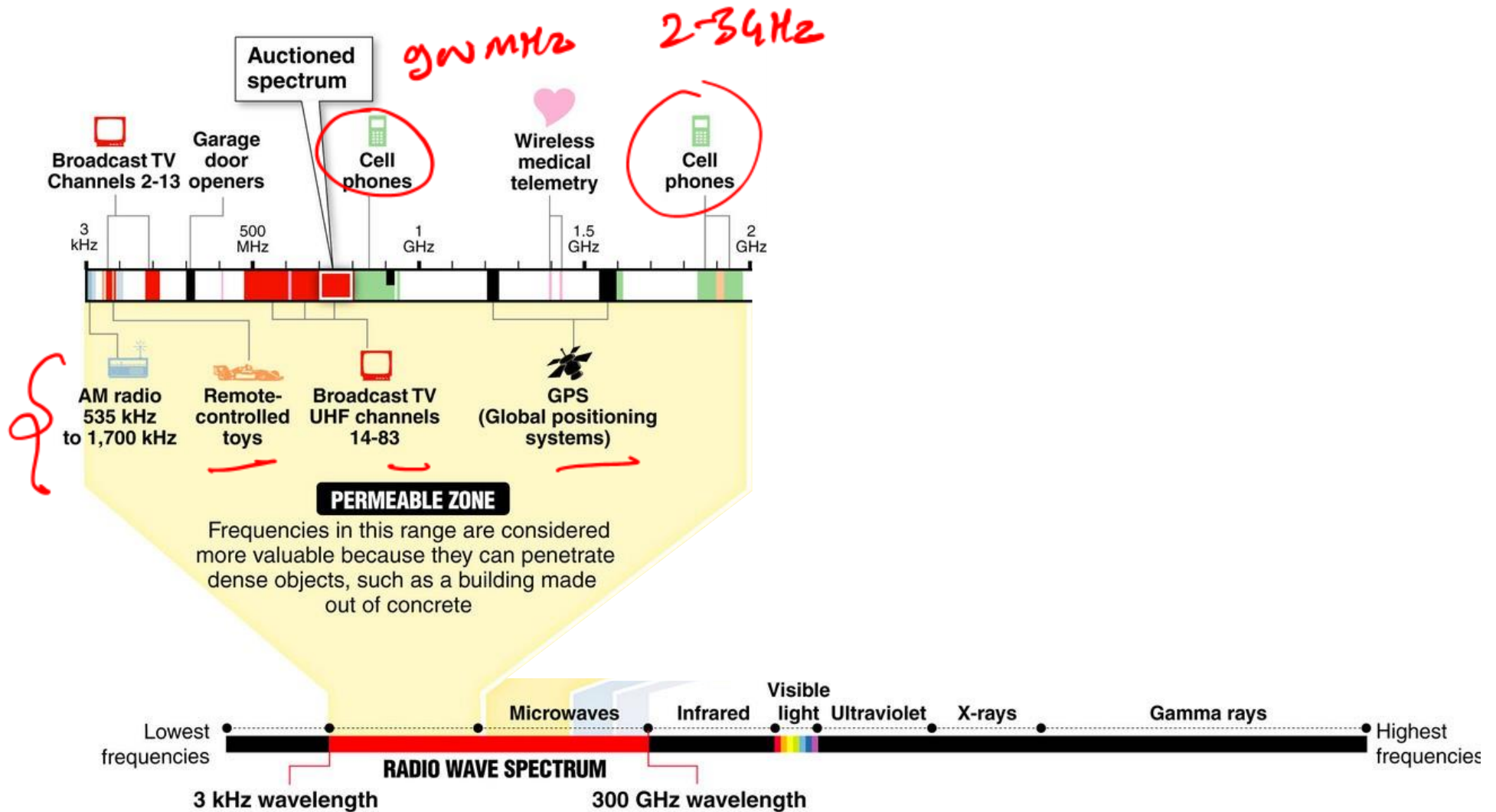
Today's Class: Wireless Overview

- Baseband/Passband Communication
- Up-Conversion and Down conversion
- Modulation
- What is SNR?
- Higher order modulation and its relation to SNR
- BER-SNR curves
- Capacity

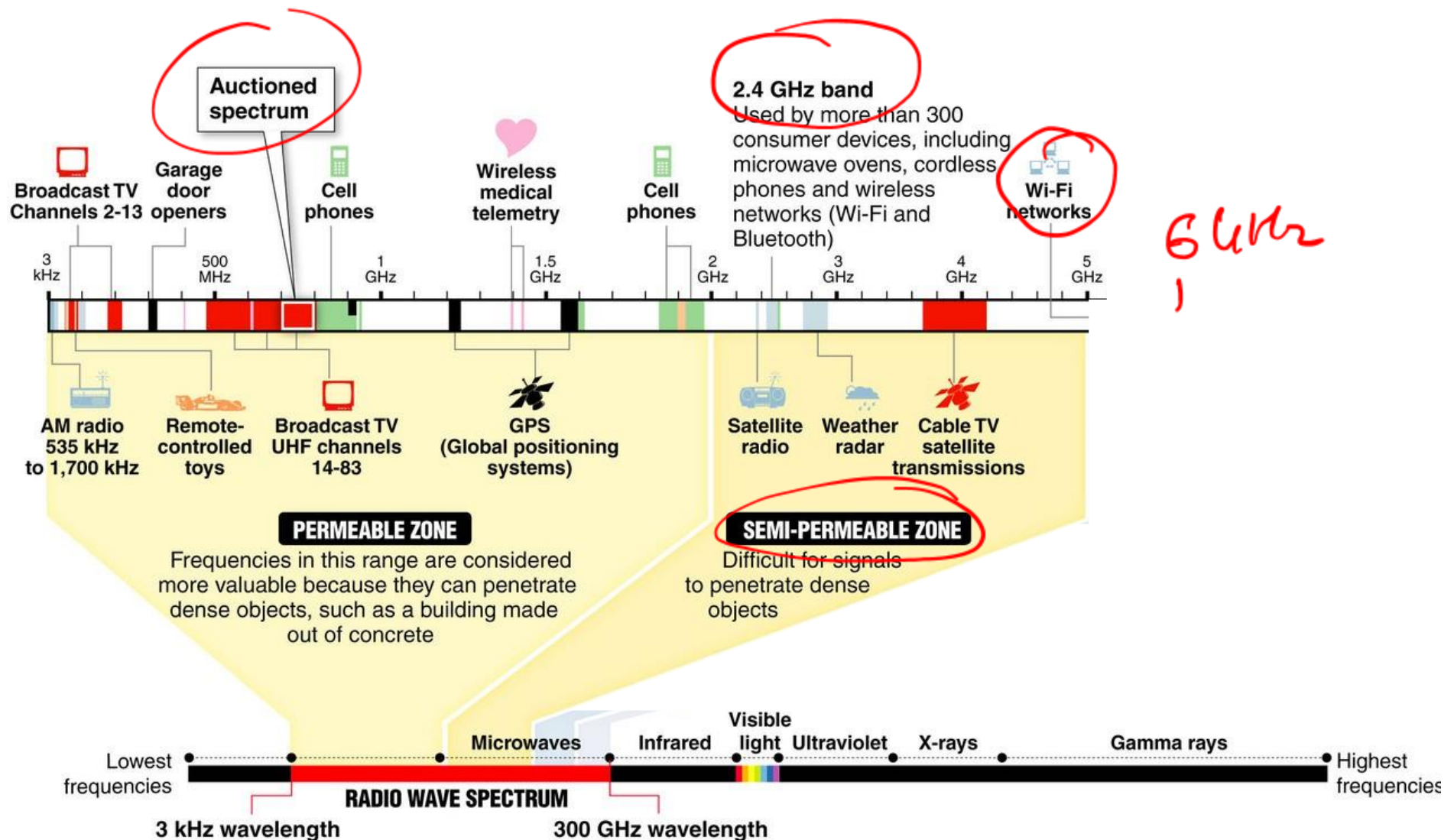
The Wireless Spectrum



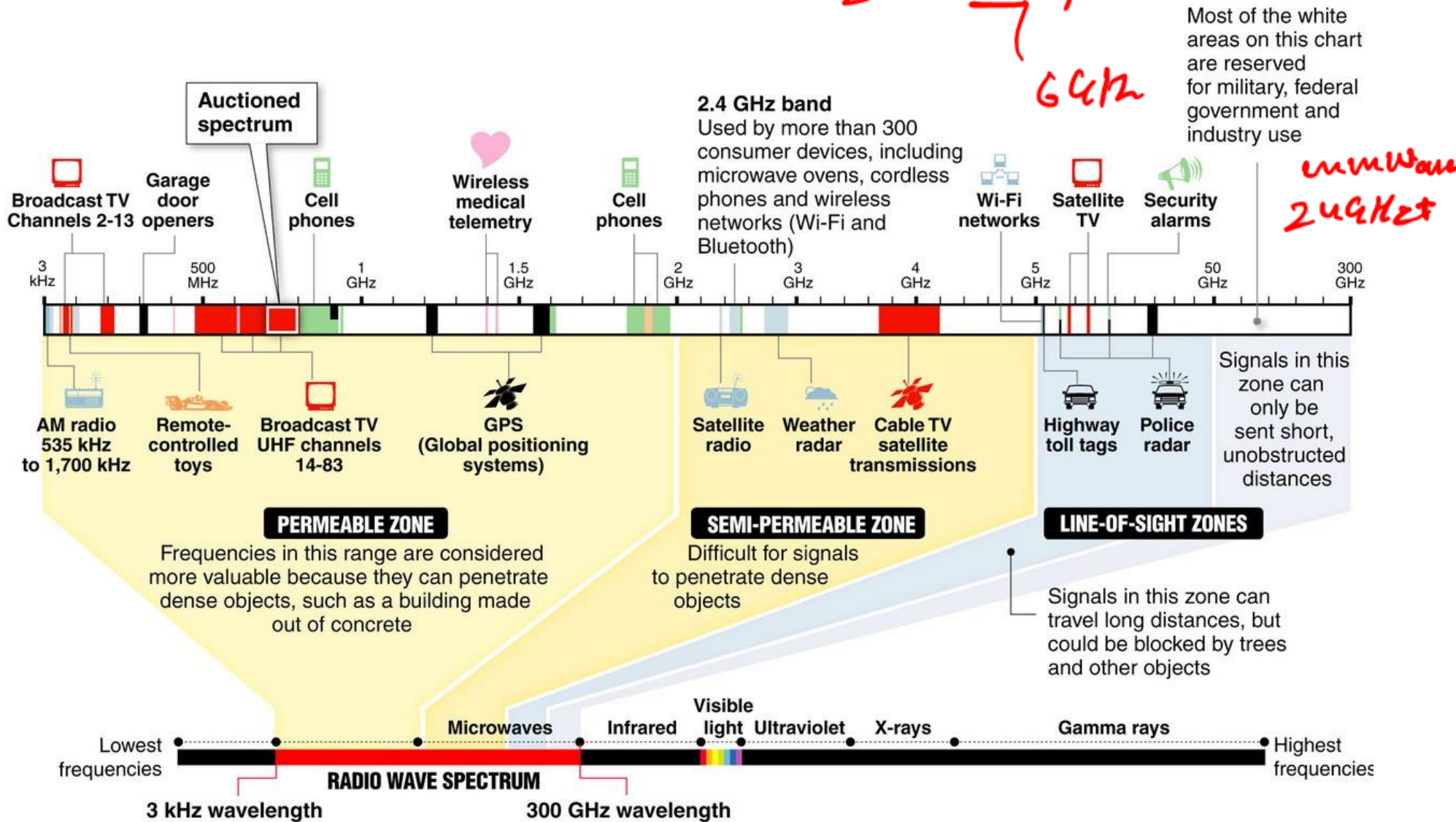
The Wireless Spectrum



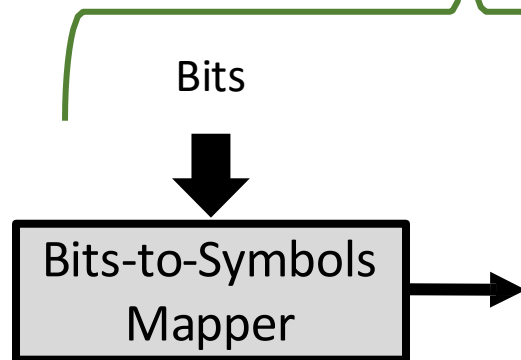
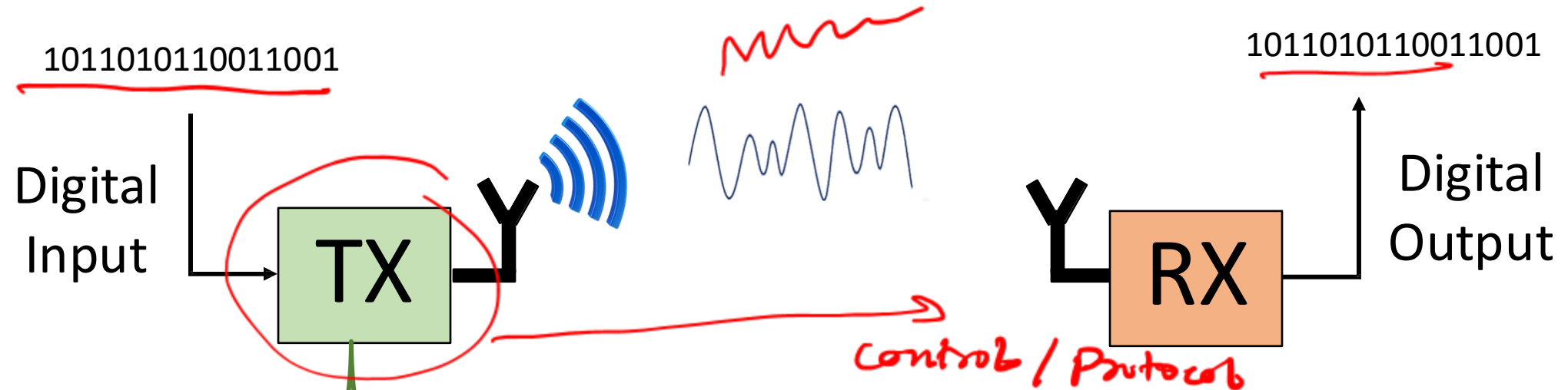
The Wireless Spectrum



The Wireless Spectrum

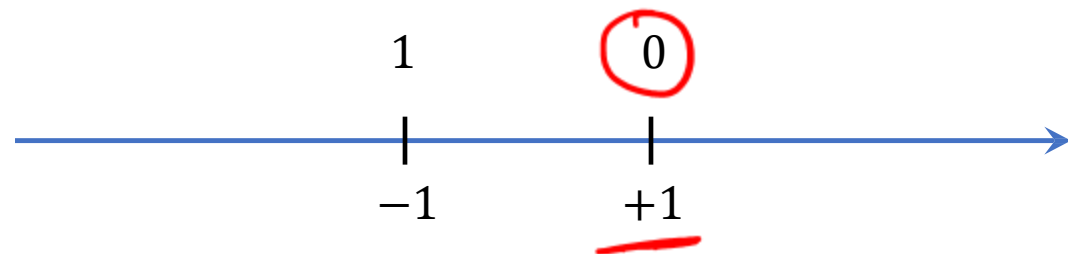


Wireless Communication System



Modulation: Mapping of Bits to Symbols

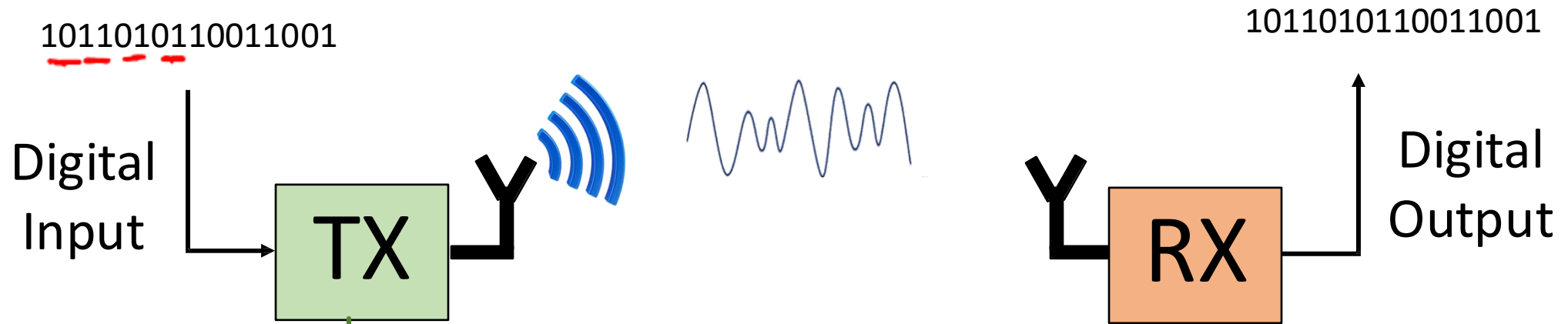
BPSK: 1 bits/symbol



Modulation
(Encoding)

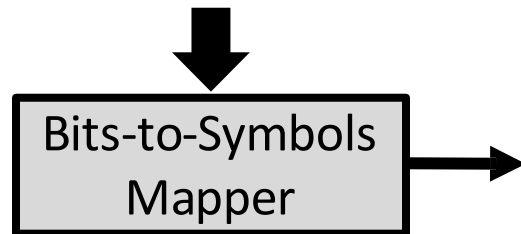


Wireless Communication System

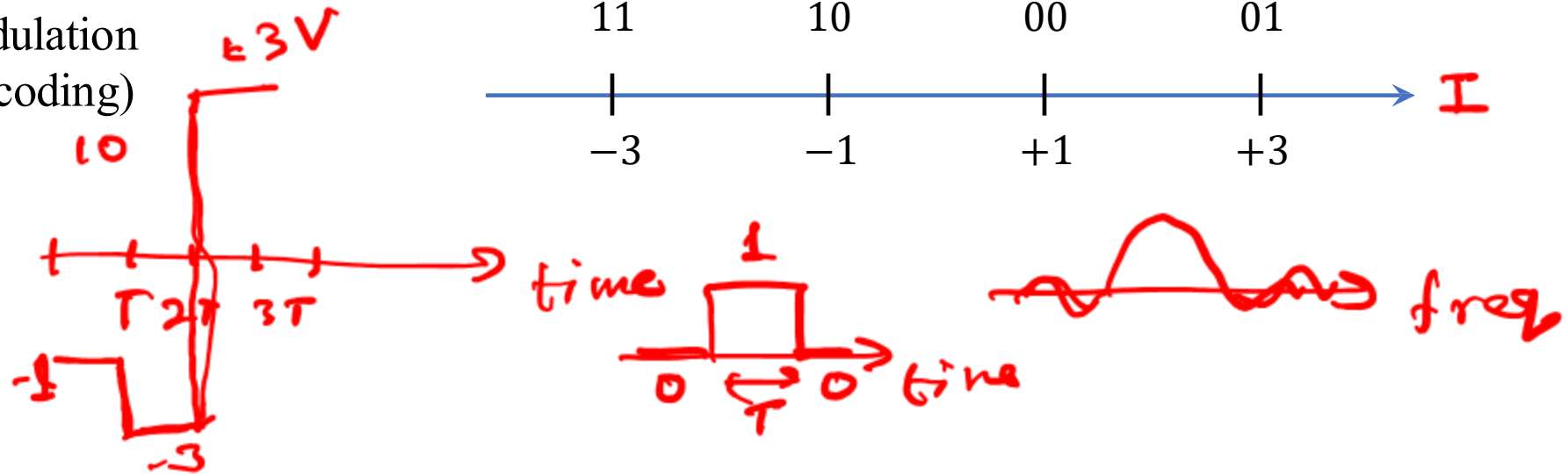


Modulation: Mapping of Bits to Symbols

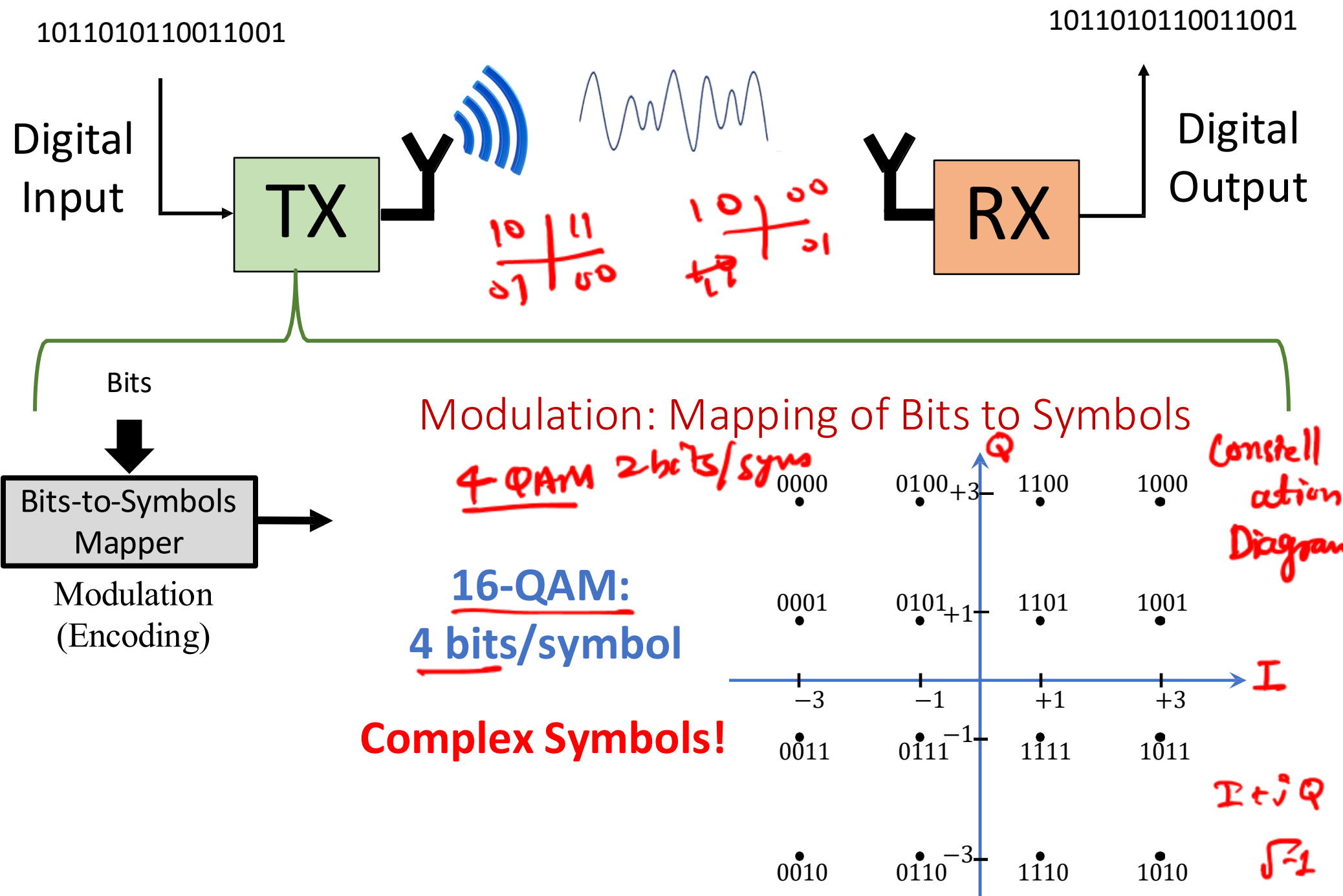
PAM: 2 bits/symbol



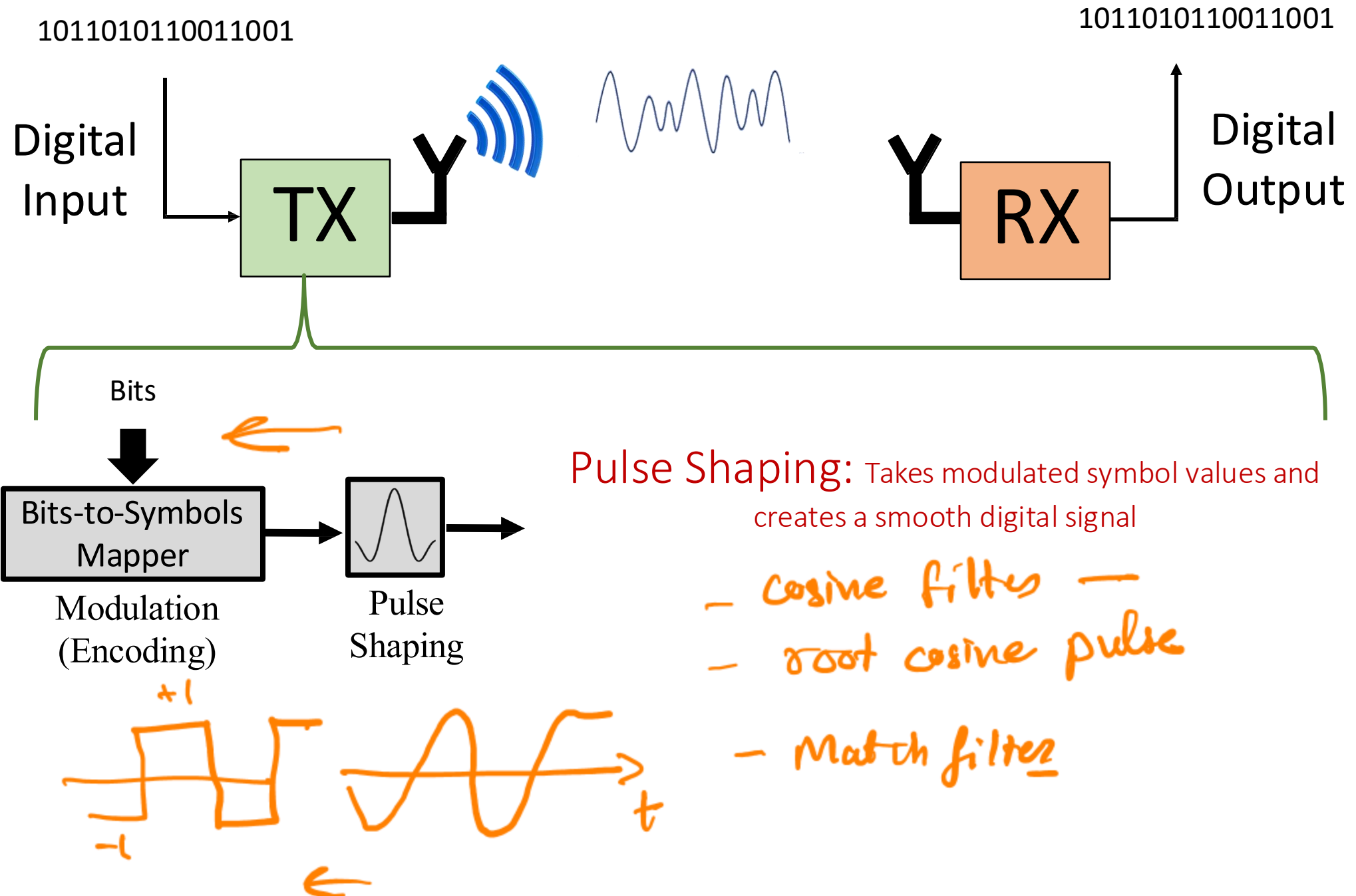
Modulation (Encoding)



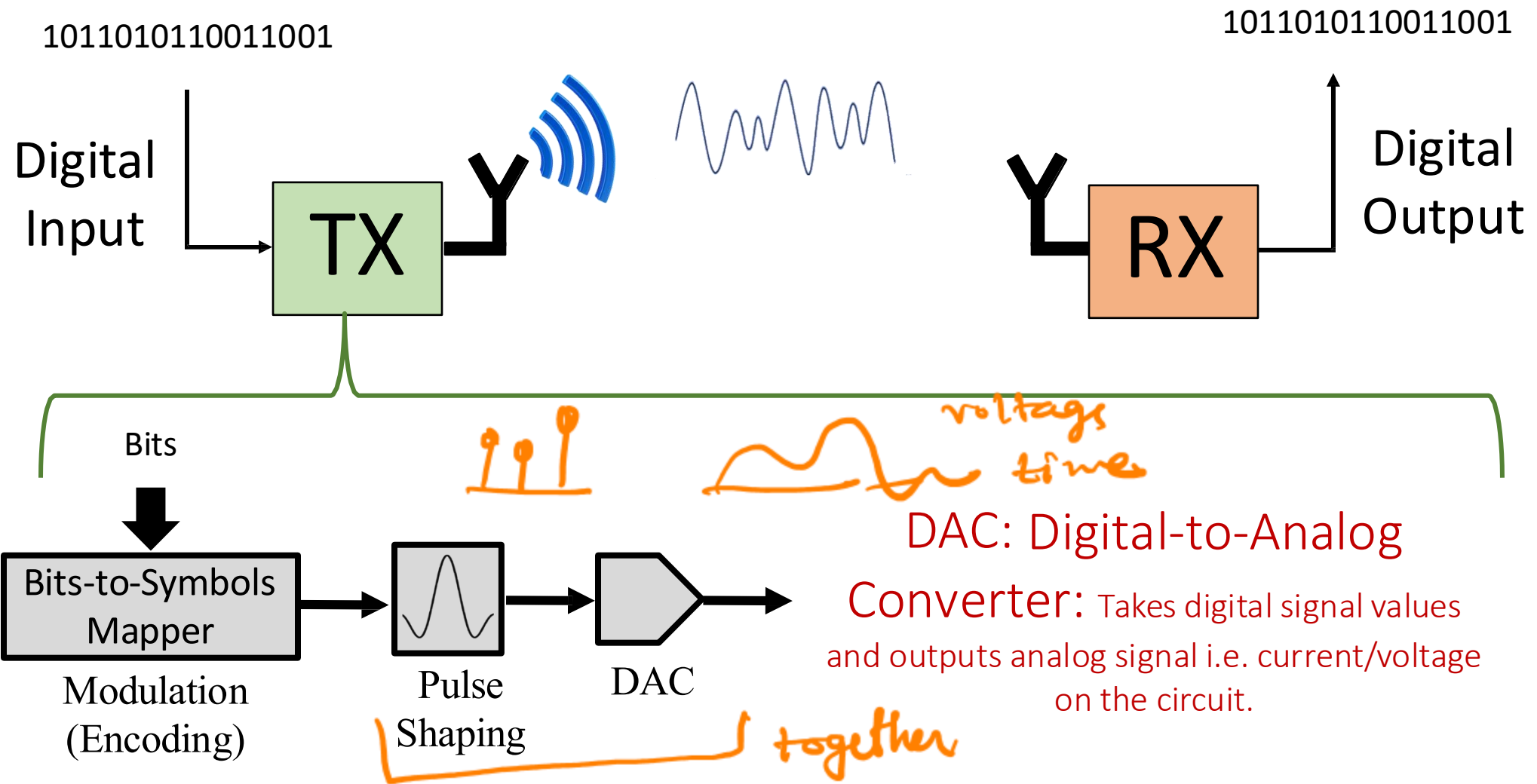
Wireless Communication System



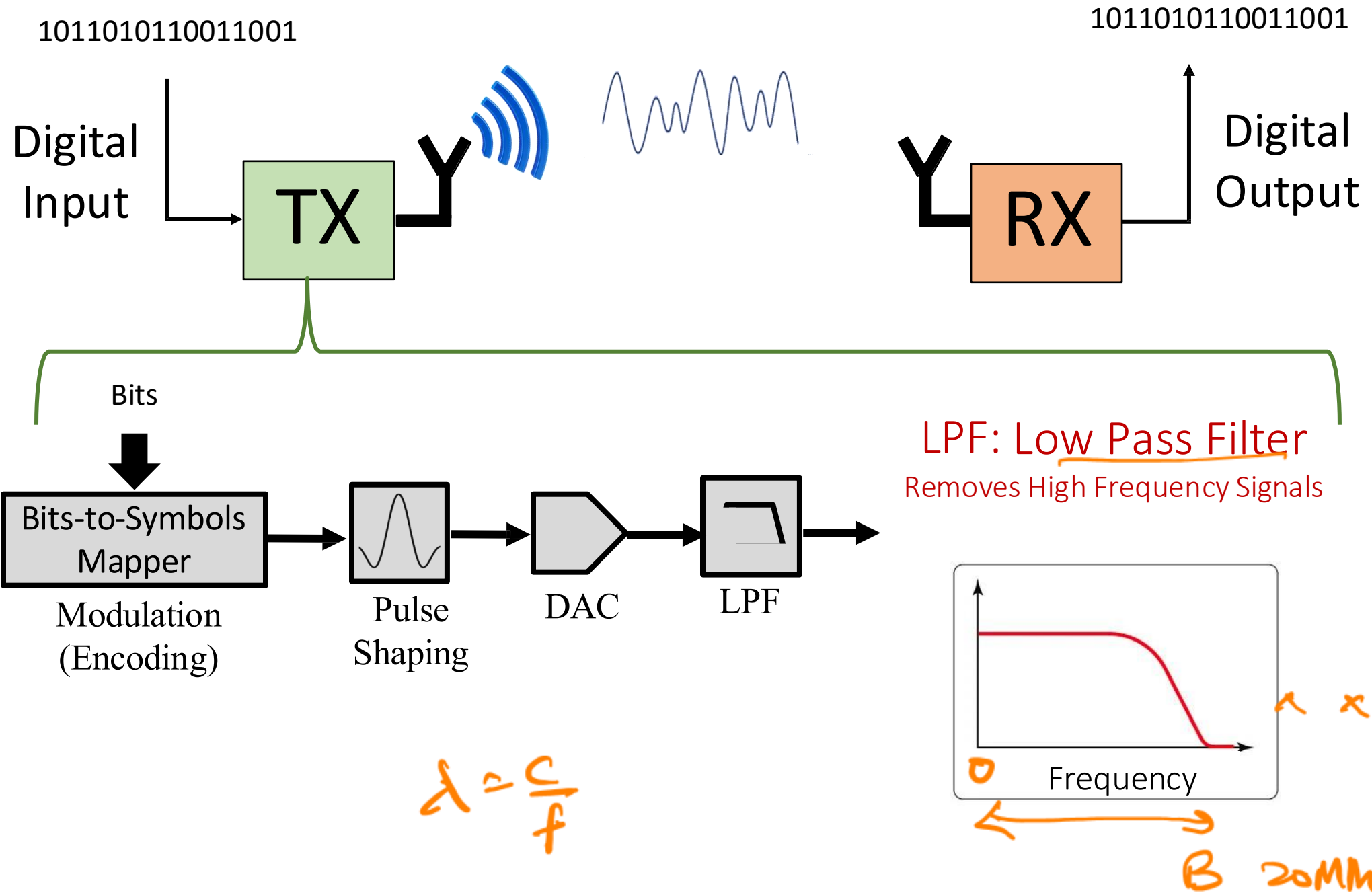
Wireless Communication System



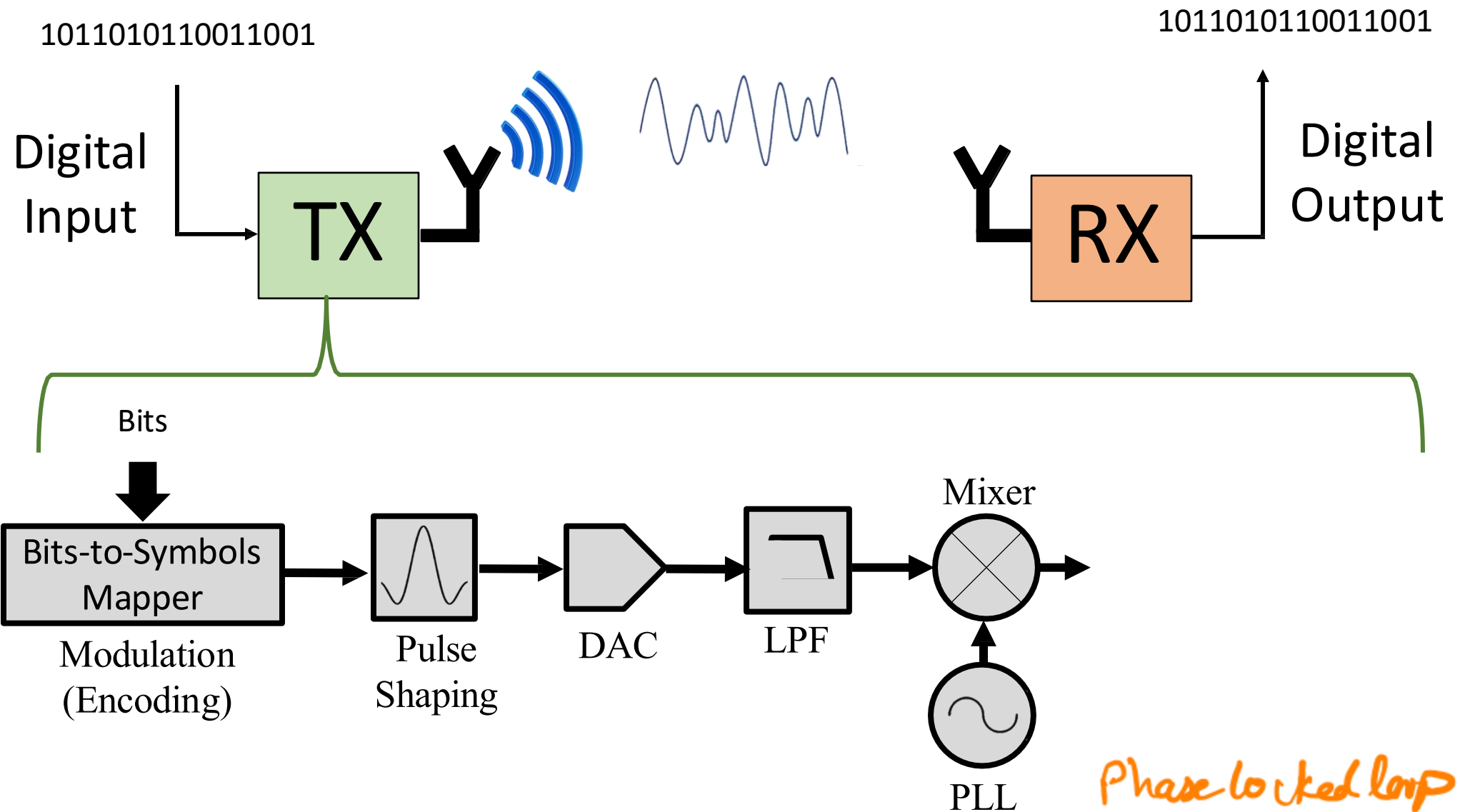
Wireless Communication System



Wireless Communication System

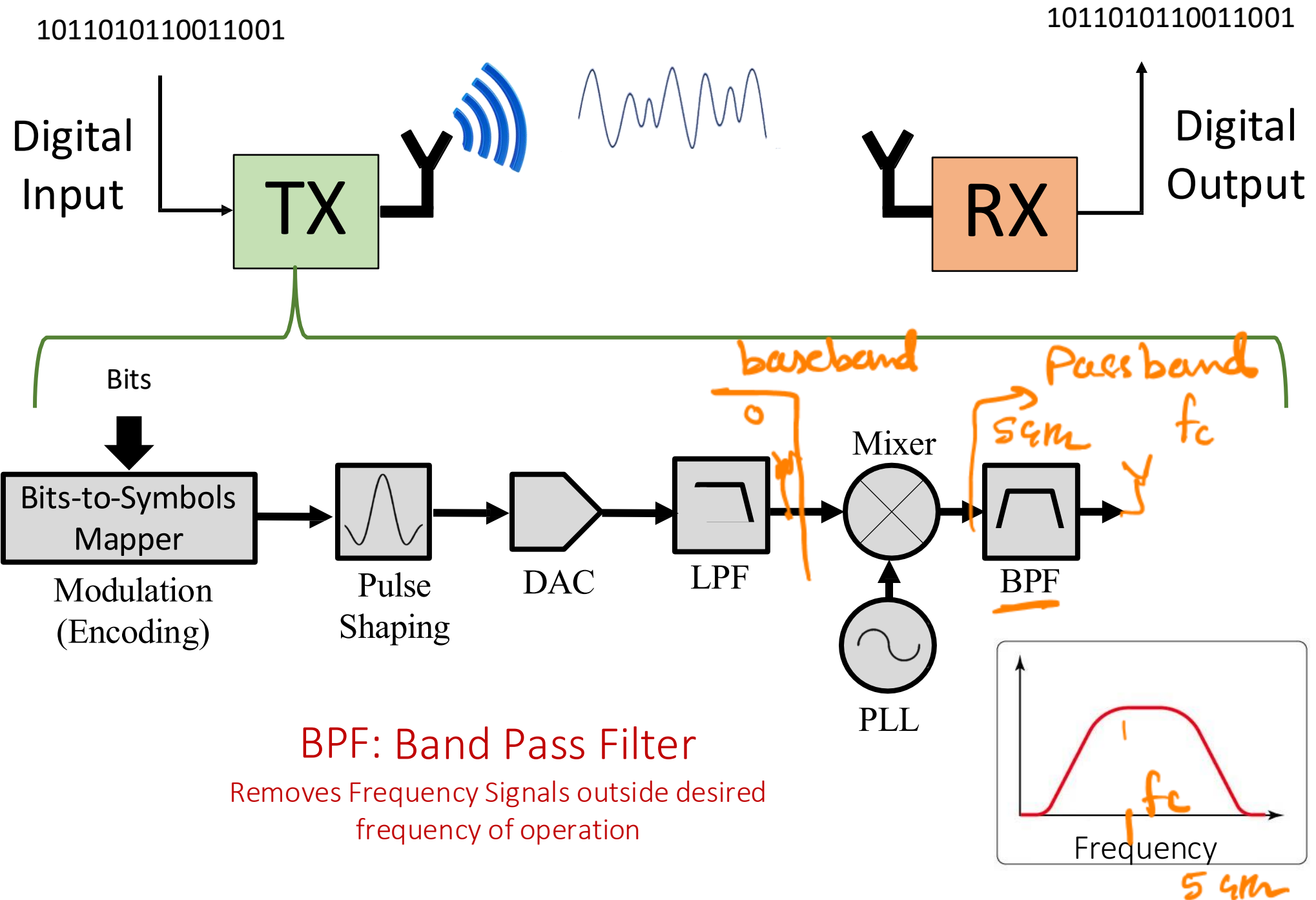


Wireless Communication System

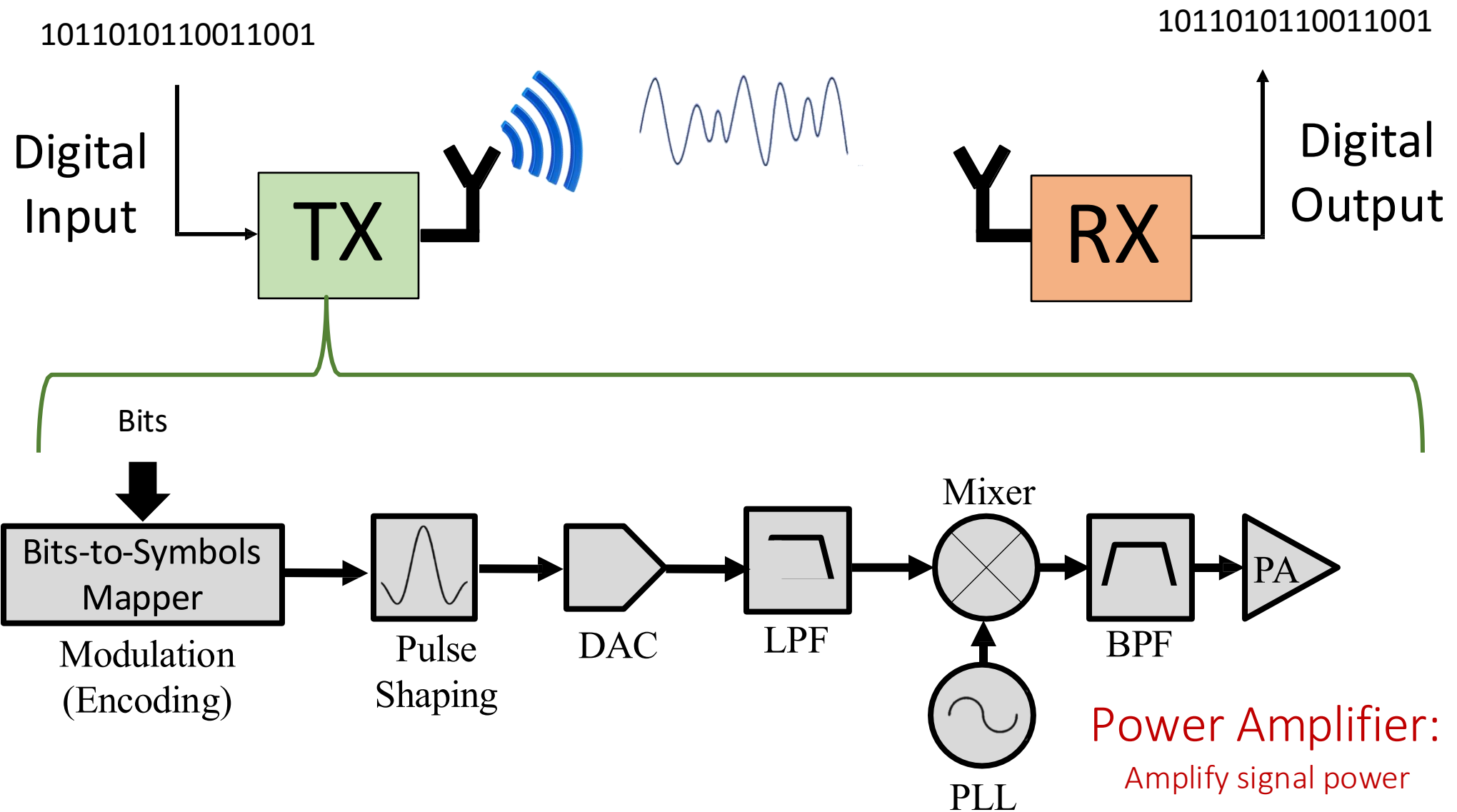


Mixer/PLL: Move signal to desired frequency of operation

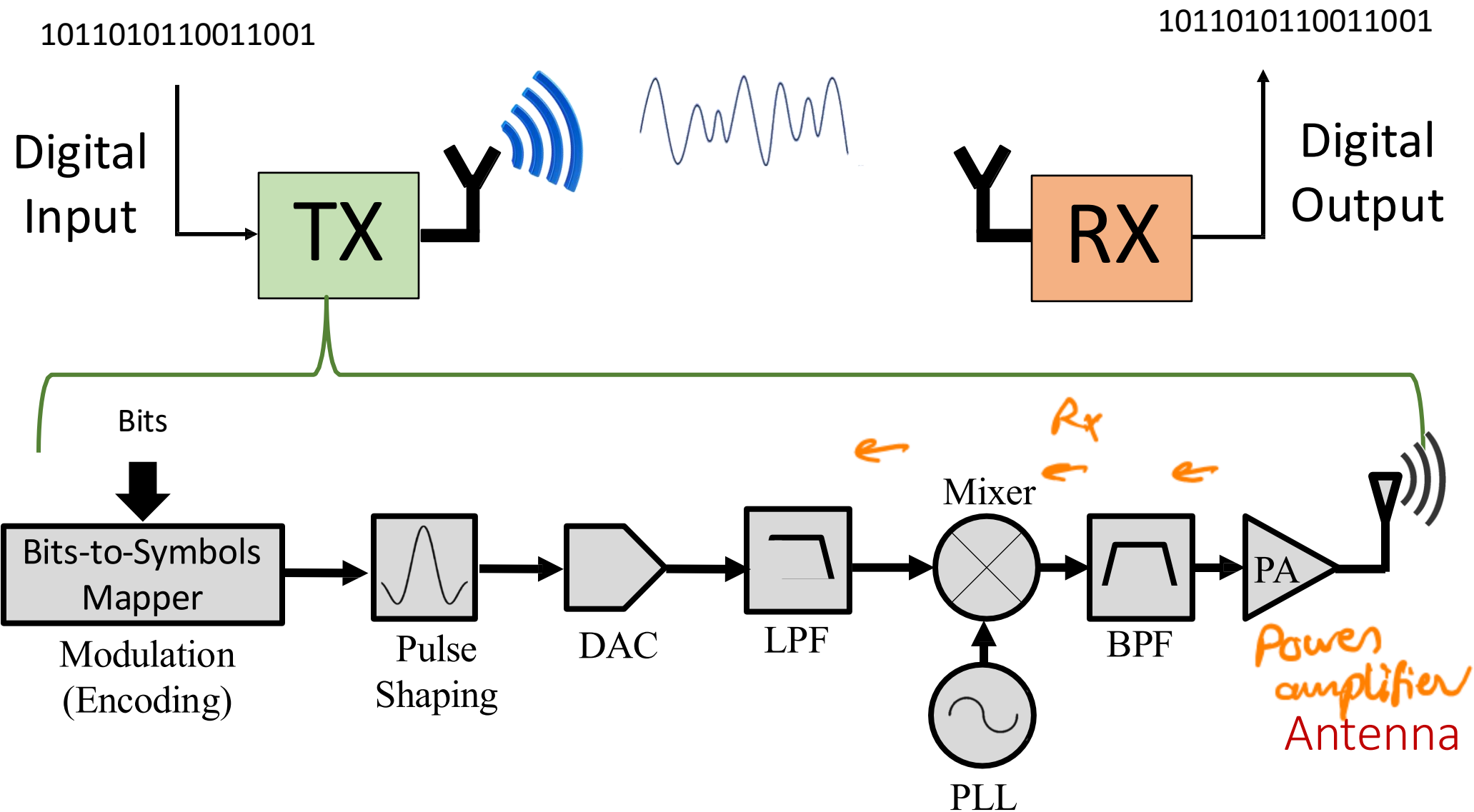
Wireless Communication System



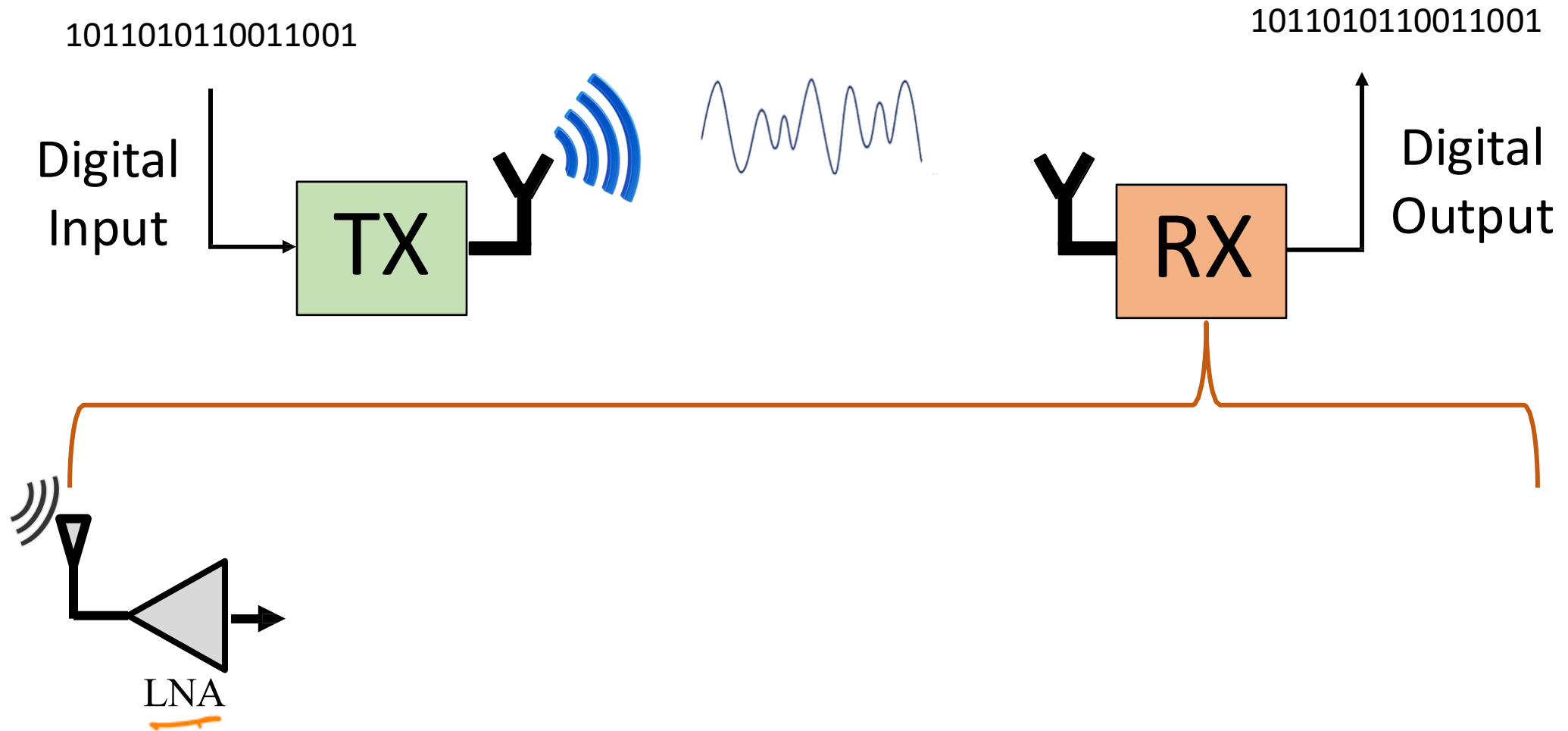
Wireless Communication System



Wireless Communication System

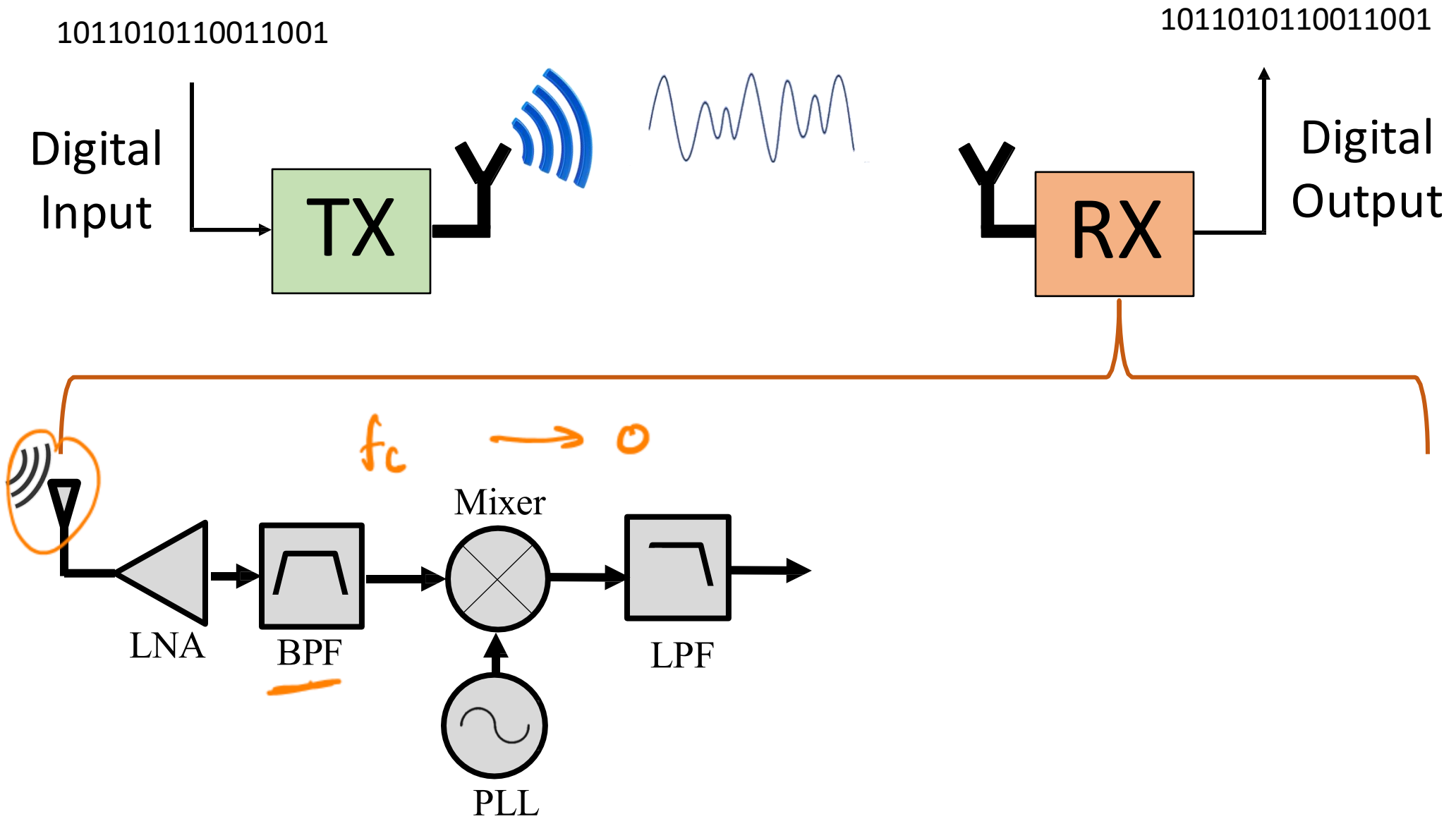


Wireless Communication System

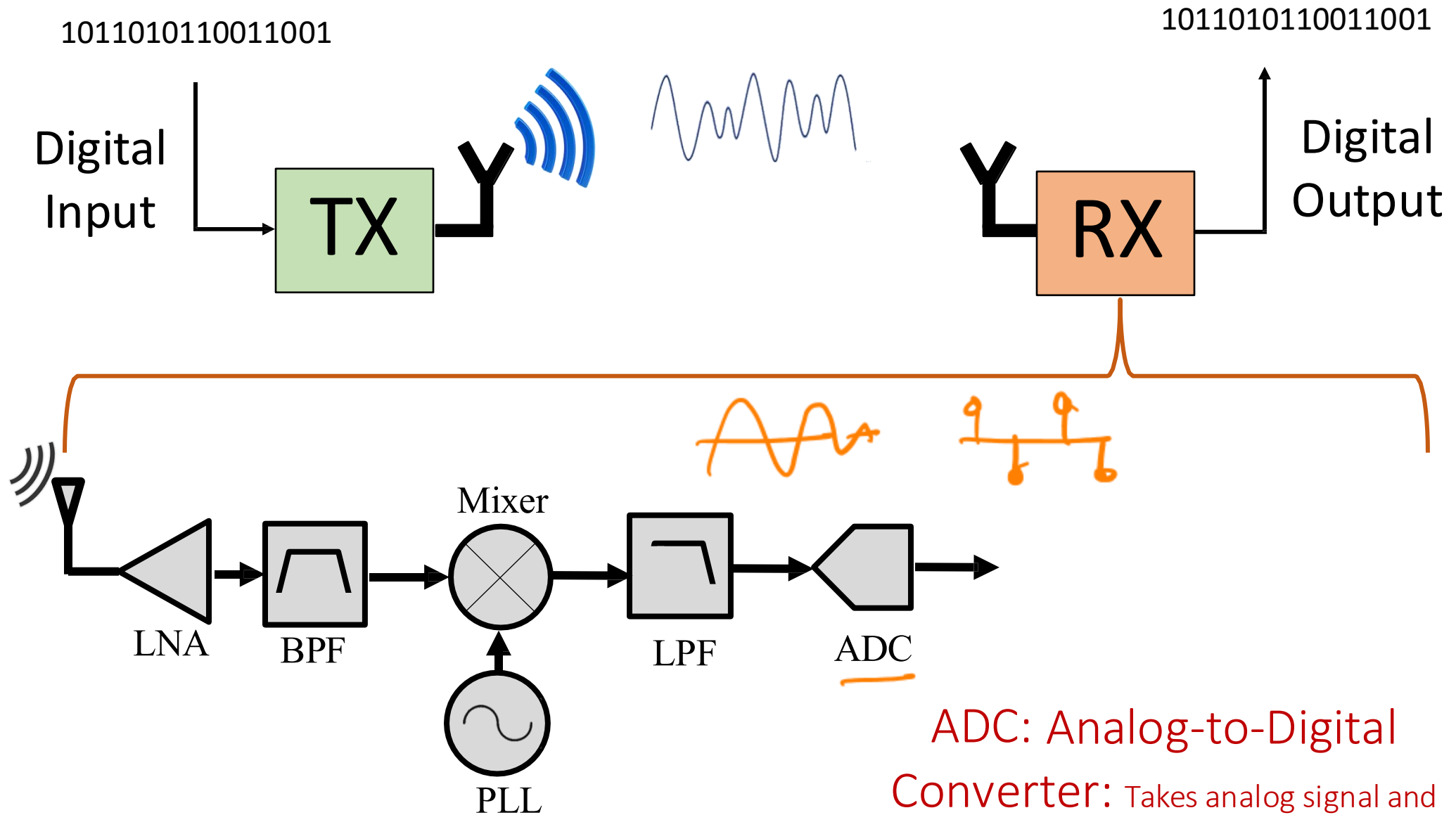


Low Noise Amplifier:
Amplify signal power with minimal
noise

Wireless Communication System

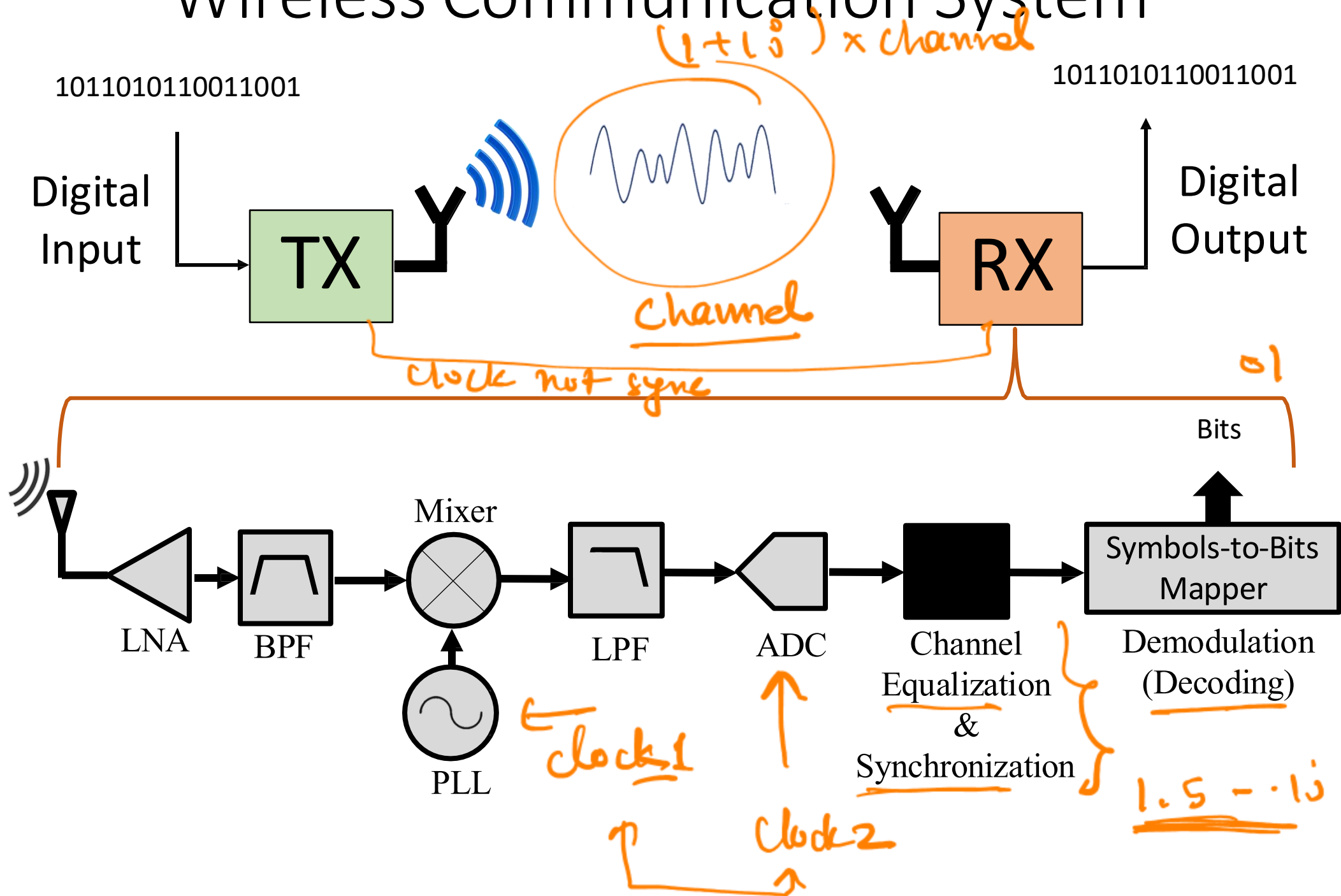


Wireless Communication System



ADC: Analog-to-Digital Converter: Takes analog signal and digitizes it, i.e. samples and quantizes the signal values.

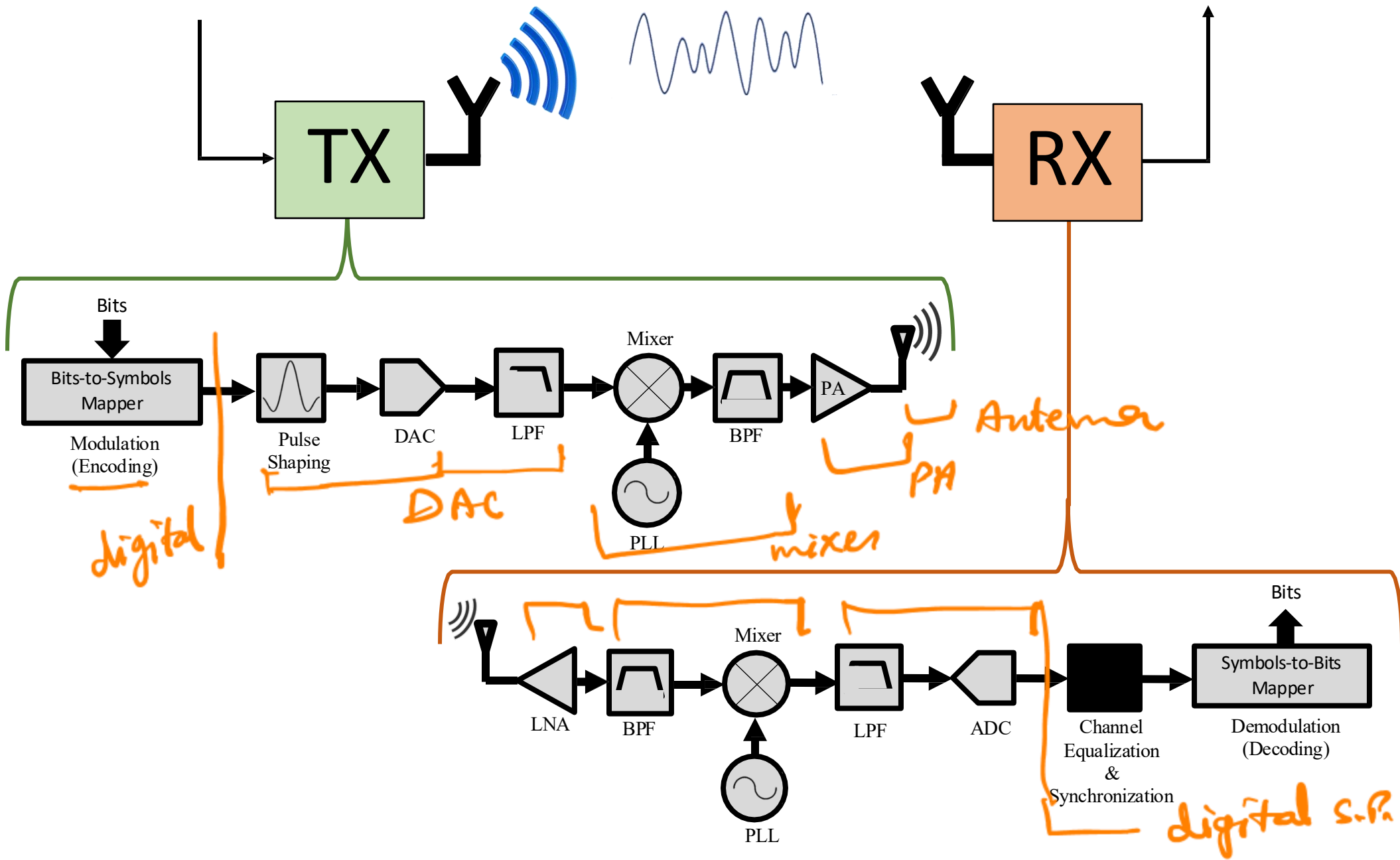
Wireless Communication System



Wireless Communication System

1011010110011001

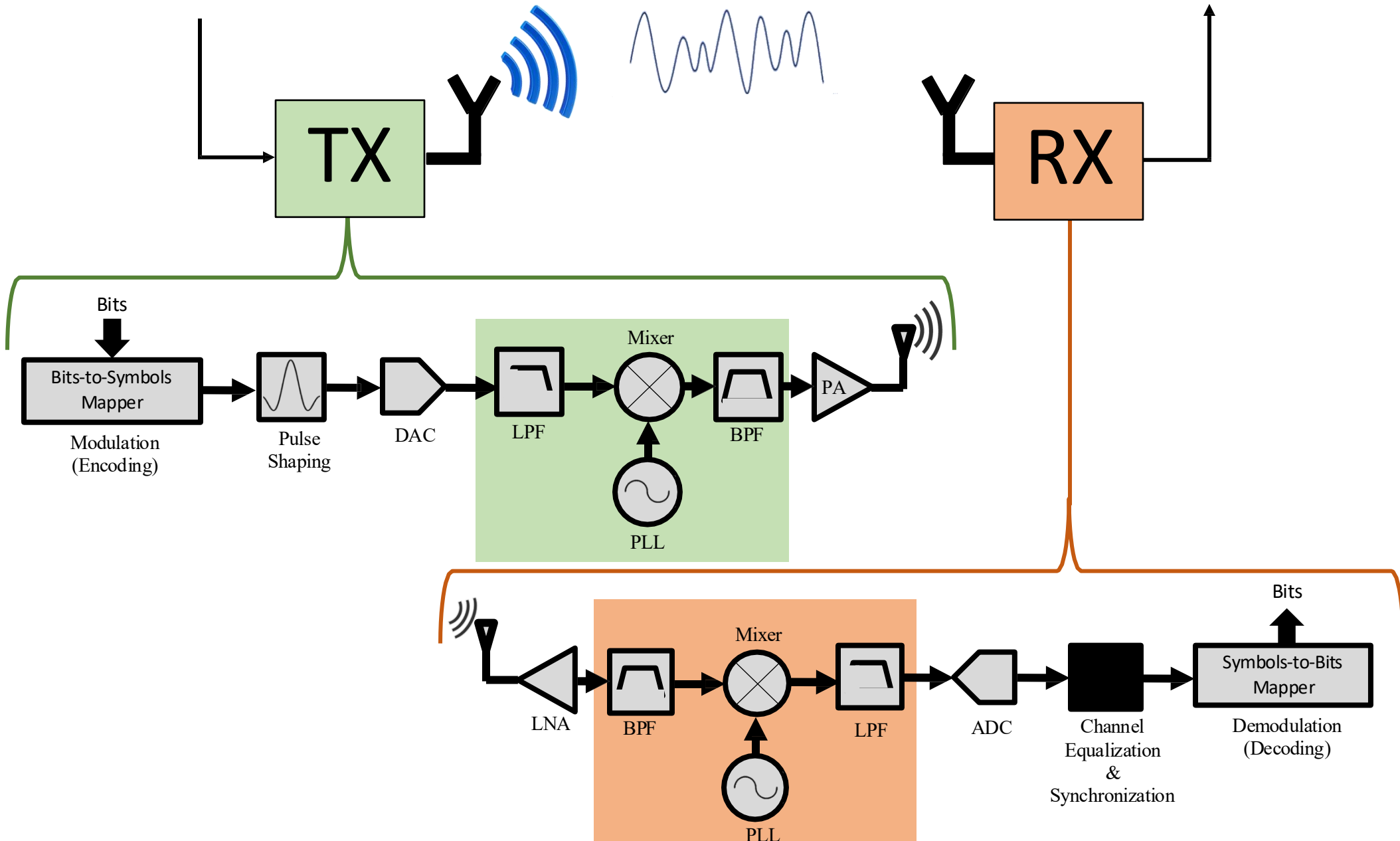
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Wireless Communication System

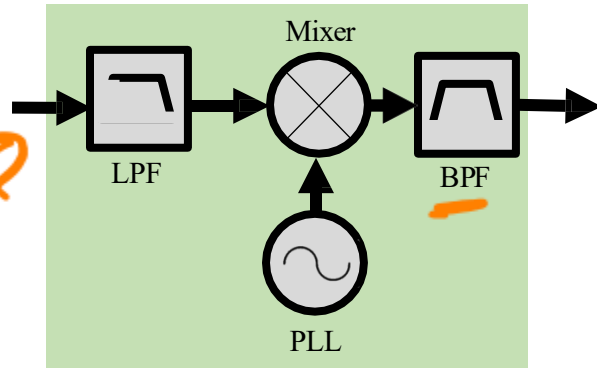
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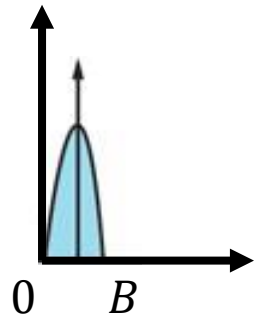


Up/Down Conversion

Transmitter
Up-conversion



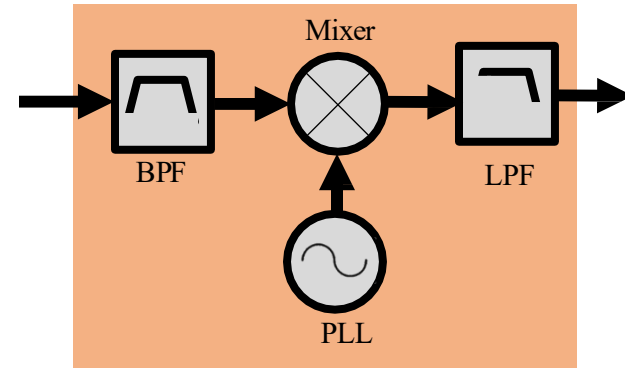
Baseband



Passband

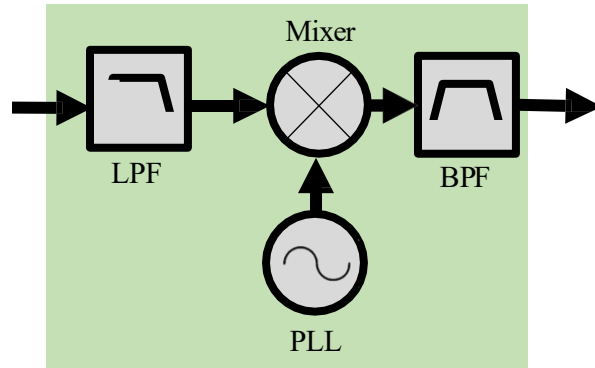


Receiver
Down-conversion

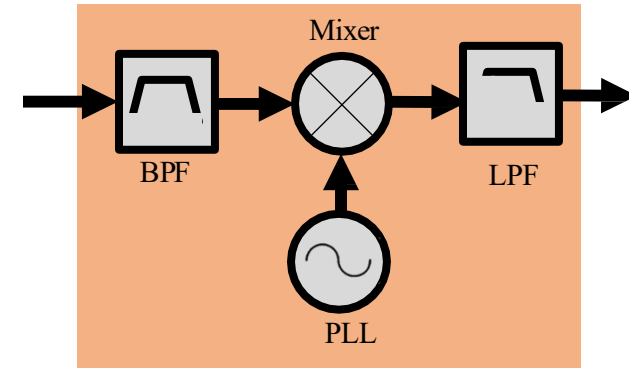


Up/Down Conversion

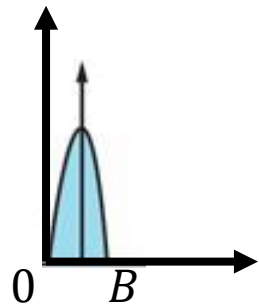
Transmitter
Up-conversion



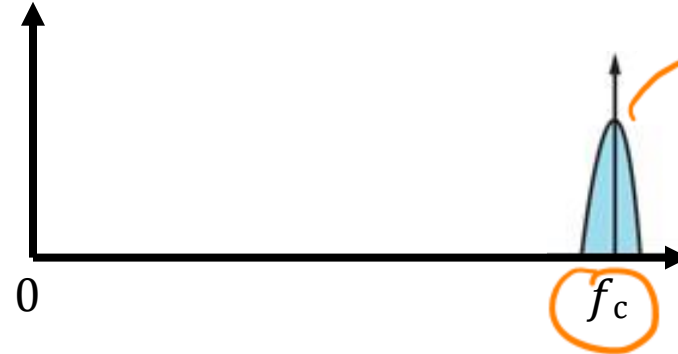
Receiver
Down-conversion



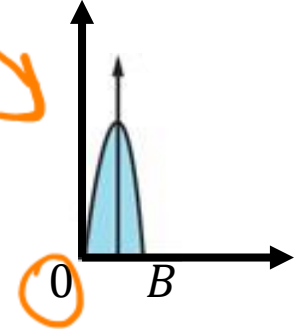
Baseband



Passband



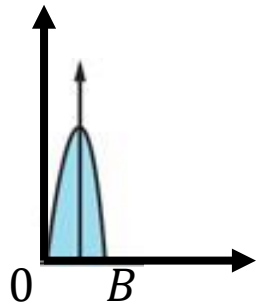
Baseband



Why do we need to up/down conversion?

Up/Down Conversion

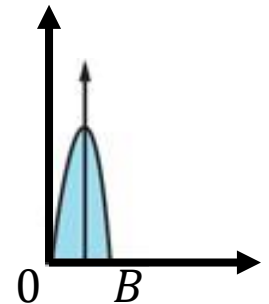
Baseband



Passband

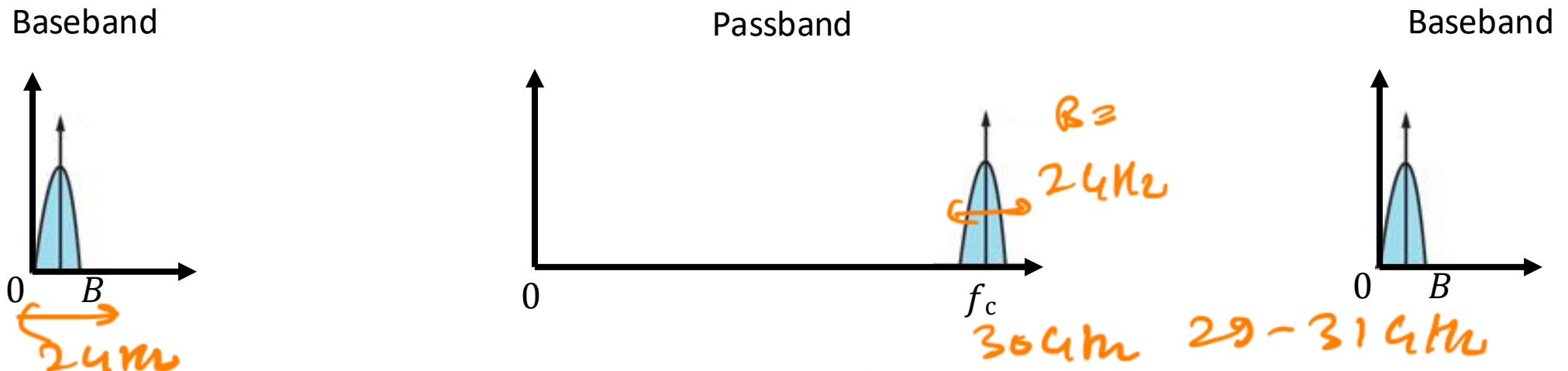


Baseband



Why do we need to up/down conversion?

Up/Down Conversion



Why do we need to up/down conversion?

■ Why not transmit everything at lower frequencies?

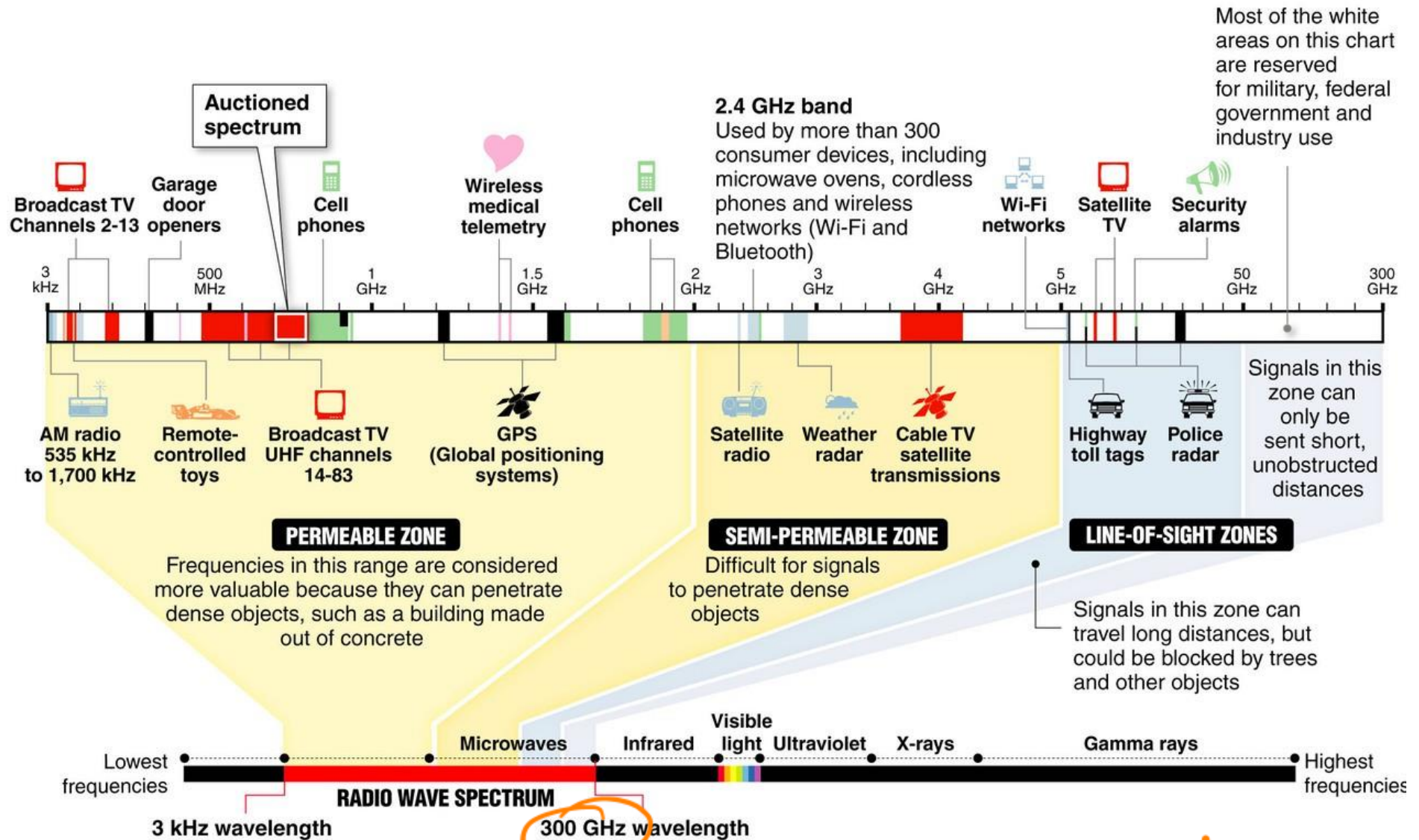
Not enough bandwidth:

Data rate $\propto$ bandwidth

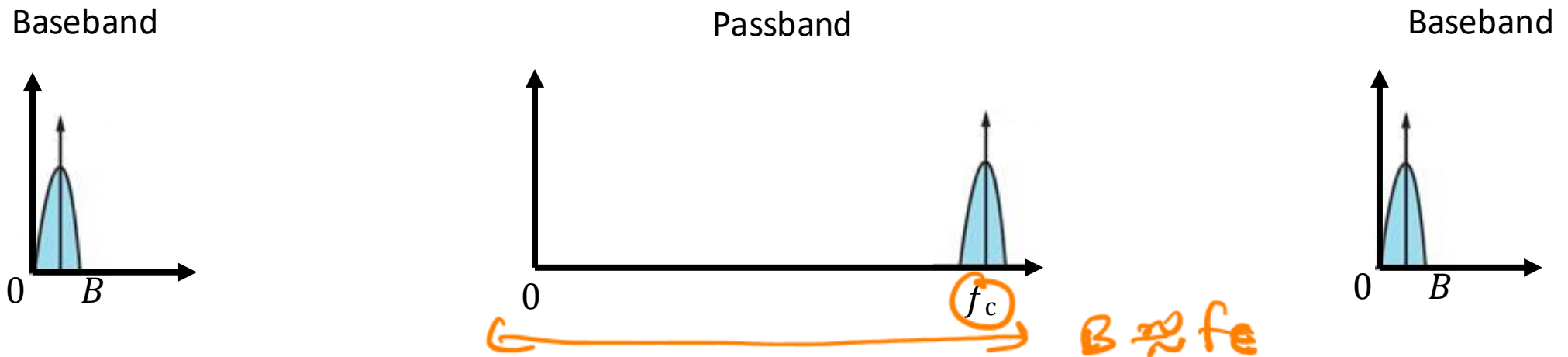
Different Technologies

Antenna size $\propto$ wavelength

The Wireless Spectrum



Up/Down Conversion



Why do we need to up/down conversion?

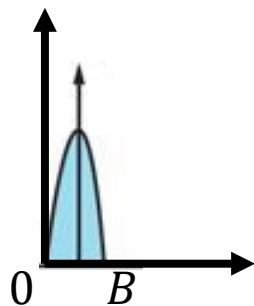
- Cannot transmit everything at low frequencies.
- Why not transmit & receive directly at high frequency?

Nyquist!

Nyquist Theorem

To recover the signal properly, we need to sample at twice the highest frequency, i.e. $2f_{\max}$

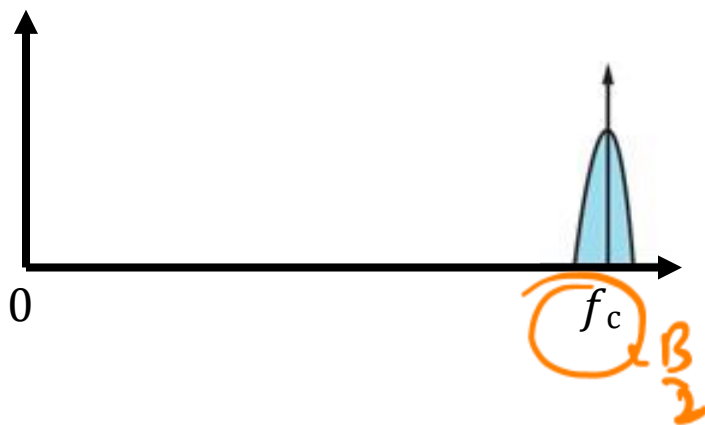
Baseband



➡ Sample at $2B$

e.g. WiFi 802.11b ≈ 40 MS/s

Passband



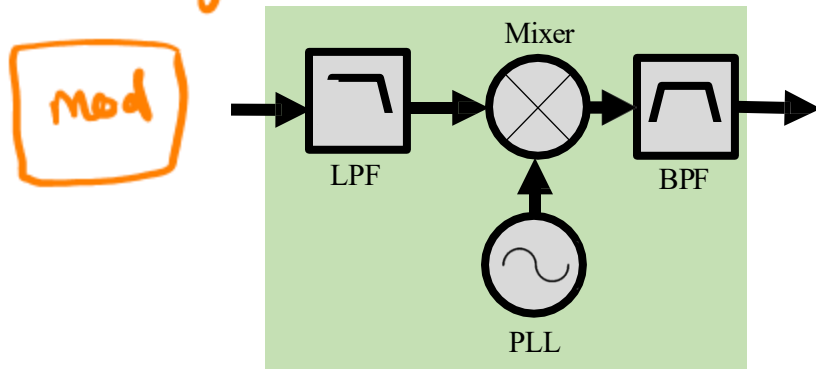
➡ Sample at $2f_c + B$

e.g. WiFi 802.11b ≈ 5 GS/s

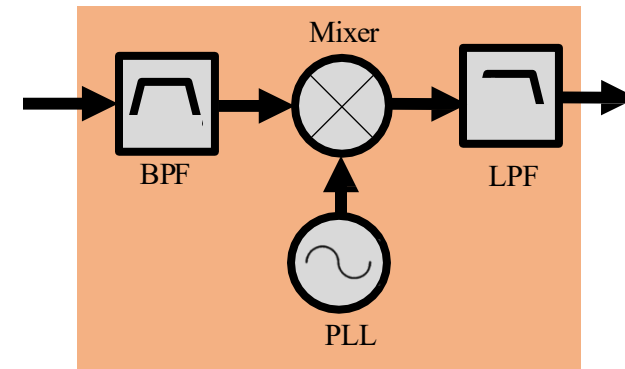
Up/Down Conversion

0101 $s(t)$ symbol
mod

Transmitter
Up-conversion



Receiver
Down-conversion

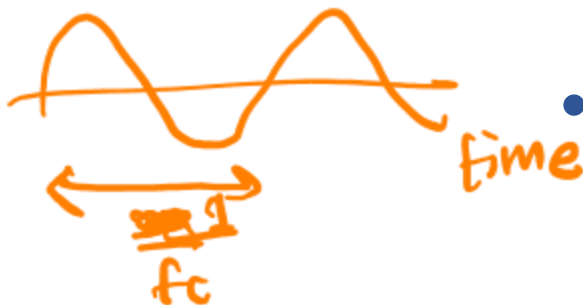


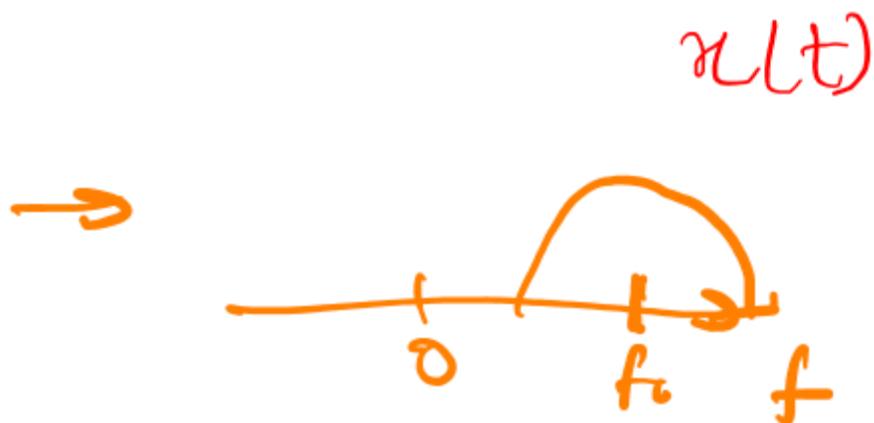
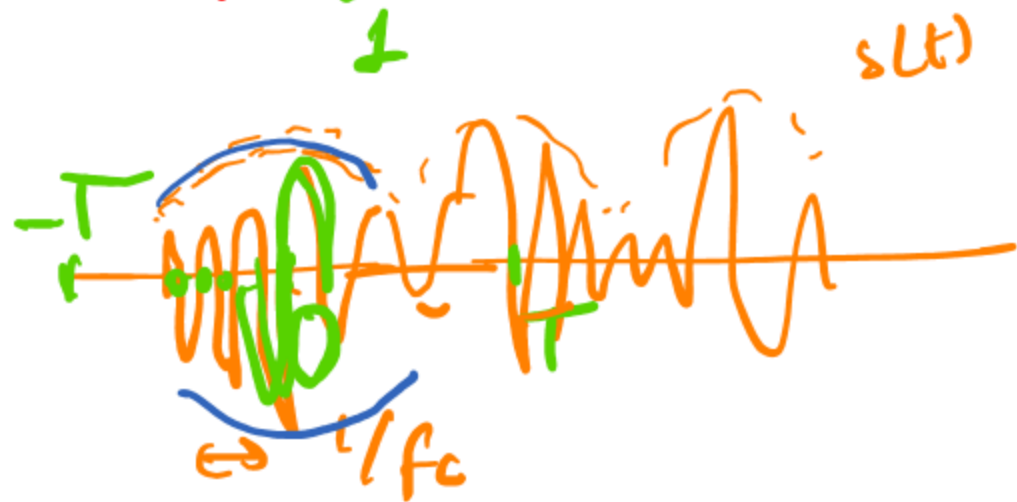
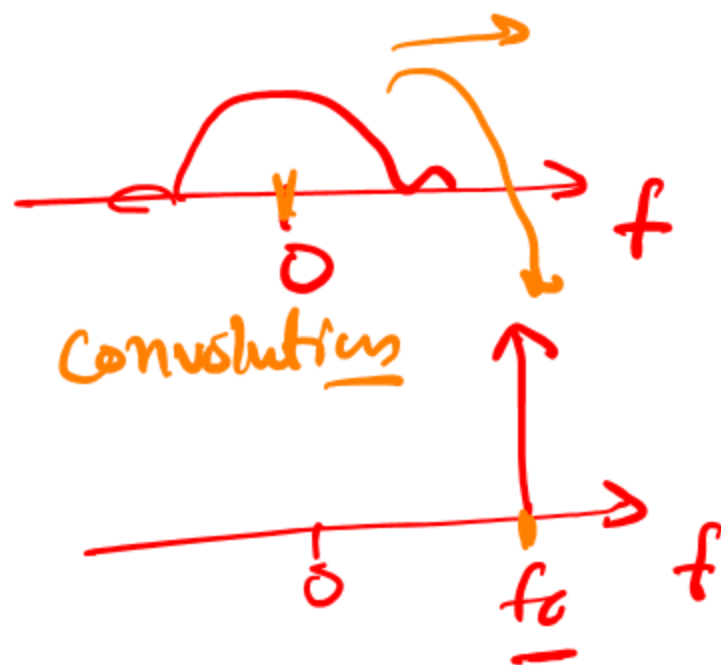
How to we do up/down conversion?

$$\underline{s(t)} \rightarrow \times \rightarrow \underline{s(t) \cos(2\pi f_c t)}$$

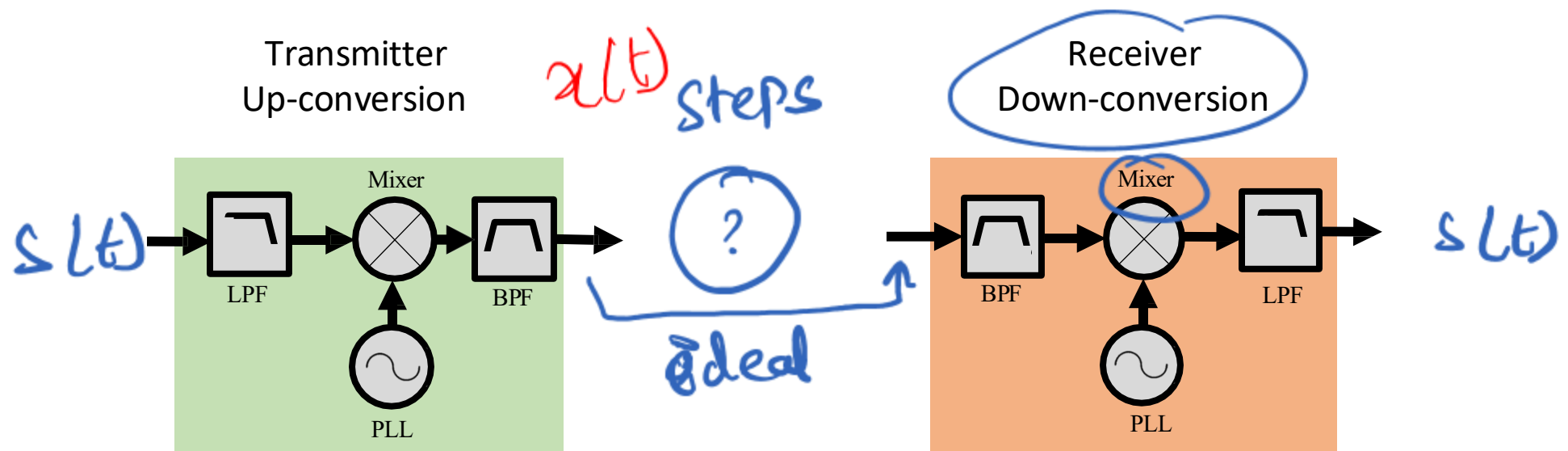
$$\cos(2\pi f_c t)$$

- PLL (Phased Lock Loop): create the cosine wave signal $\cos(2\pi f_c t)$
- Mixer: multiplies the two signals





Up/Down Conversion



How to we do up/down conversion?

$$\begin{aligned}
 \underbrace{s(t)}_{\text{circled in blue}} &\xrightarrow{\uparrow \cos(2\pi f_c t)} \times \rightarrow \underbrace{s(t) \cos(2\pi f_c t)}_{x(t) \text{ (handwritten in red)}} \xrightarrow{\uparrow \cos(2\pi f_c t) \text{ (circled in blue)}} \times \rightarrow \underbrace{s(t) \cos^2(2\pi f_c t)}_{\text{circled in blue}}
 \end{aligned}$$

$$2 \cos^2 \theta = 1 + \cos 2\theta$$

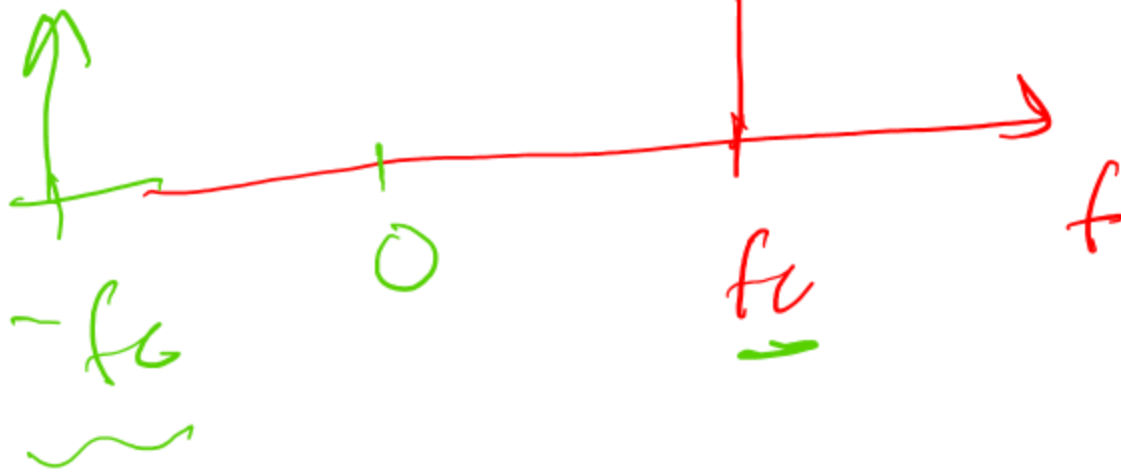
$$\underbrace{s(t) \cos^2(2\pi f_c t)}_{\text{circled in blue}} = \underbrace{s(t)}_{\text{circled in blue}} \left(\frac{1}{2} + \frac{1}{2} \cos(2\pi \underbrace{2f_c t}_{\text{circled in blue}}) \right)$$

Rx chain $X(f)$



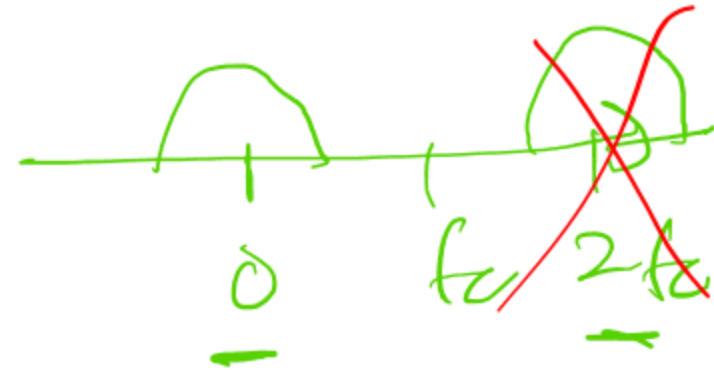
(conv.)

$\cos 2\pi f_c t$



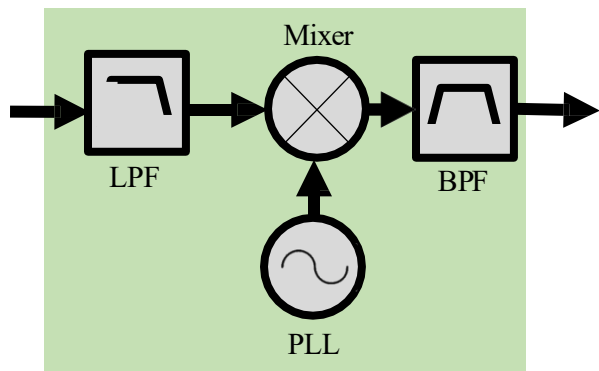
desired

not
desired

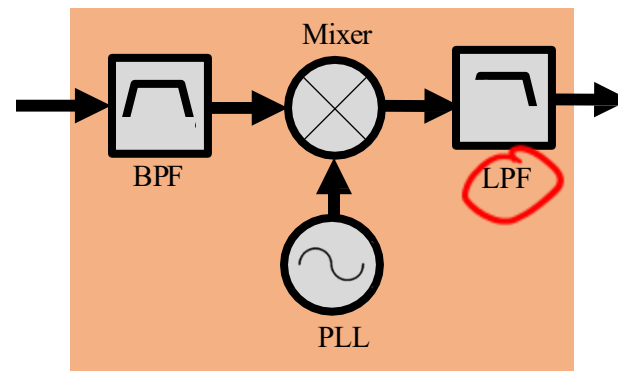


Up/Down Conversion

Transmitter
Up-conversion



Receiver
Down-conversion

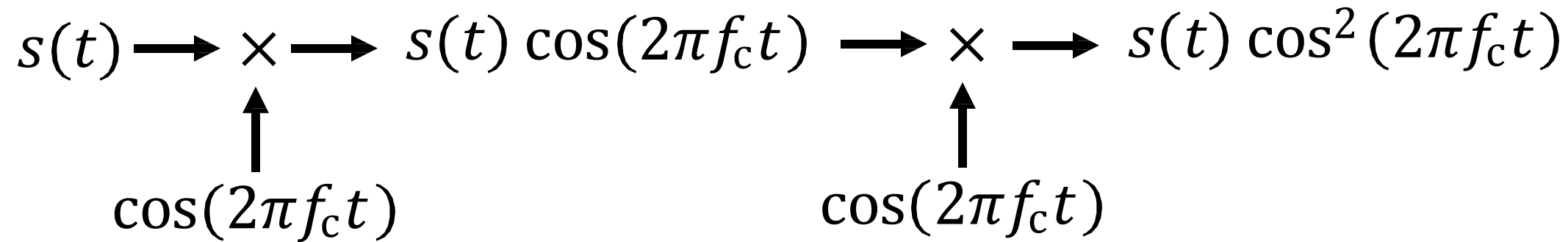


How to we do up/down conversion?

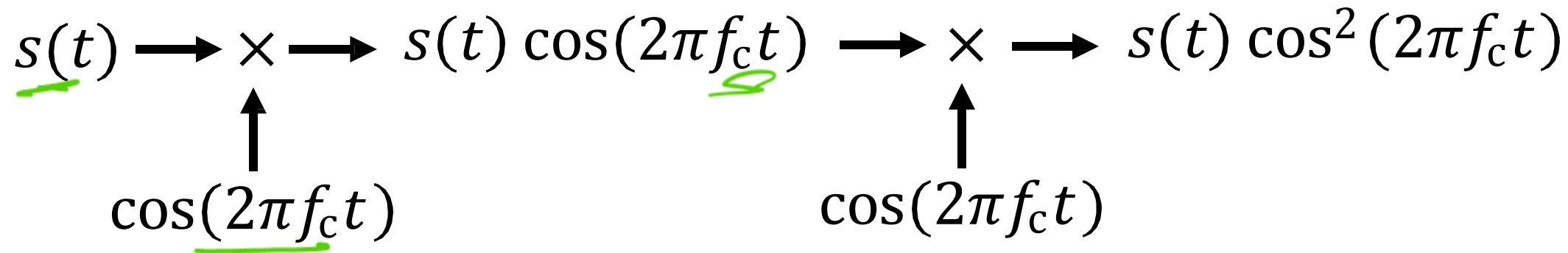
$$s(t) \xrightarrow{\uparrow \frac{1}{\sqrt{2}} \cos(2\pi f_c t)} \times \rightarrow s(t) \cos(2\pi f_c t) \xrightarrow{\uparrow \frac{1}{\sqrt{2}} \cos(2\pi f_c t)} \times \rightarrow s(t) \cos^2(2\pi f_c t)$$

$$s(t) \cos^2(2\pi f_c t) = s(t) \left(\frac{1}{2} + \frac{1}{2} \cos(4\pi f_c t) \right) \xrightarrow{\text{LPF Low Pass Filter}} \underline{\underline{\frac{1}{2} s(t)}}$$

Up/Down Conversion



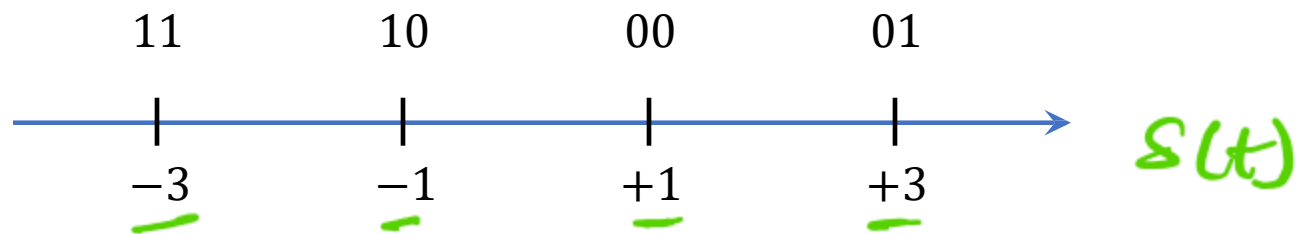
Up/Down Conversion



Consider using PAM: Pulse Amplitude Modulation

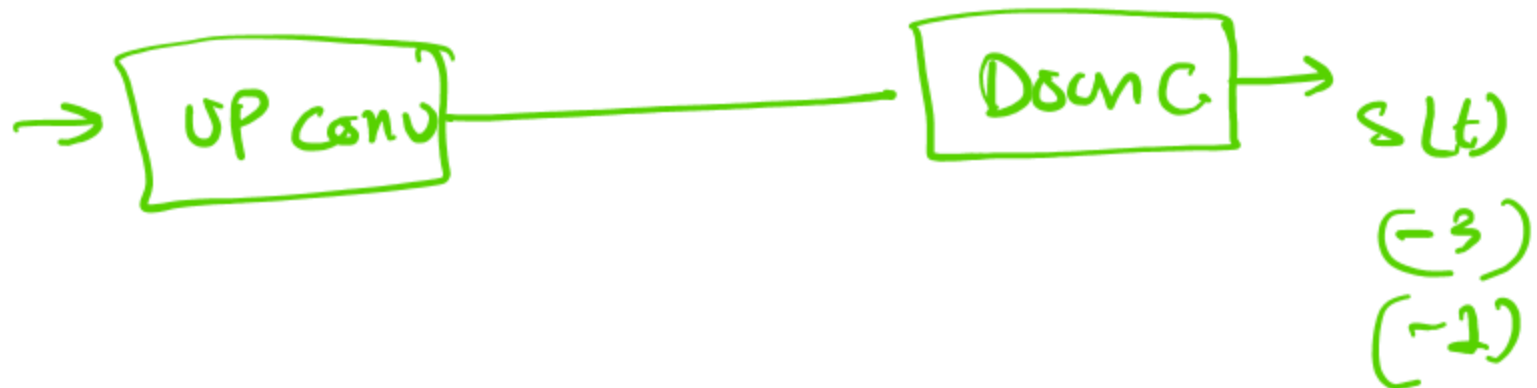
$2^2 = 4$

4 PAM:
2 bits/symbol

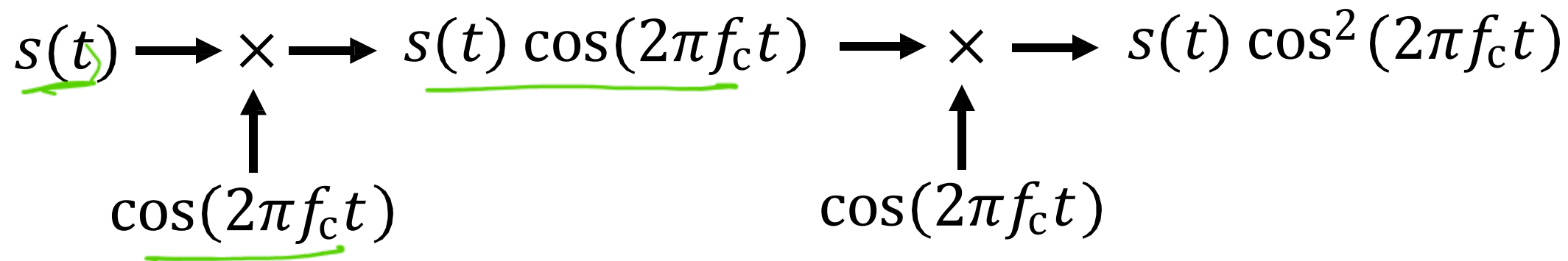


red

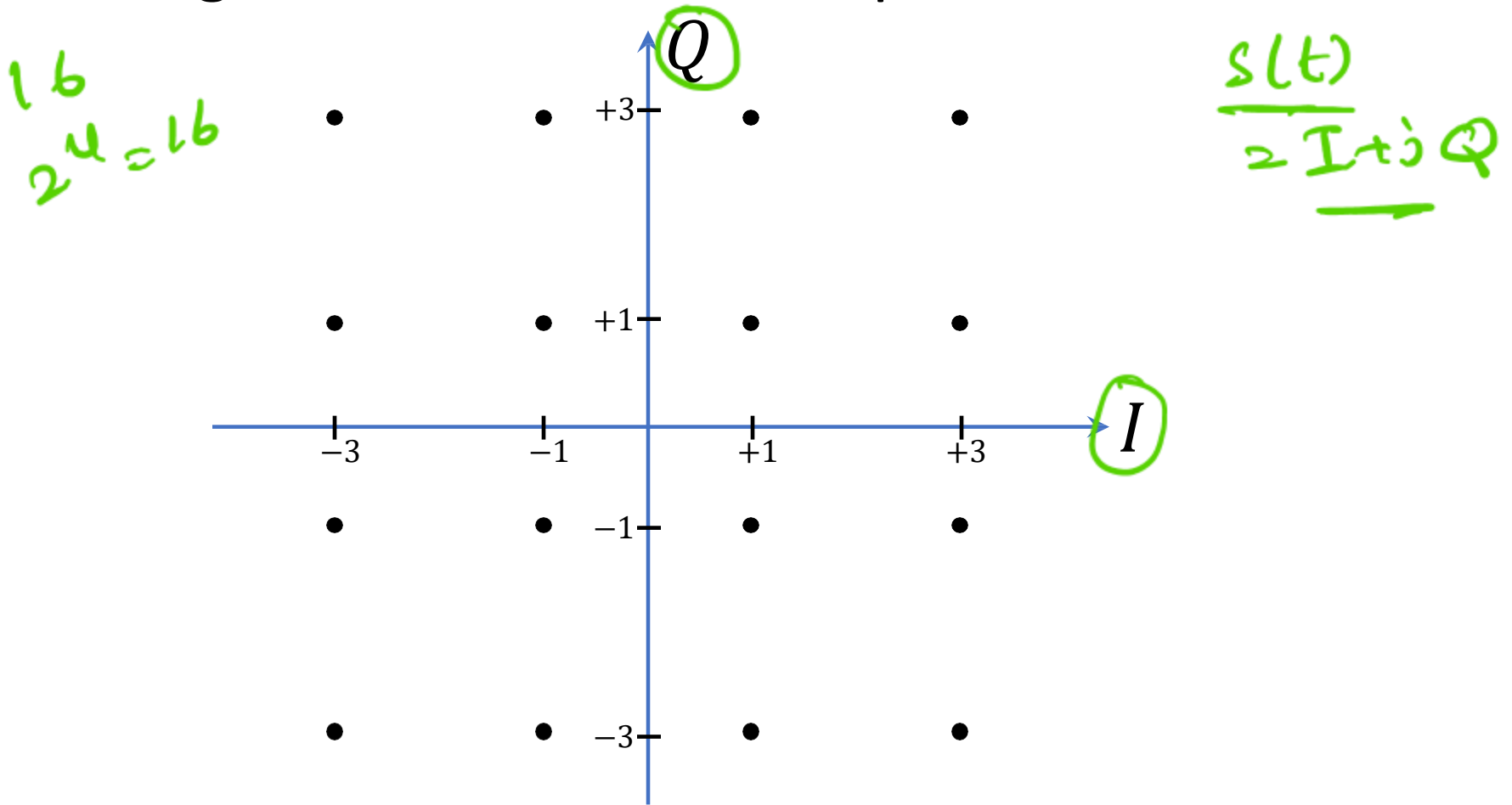
$s(t)$
 (-3)
 (-1)



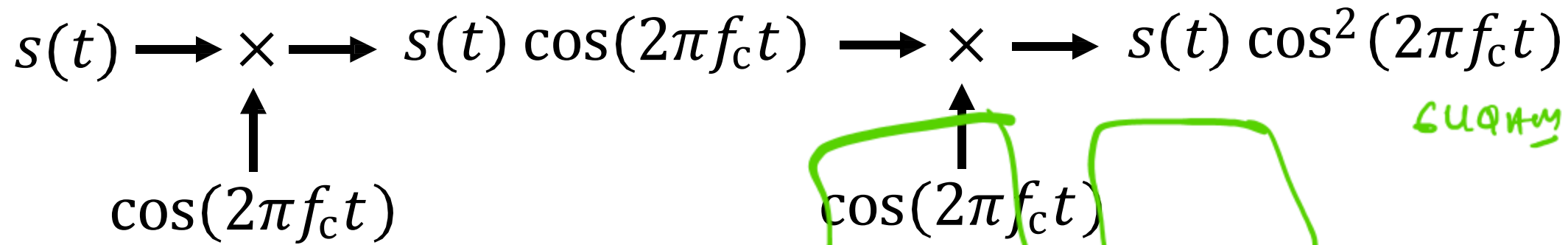
Up/Down Conversion



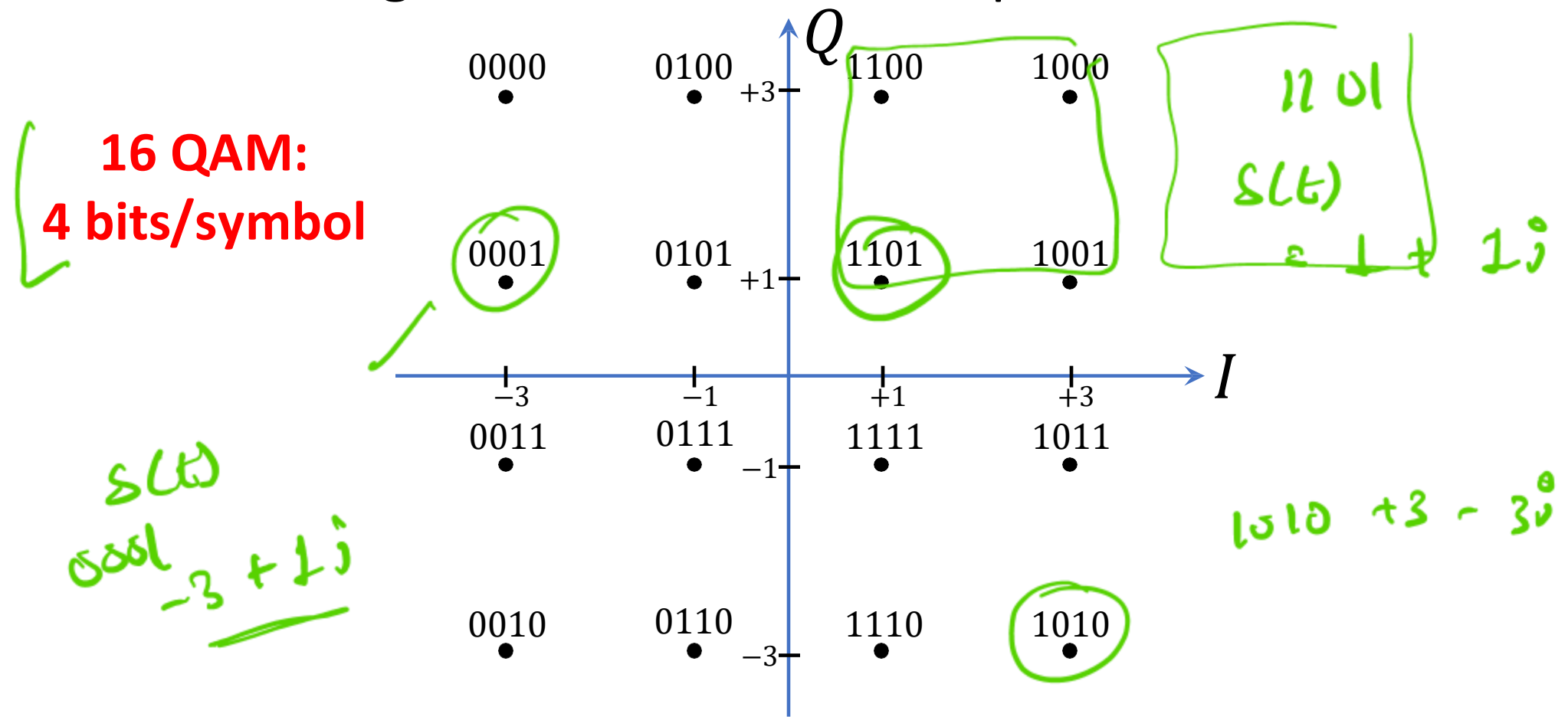
Consider using QAM: Quadrature Amplitude Modulation



Up/Down Conversion



Consider using QAM: Quadrature Amplitude Modulation



Up/Down Conversion

$$s(t) \xrightarrow{\uparrow \cos(2\pi f_c t)} \times \longrightarrow \underline{s(t) \cos(2\pi f_c t)} \xrightarrow{\uparrow \cos(2\pi f_c t)} \times \longrightarrow s(t) \cos^2(2\pi f_c t)$$

Consider using QAM: Quadrature Amplitude Modulation

$s(t)$ is complex: $s(t) = I + jQ$

Up Conversion

$$\underline{s(t)} \longrightarrow \times \longrightarrow s(t) \cos(2\pi f_c t)$$

$\uparrow$
 $\cos(2\pi f_c t)$

Traditional real chain

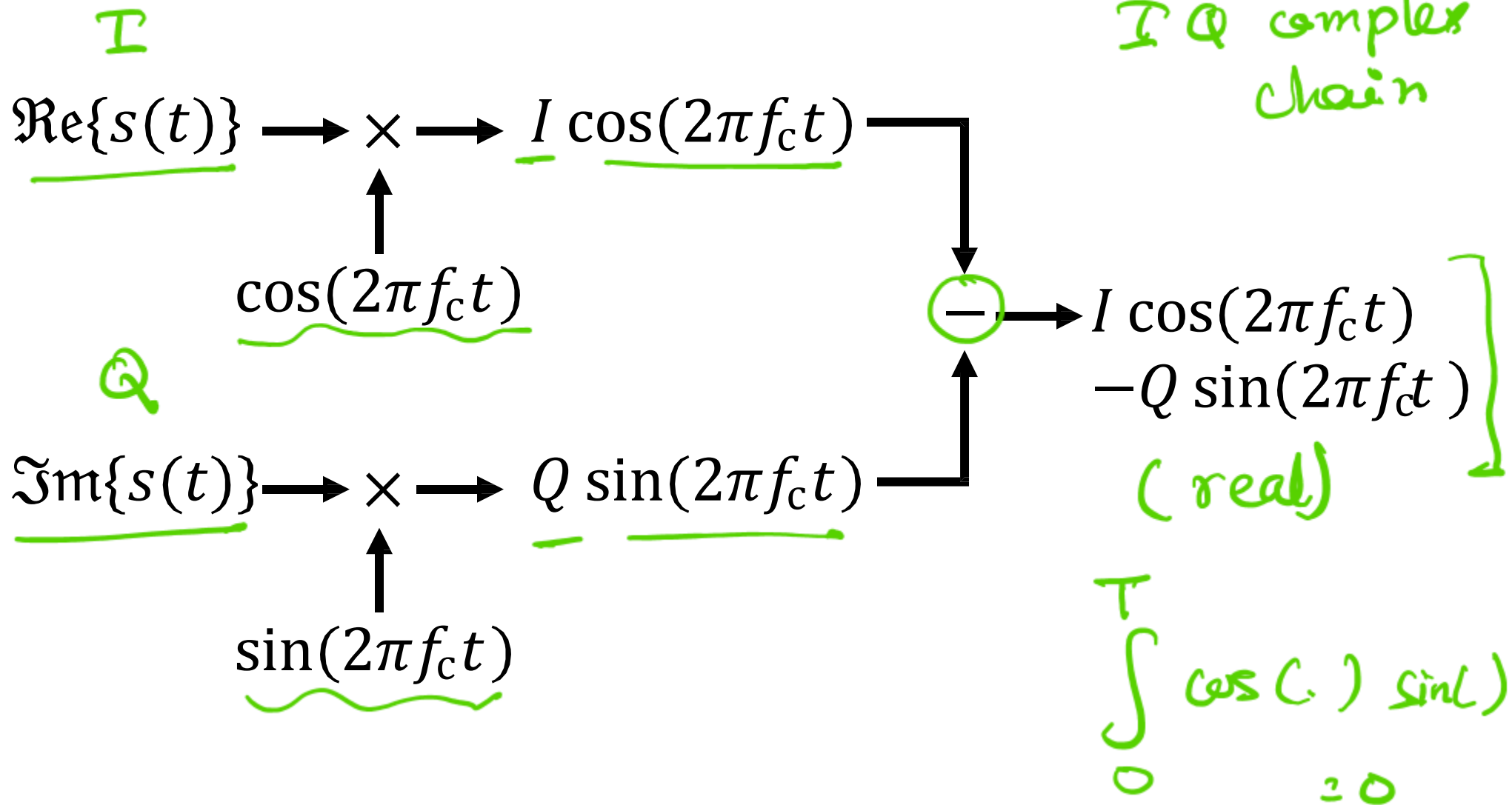
Consider using QAM: Quadrature Amplitude Modulation

$$s(t) \text{ is complex: } s(t) = I + jQ$$

Up Conversion

modified

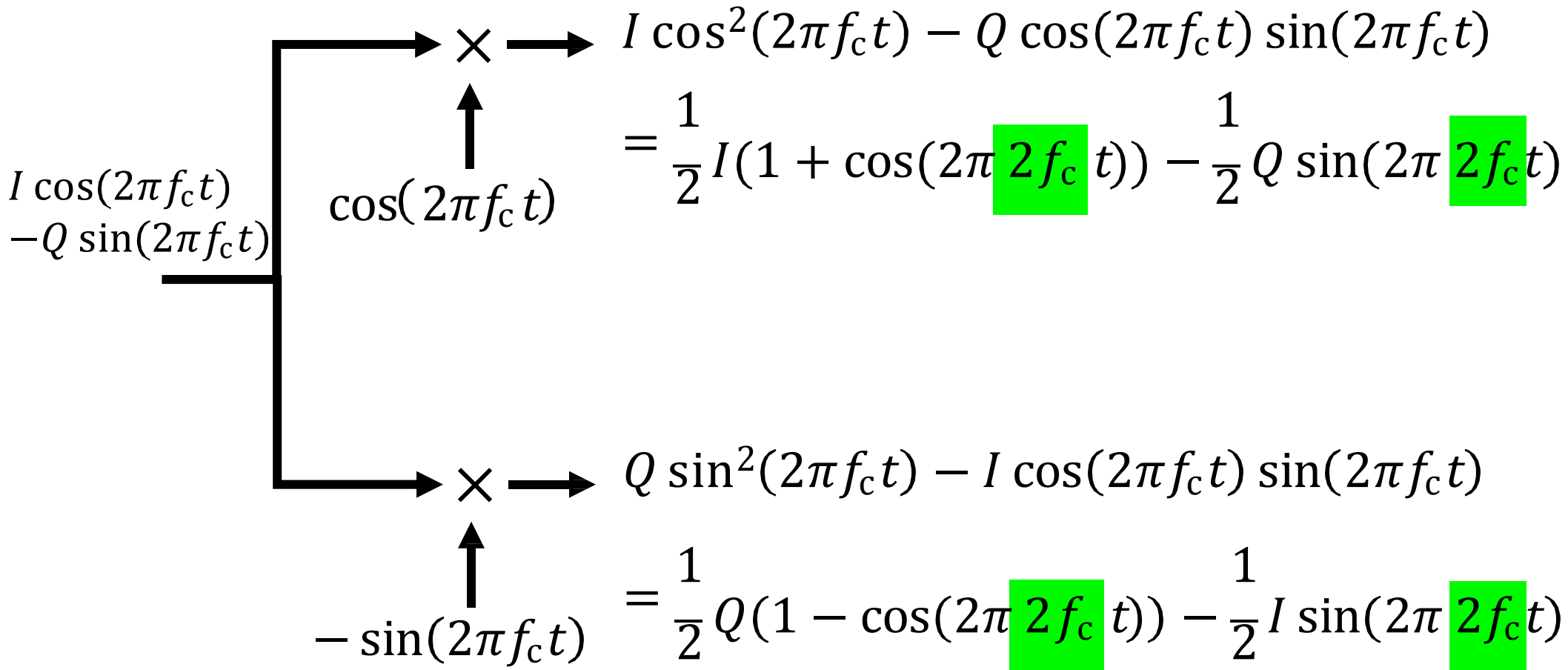
I Q complex chain



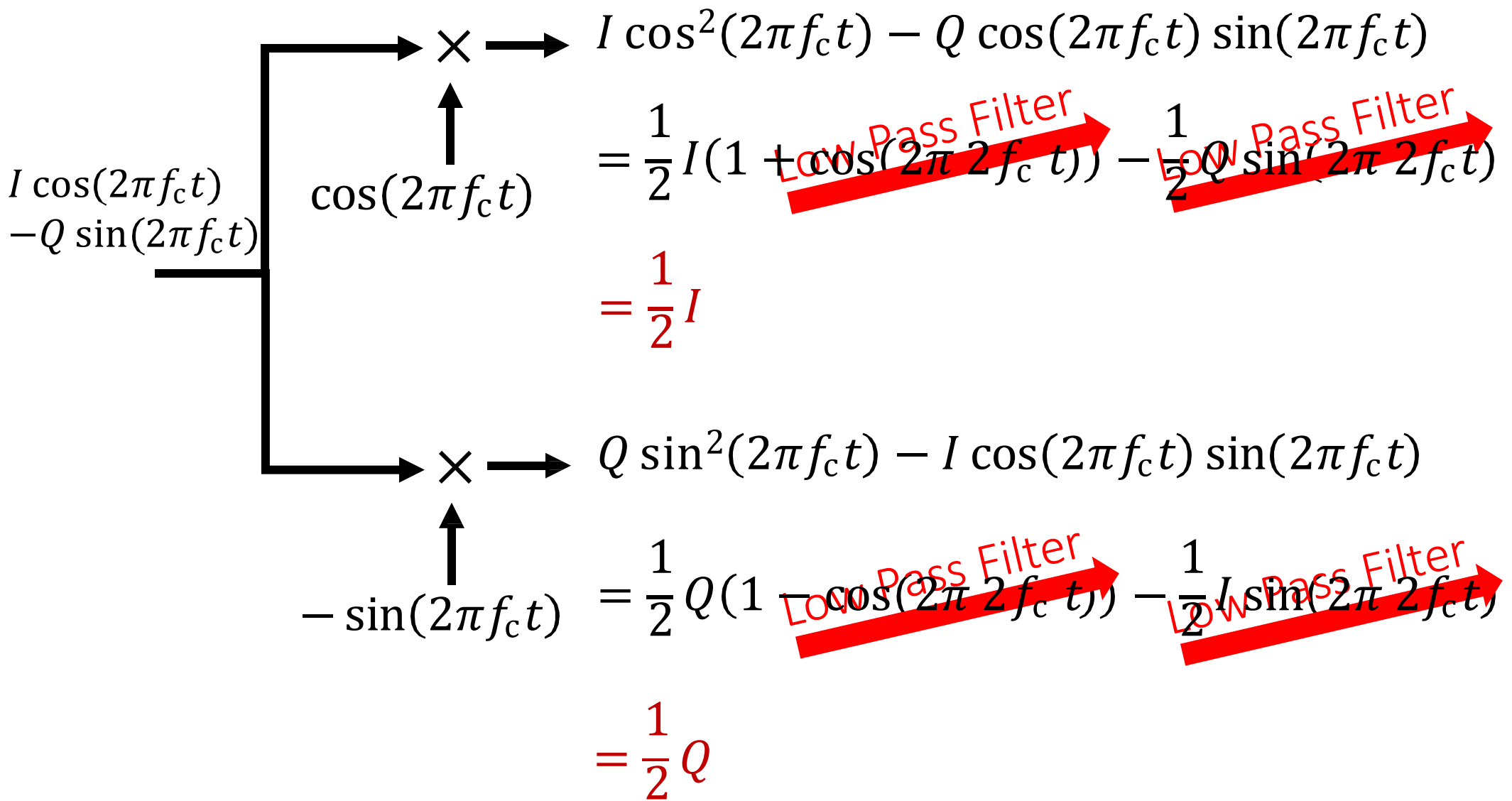
Consider using QAM: Quadrature Amplitude Modulation

$s(t)$ is complex: $s(t) = I + jQ$

Down Conversion

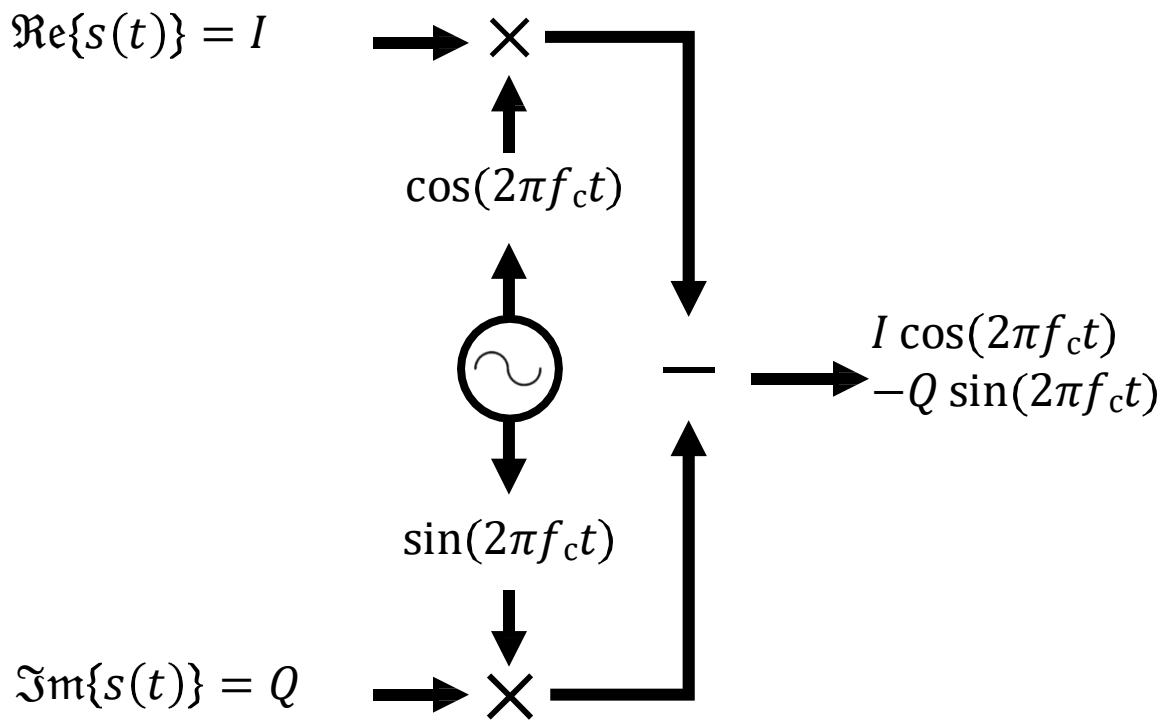


Down Conversion

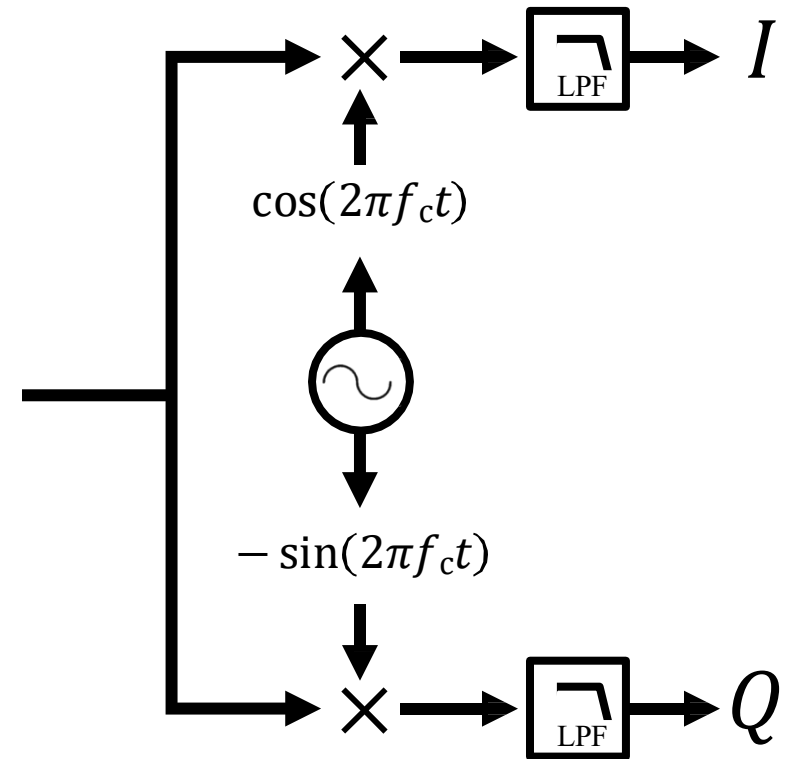


Up/Down Conversion

Transmitter



Receiver



$$s(t) \times e^{j2\pi f_c t} \rightarrow \Re\{s(t)e^{j2\pi f_c t}\} \rightarrow \times e^{-j2\pi f_c t} \rightarrow s(t)$$

Up/Down Conversion

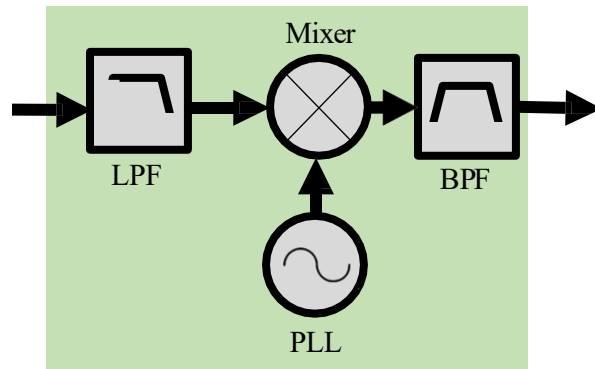
Transmitter

Receiver

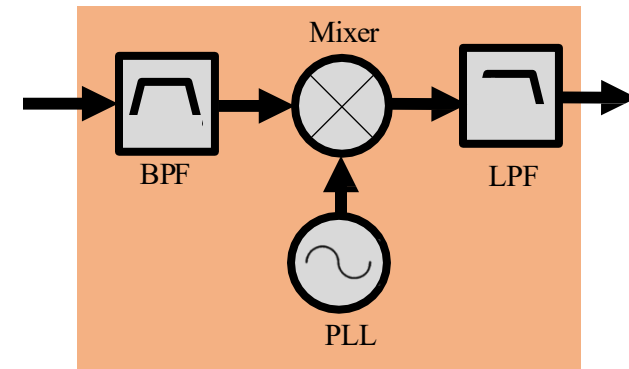
$$s(t) \times e^{j2\pi f_c t} \rightarrow \Re\{s(t)e^{j2\pi f_c t}\} \rightarrow \times e^{-j2\pi f_c t} \rightarrow s(t)$$

Up/Down Conversion

Transmitter
Up-conversion

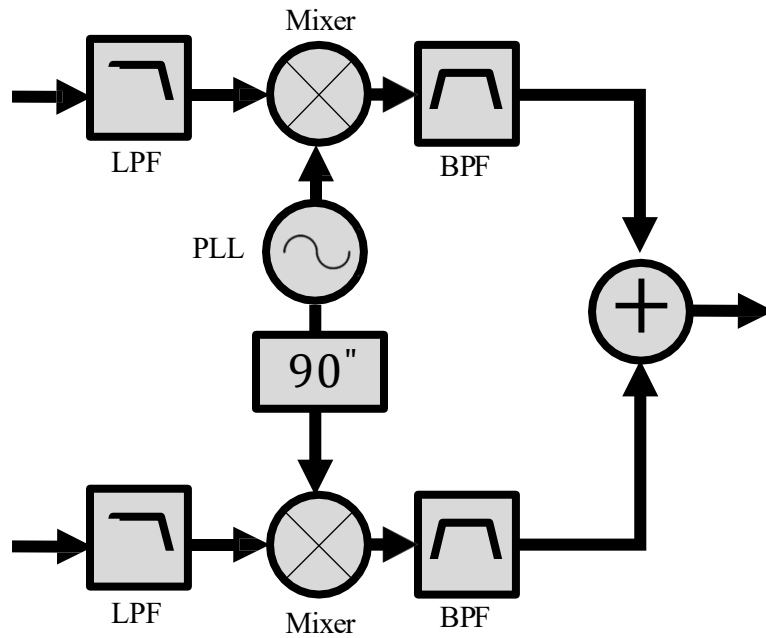


Receiver
Down-conversion

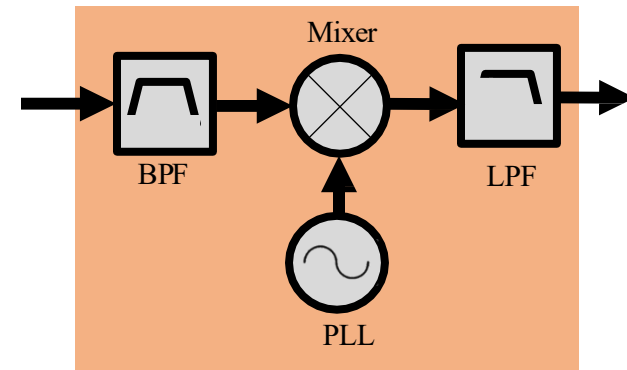


Up/Down Conversion

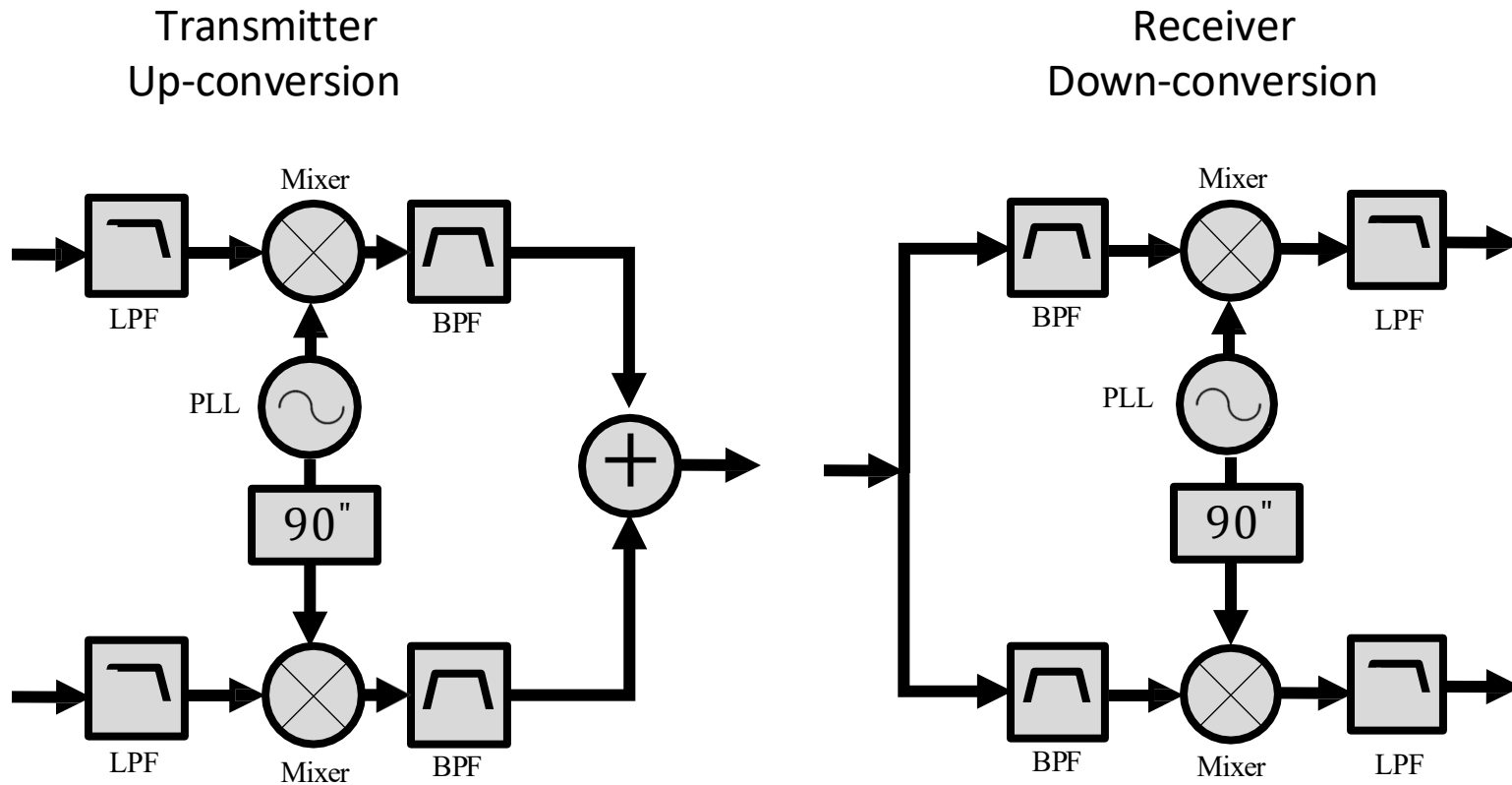
Transmitter
Up-conversion



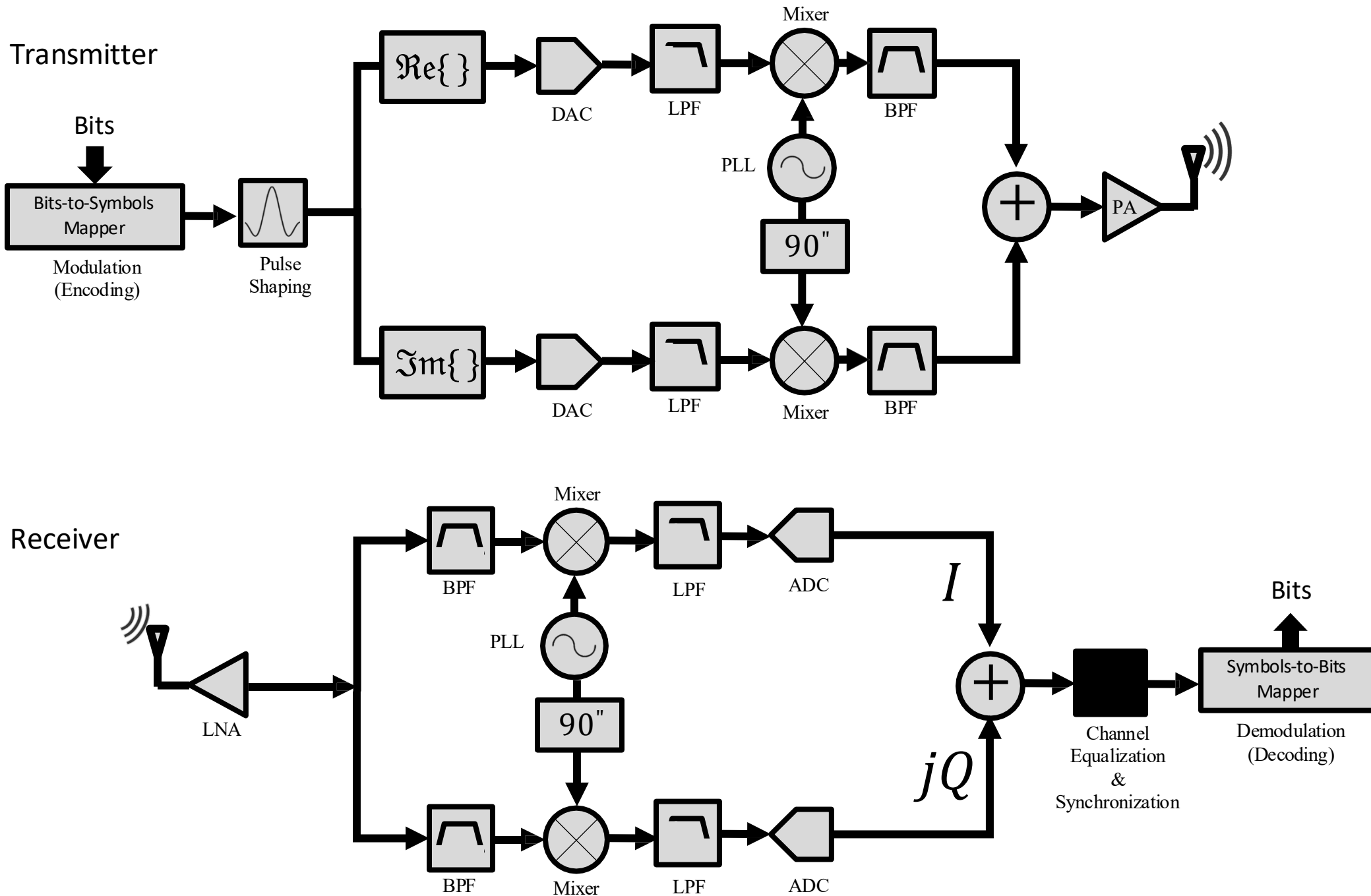
Receiver
Down-conversion



Up/Down Conversion

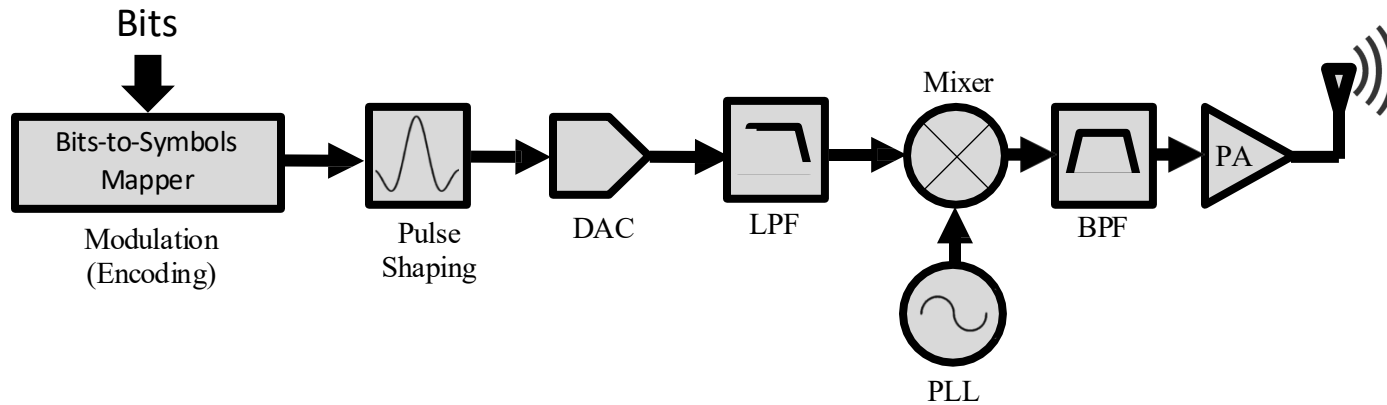


Wireless Communication System

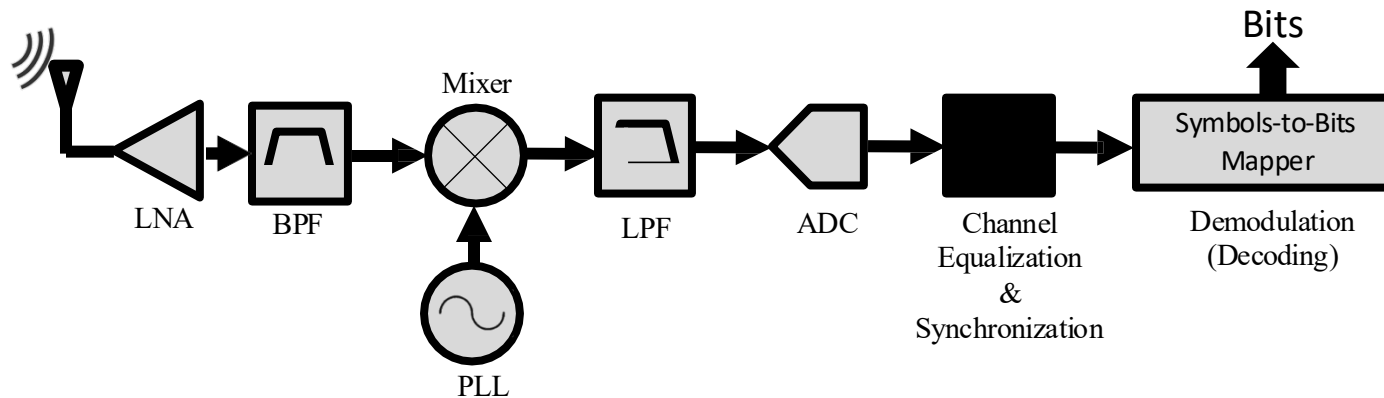


Wireless Communication System

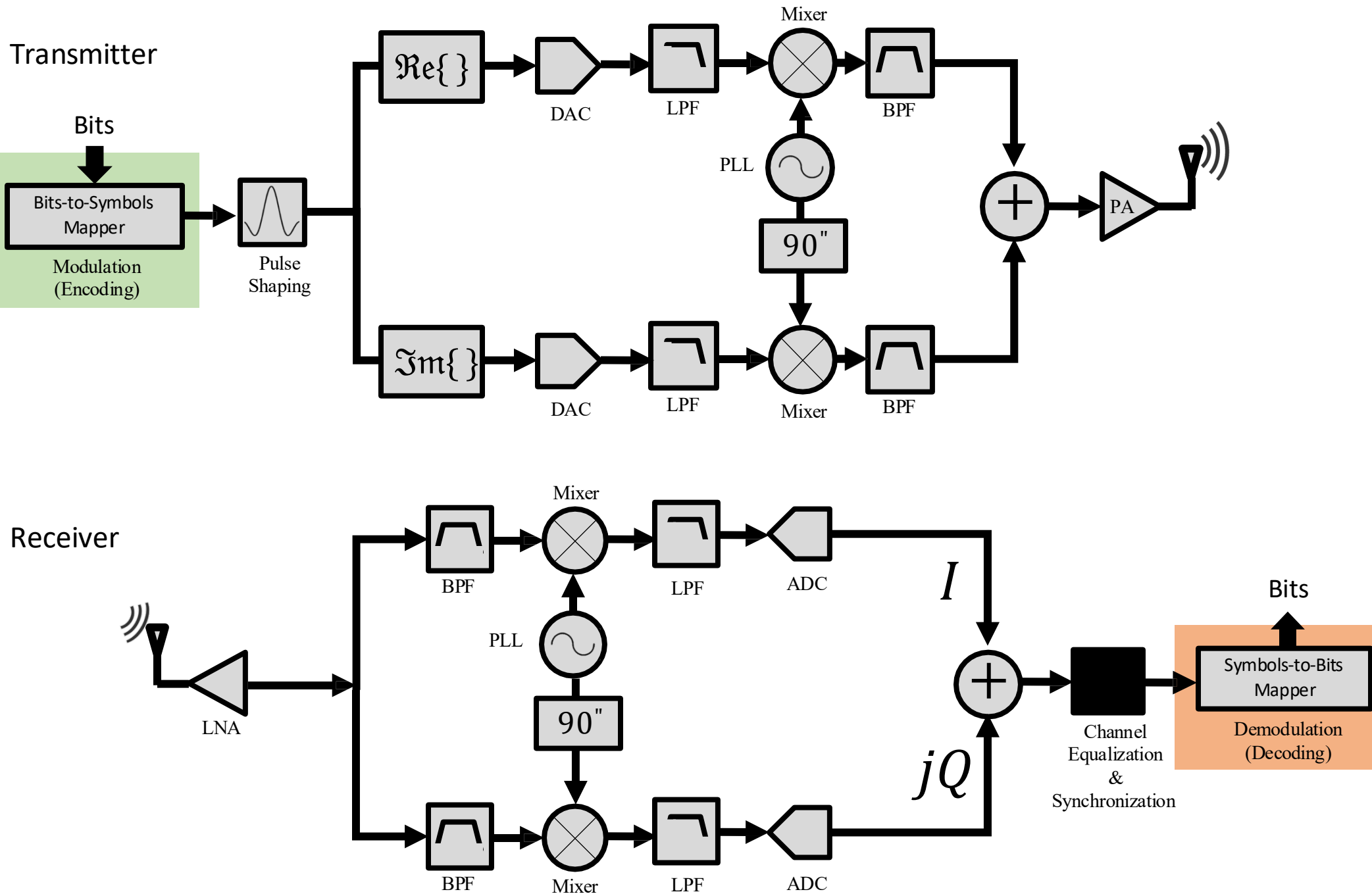
Transmitter



Receiver

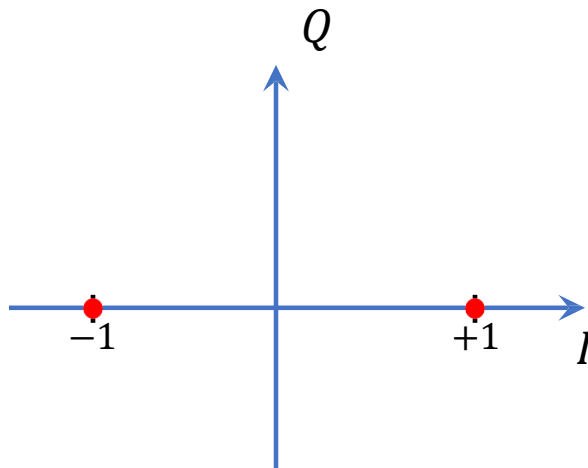


Wireless Communication System



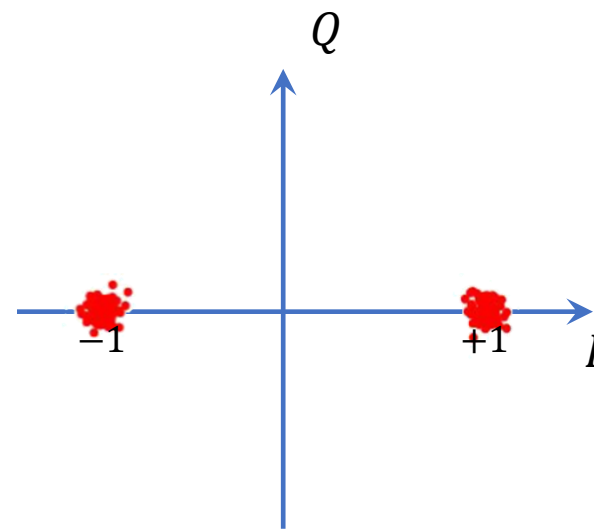
BPSK Modulation

$$\begin{aligned} 0 &\rightarrow -1 \\ 1 &\rightarrow +1 \end{aligned}$$



$$x(t)$$

Transmitted Constellation

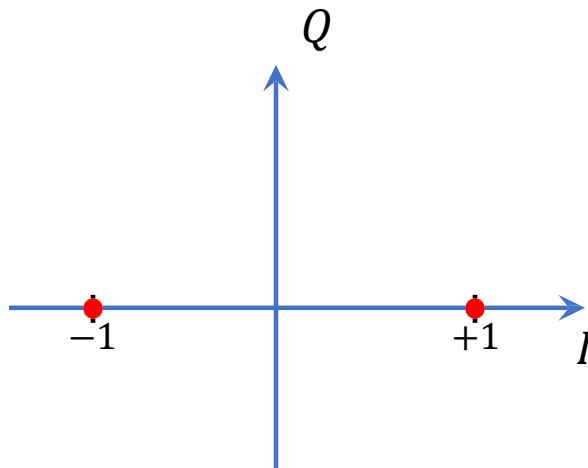


$$y(t) = x(t) + v(t)$$

Received Constellation

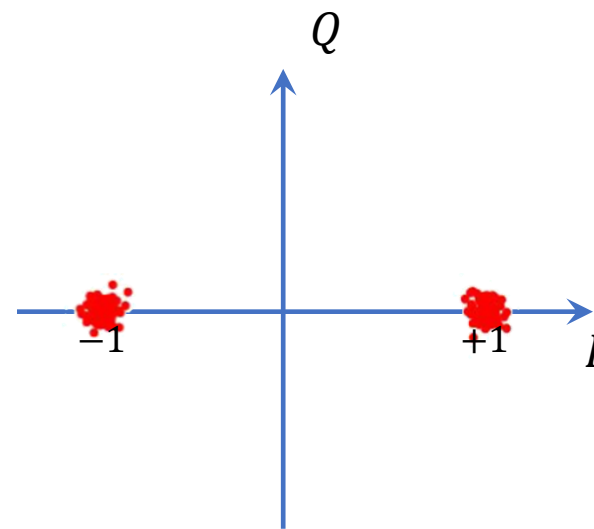
BPSK Modulation

$$\begin{aligned} 0 &\rightarrow -1 \\ 1 &\rightarrow +1 \end{aligned}$$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

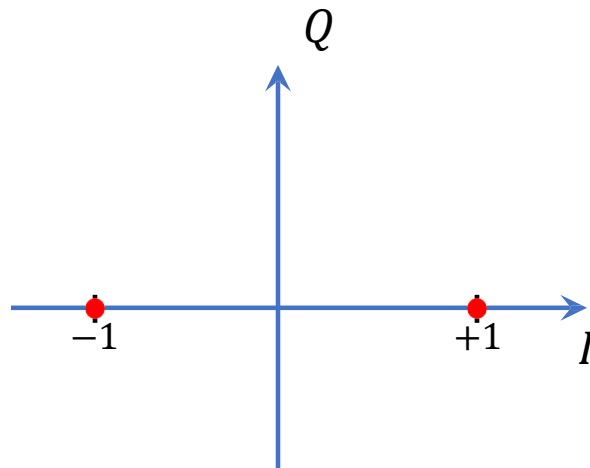
Received Constellation

How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E[|x(t)|^2]}{E[|v(t)|^2]} = \frac{E_s}{N_0}$$

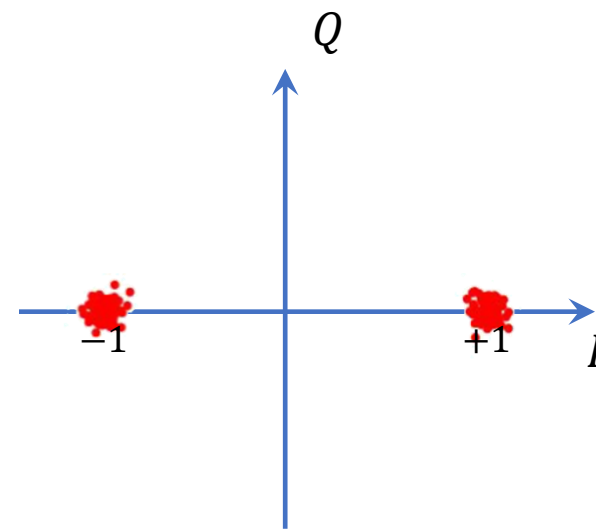
BPSK Modulation

$$\begin{aligned} 0 &\rightarrow -1 \\ 1 &\rightarrow +1 \end{aligned}$$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

Received Constellation

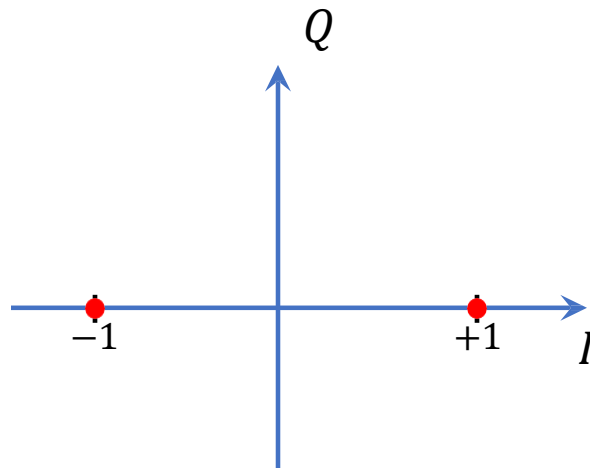
AWGN

How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E_s}{N_0} = 25 \text{ dB}$$

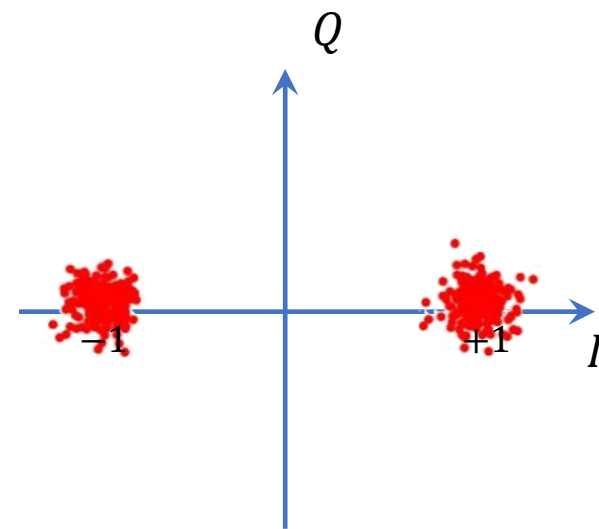
BPSK Modulation

$$\begin{aligned} 0 &\rightarrow -1 \\ 1 &\rightarrow +1 \end{aligned}$$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

Received Constellation

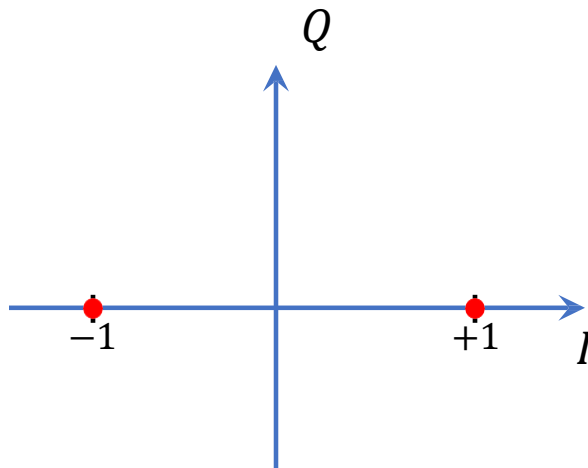
AWGN

How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E_s}{N_0} = 19 \text{ dB}$$

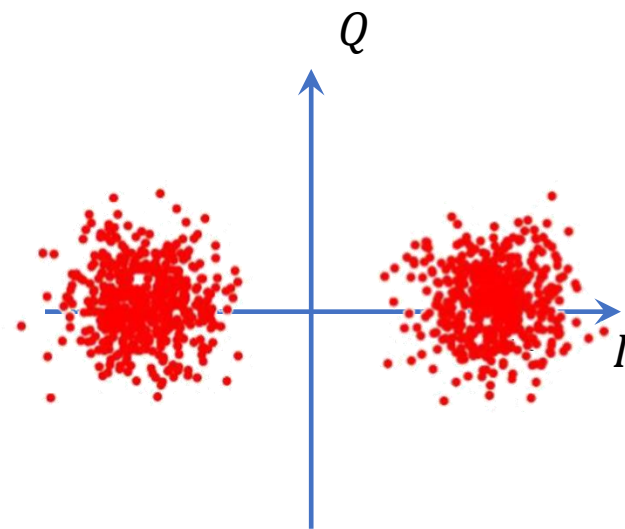
BPSK Modulation

$$\begin{aligned} 0 &\rightarrow -1 \\ 1 &\rightarrow +1 \end{aligned}$$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

Received Constellation

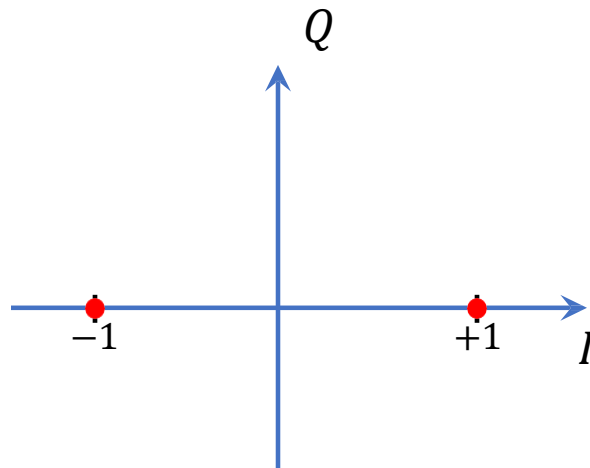
AWGN

How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E_s}{N_0} = 13 \text{ dB}$$

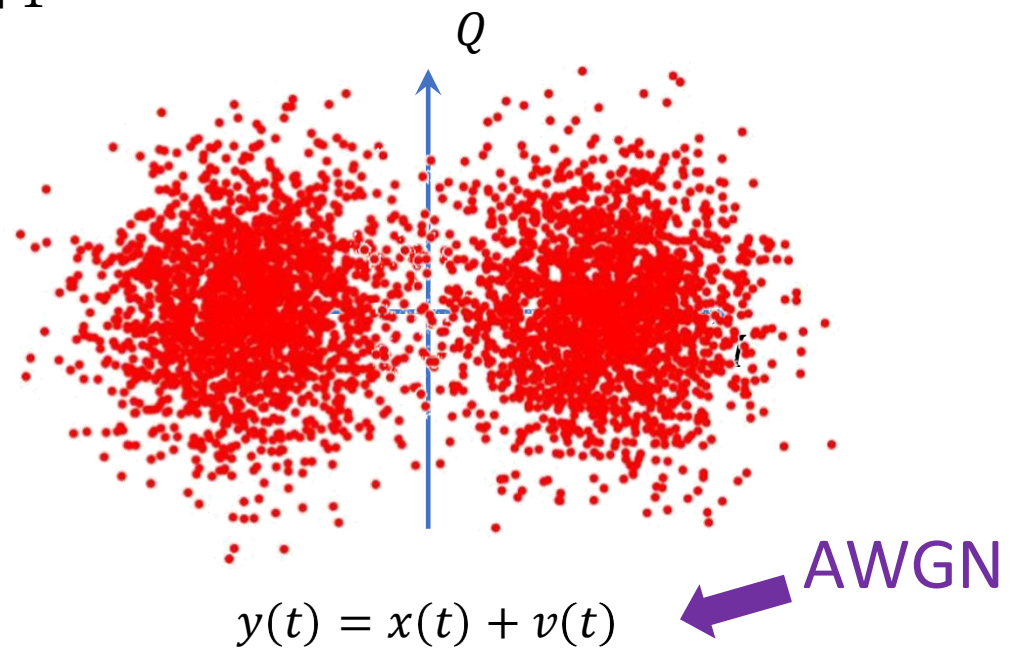
BPSK Modulation

$0 \rightarrow -1$
 $1 \rightarrow +1$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

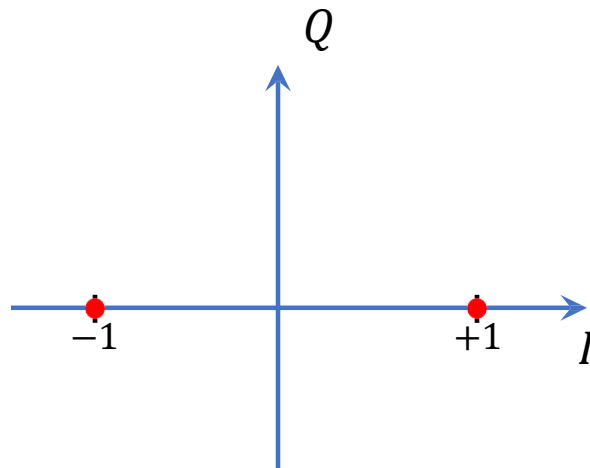
Received Constellation

How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E_s}{N_0} = 7 \text{ dB}$$

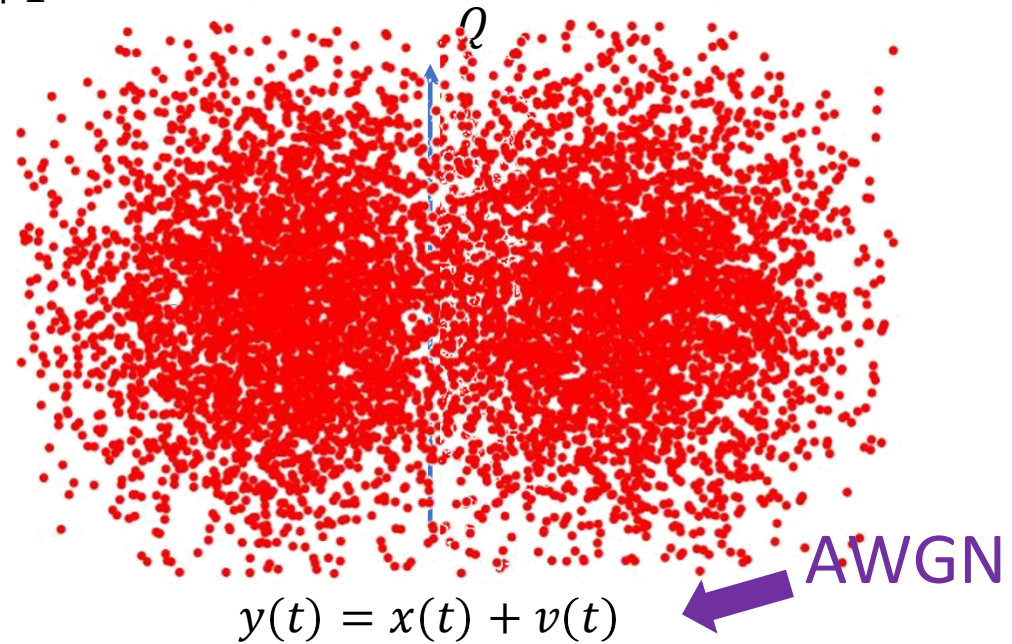
BPSK Modulation

$0 \rightarrow -1$
 $1 \rightarrow +1$



$x(t)$

Transmitted Constellation

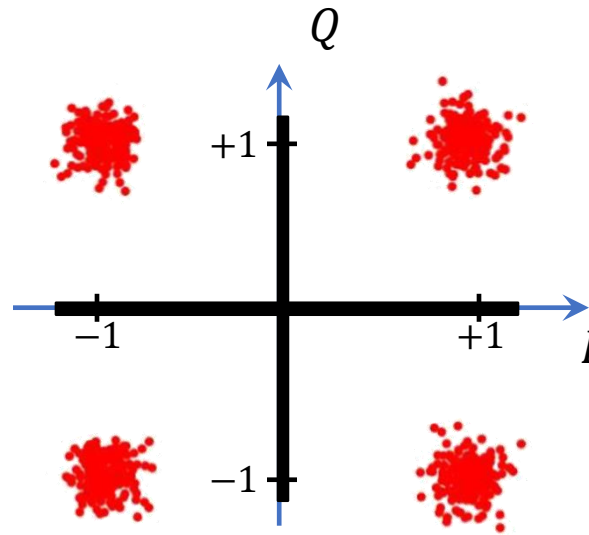


Received Constellation

How well we can decode depends on SNR?

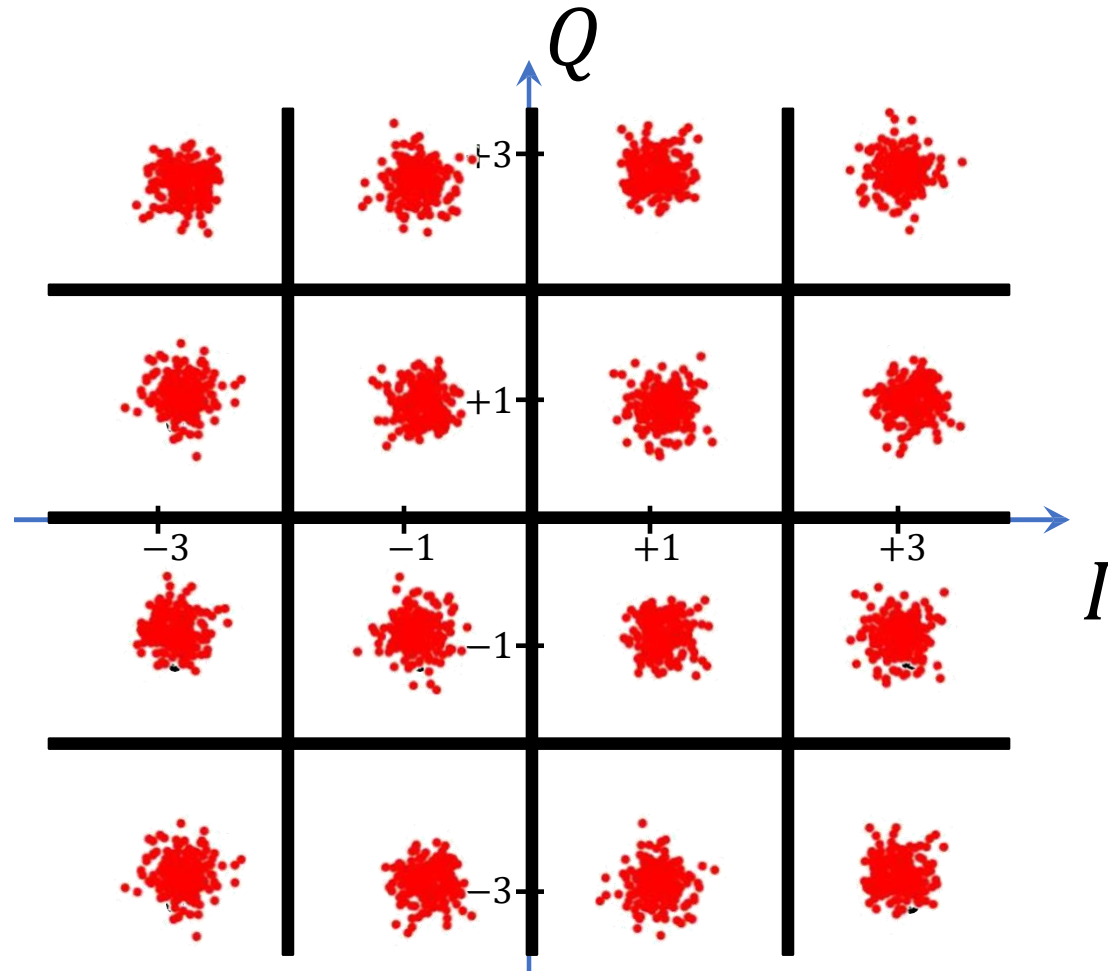
$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E_s}{N_0} = 3.5 \text{ dB}$$

4-QAM



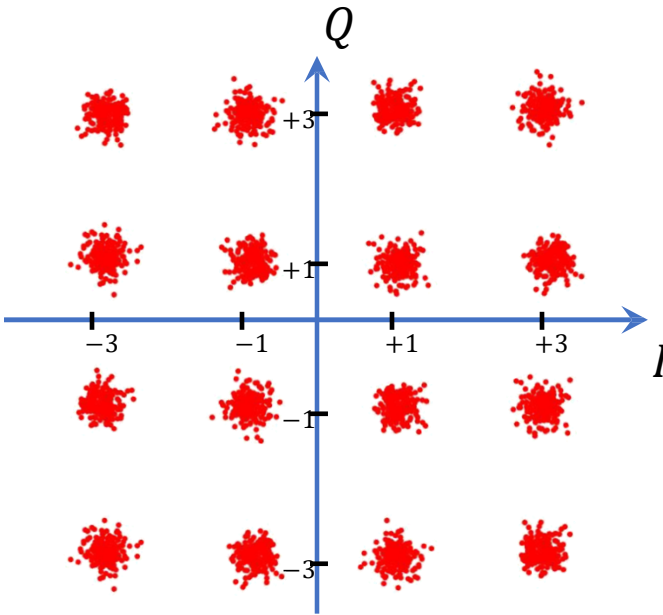
Encodes 2 bits per symbol resulting in Higher Data Rate

16-QAM

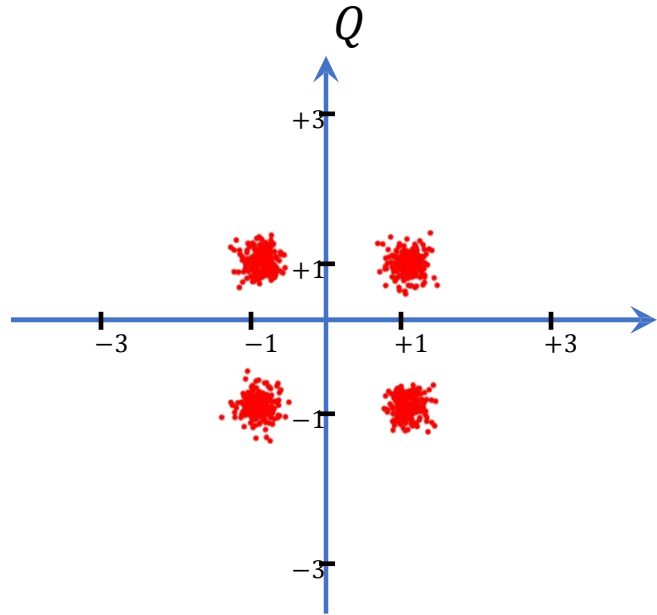


Encodes 4 bits per symbol resulting in Higher Data Rate

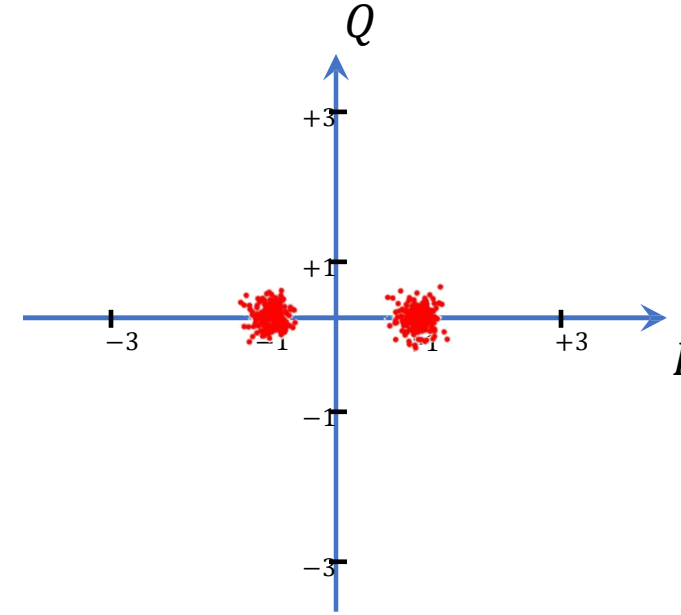
16-QAM



4-QAM



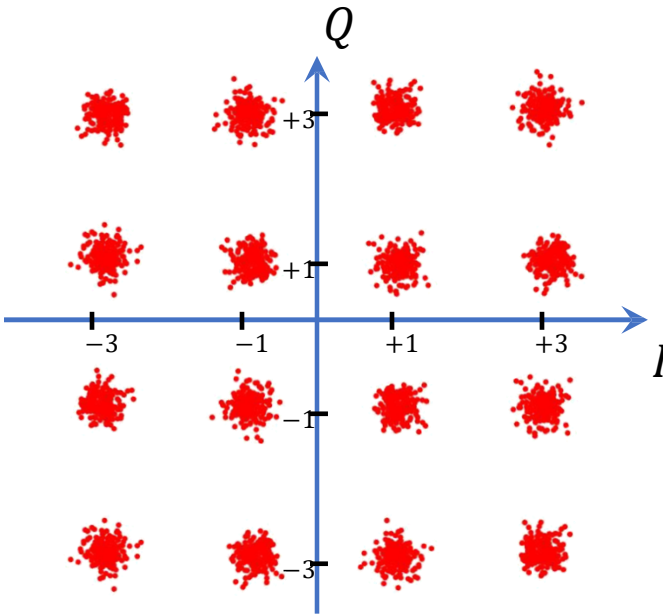
BPSK



Is the SNR the same in these 3 constellations?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E[|x(t)|^2]}{E[|v(t)|^2]} = \frac{E_s}{N_0}$$

16-QAM

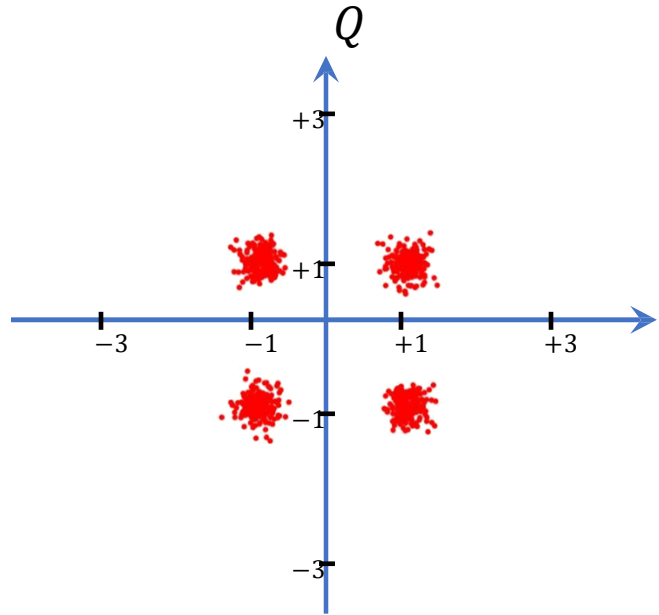


$$SNR = \frac{(4 \times 2 + 8 \times 10 + 4 \times 18) / 16}{N_0}$$

$$= \frac{10}{N_0}$$

>

4-QAM



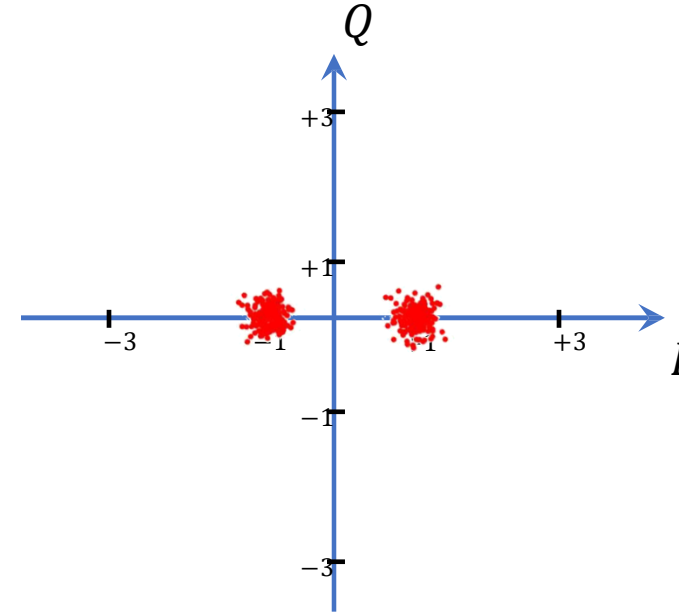
$$SNR = \frac{E[|x(t)|^2]}{N_0}$$

$$SNR = \frac{(2 + 2 + 2 + 2) / 4}{N_0}$$

$$= \frac{2}{N_0}$$

>

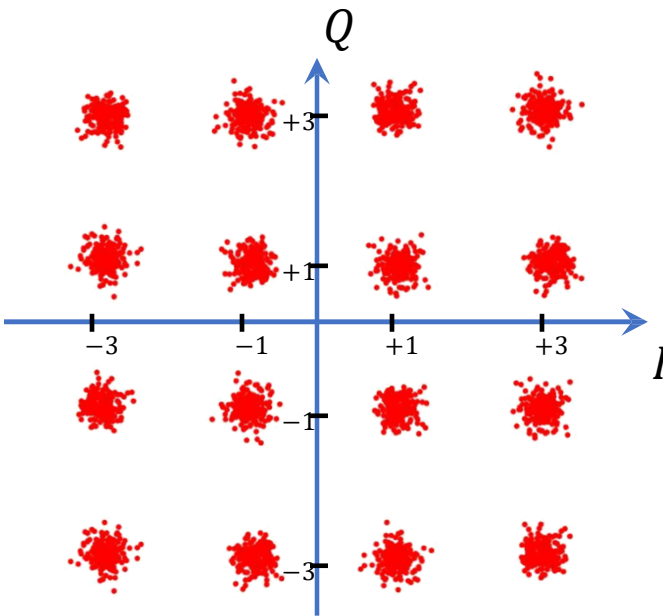
BPSK



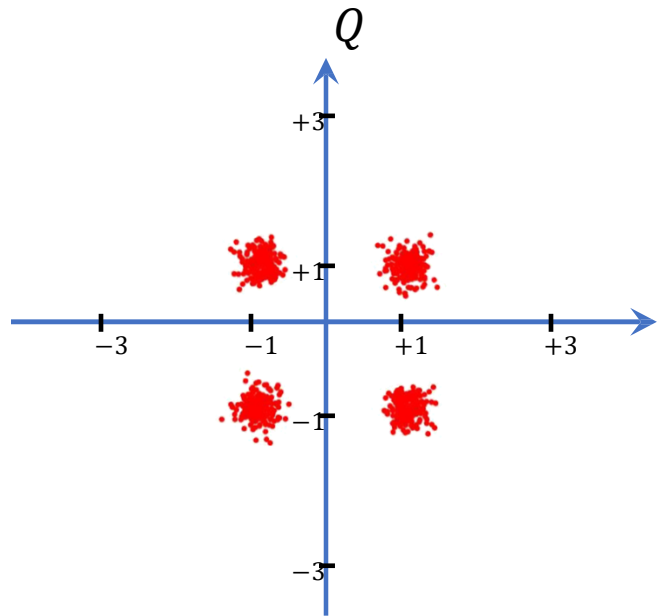
$$SNR = \frac{(1 + 1) / 2}{N_0}$$

$$= \frac{1}{N_0}$$

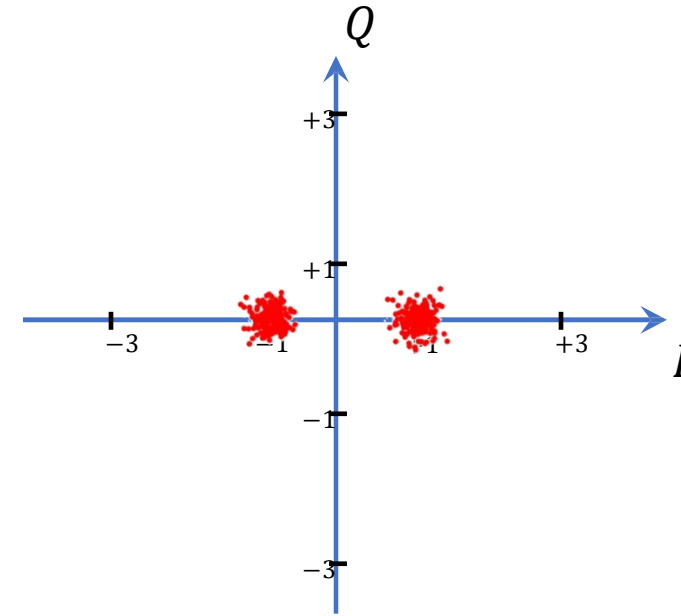
16-QAM



4-QAM



BPSK



Transmit power is not the same!

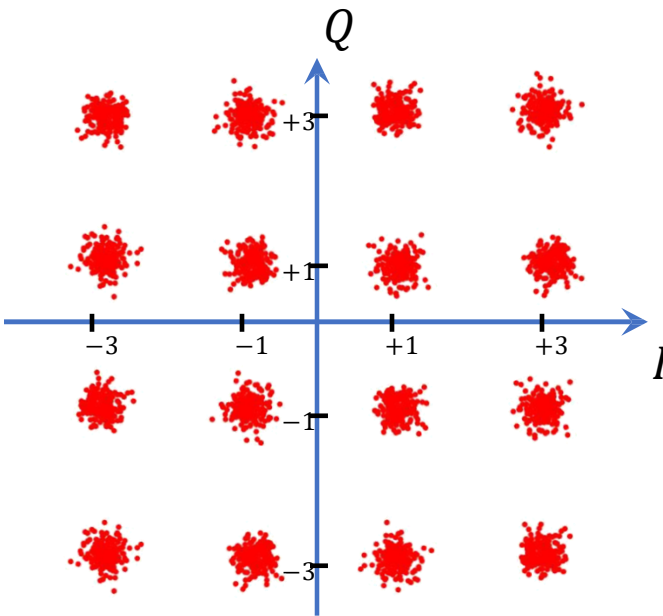
$$SNR = \frac{(4 \times 2 + 8 \times 10 + 4 \times 18)}{N_0}$$

$$= \frac{10}{N_0}$$

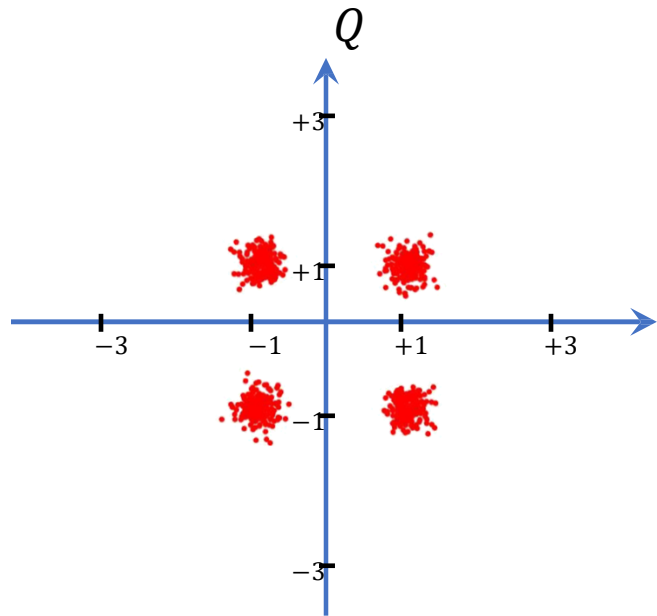
Must not increase transmit power to use higher order modulation!

$$\frac{1}{N_0}$$

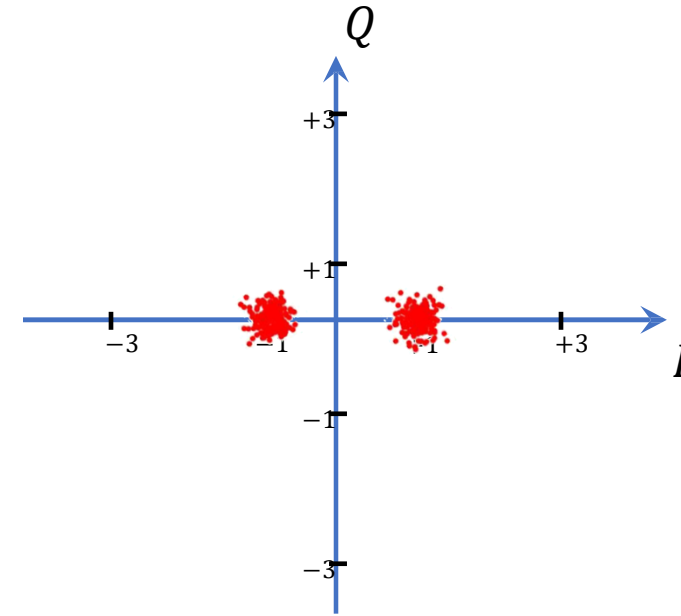
16-QAM



4-QAM



BPSK



Normalize to maintain constant transmit power.

$$SNR = \frac{(4 \times 2 + 8 \times 10 + 4 \times 18) / 16}{N_0}$$

$$= \frac{10}{N_0}$$

>

$$SNR = \frac{(2 + 2 + 2 + 2) / 4}{N_0}$$

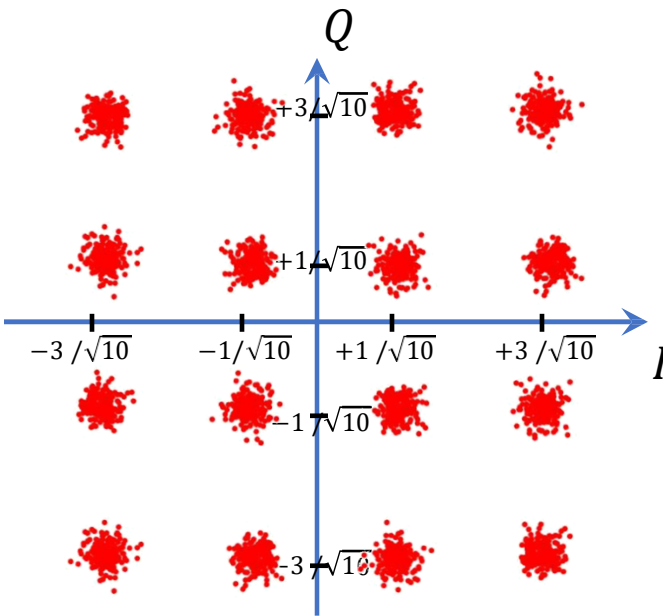
$$= \frac{2}{N_0}$$

>

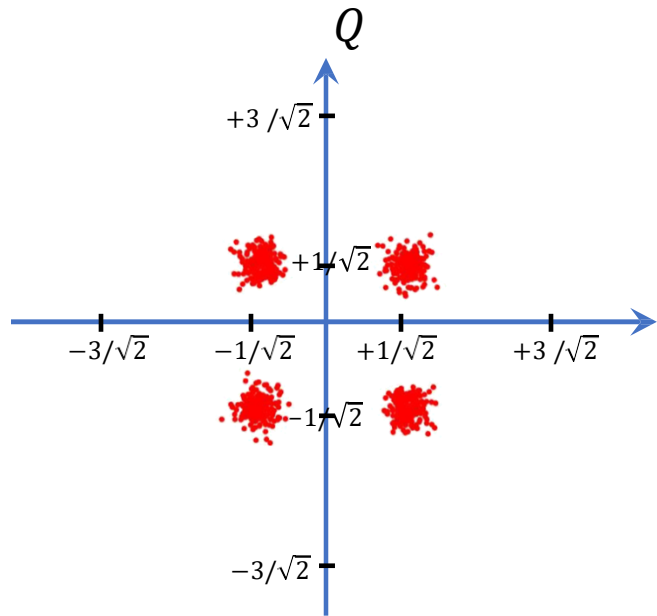
$$SNR = \frac{(1 + 1) / 2}{N_0}$$

$$= \frac{1}{N_0}$$

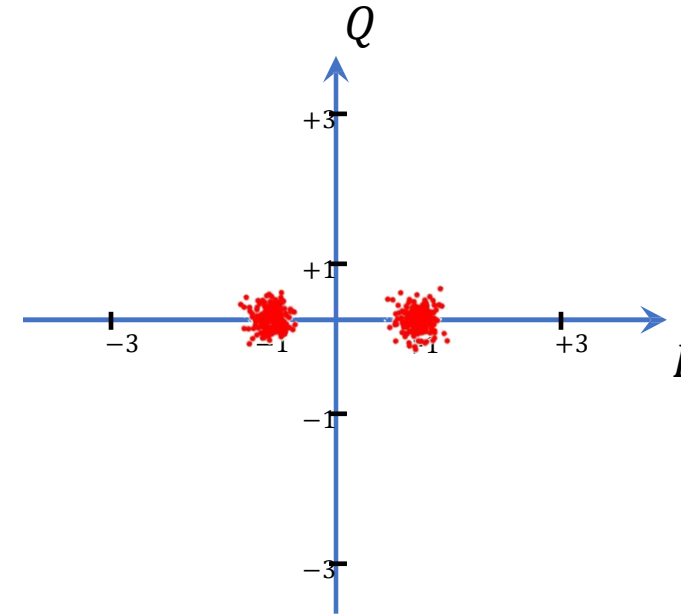
16-QAM



4-QAM



BPSK



Normalize to maintain constant transmit power.

$$SNR = \frac{(4 \times 2/10 + 8 \times 1 + 4 \times 18/10)/16}{N_0}$$

$$= \frac{1}{N_0}$$

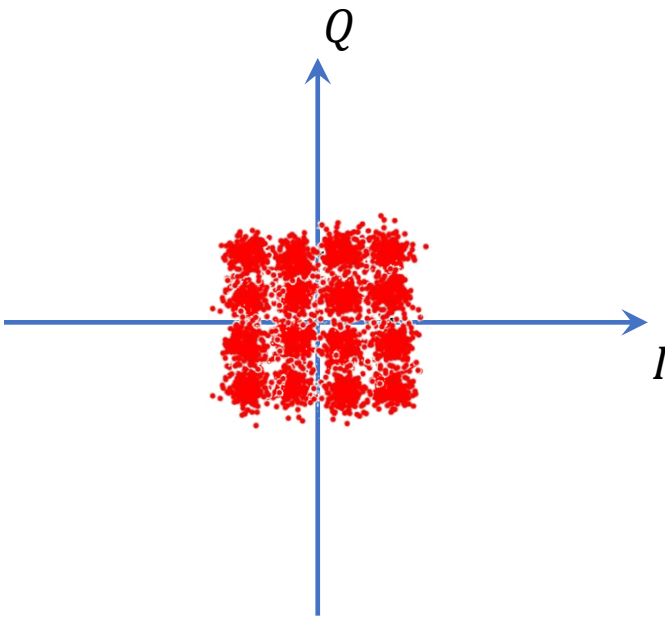
$$SNR = \frac{(1 + 1 + 1 + 1)/4}{N_0}$$

$$= \frac{1}{N_0}$$

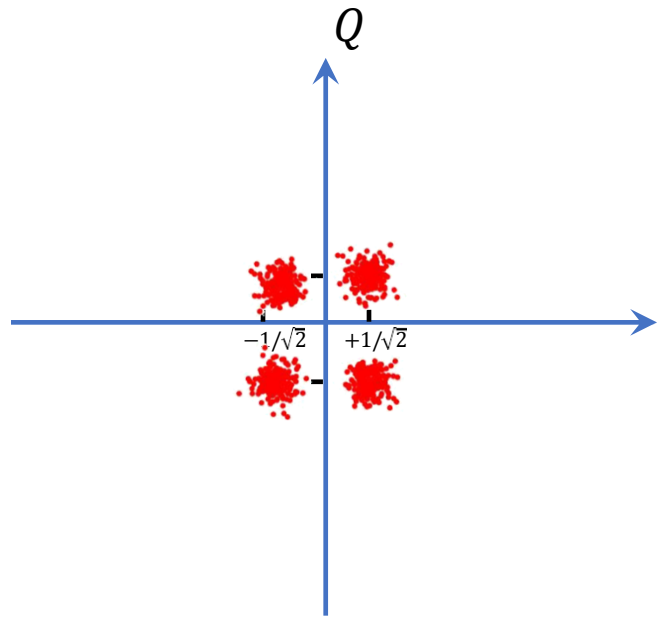
$$SNR = \frac{(1 + 1)/2}{N_0}$$

$$= \frac{1}{N_0}$$

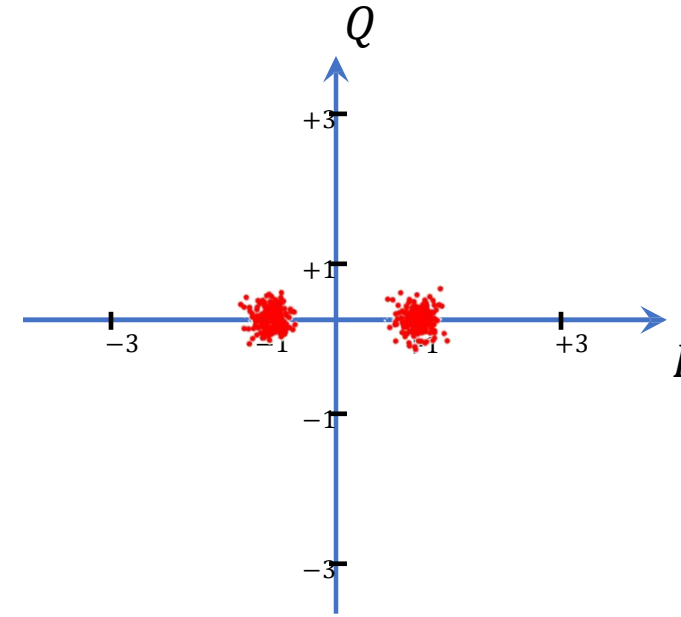
16-QAM



4-QAM



BPSK



Normalize to maintain constant transmit power.

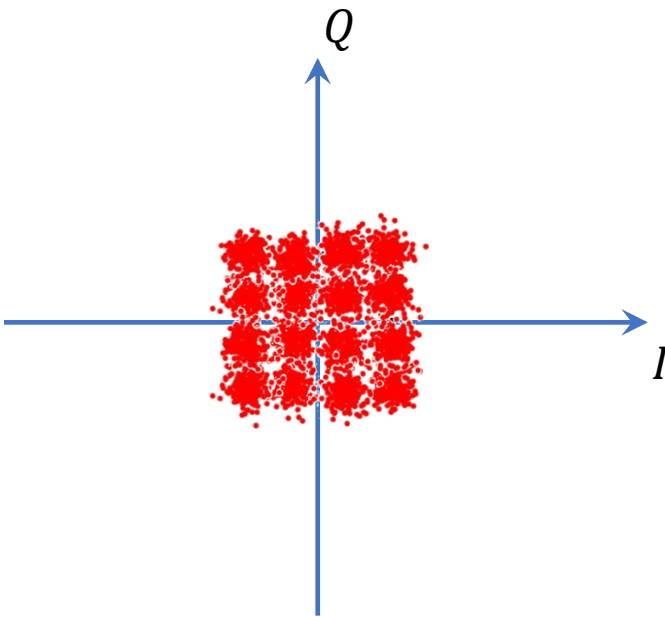
$$SNR = \frac{1}{N_0}$$

$$SNR = \frac{1}{N_0}$$

$$SNR = \frac{1}{N_0}$$

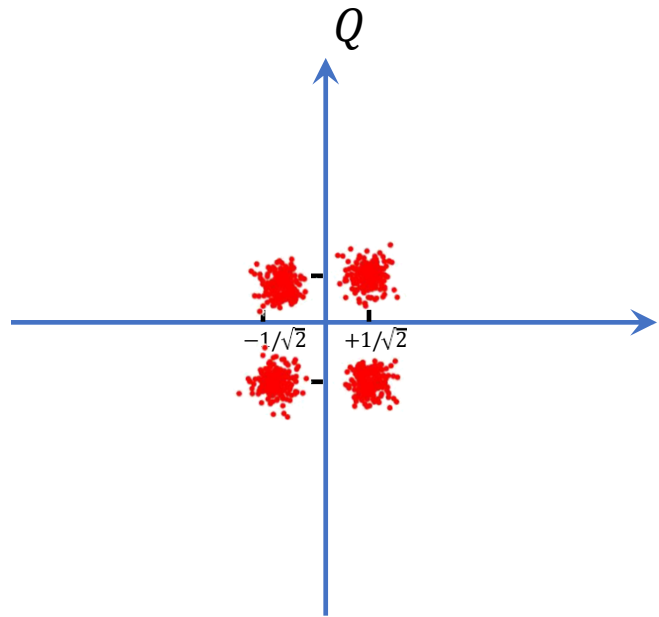
Need Higher SNR to Decode
Higher Order Modulation.

16-QAM



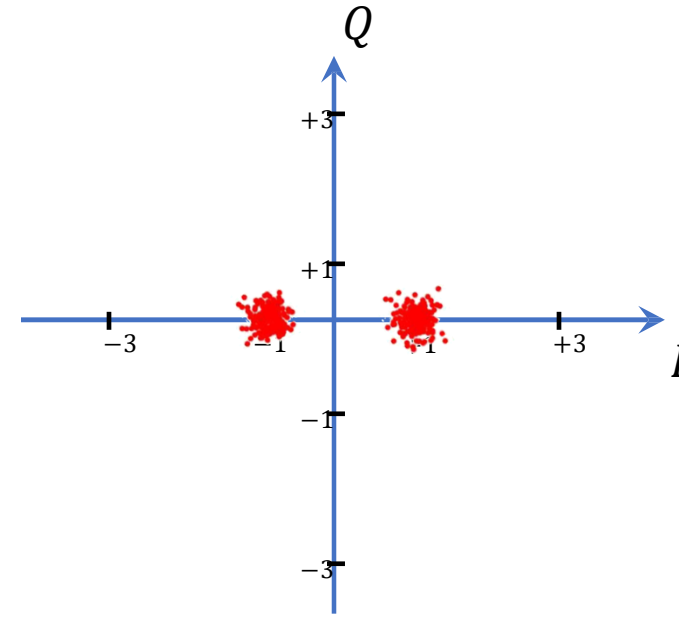
4 bits/symbol

4-QAM



2 bits/symbol

BPSK



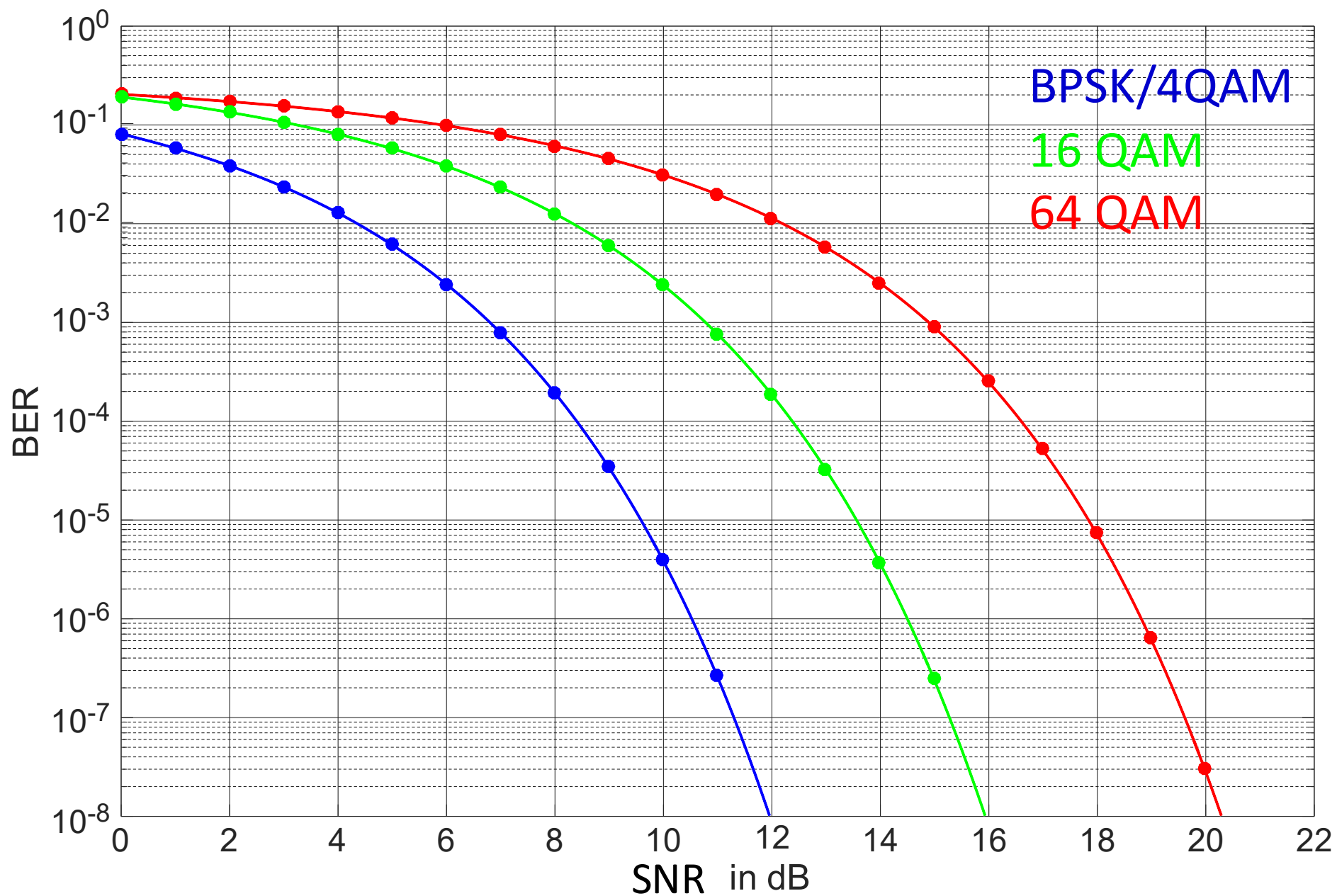
1 bit/symbol

Higher Order Modulation:

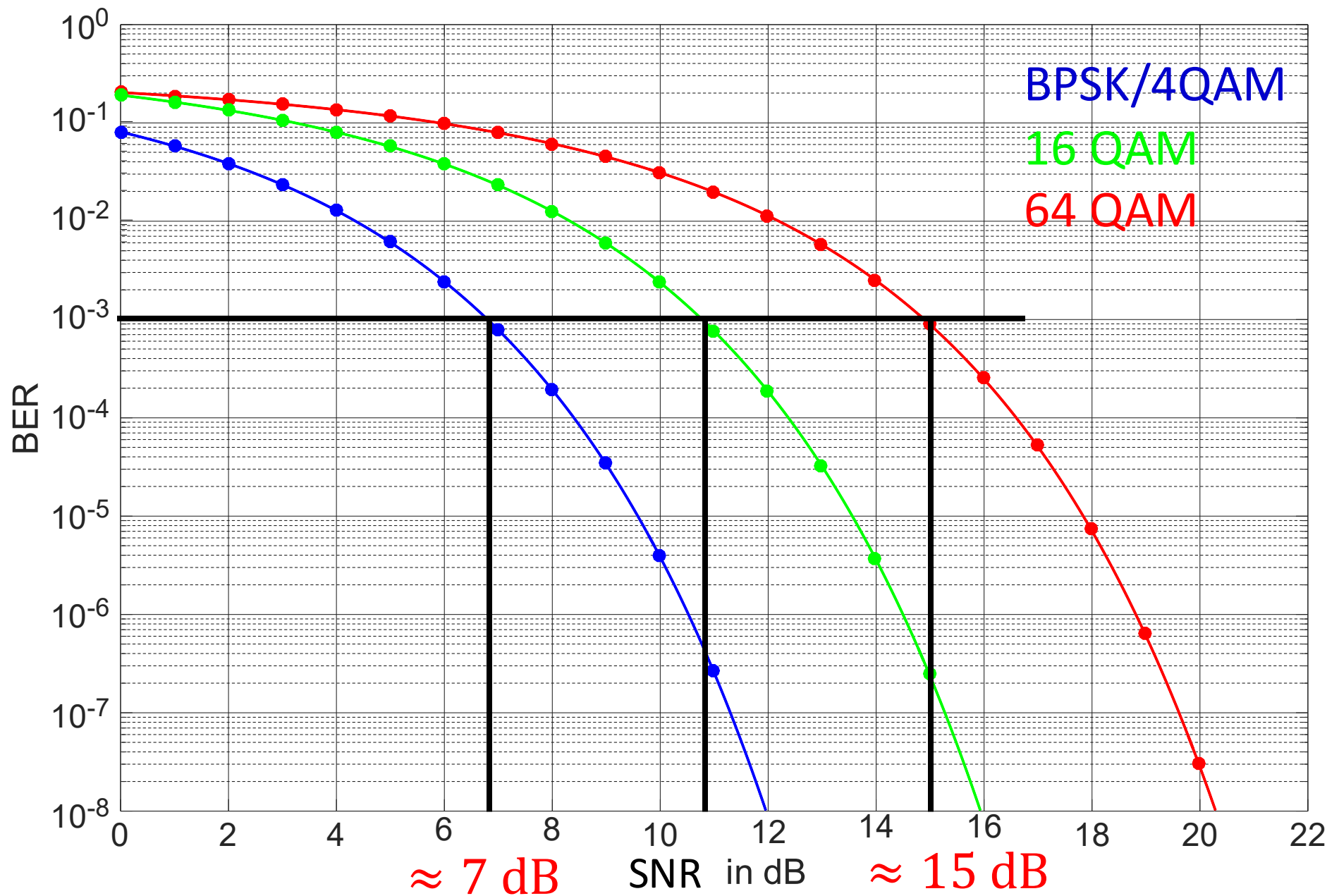
- Needs higher SNR to decode correctly.
- Achieves higher Bite Rate

Given an SNR, choose highest order modulation that guarantees minimal Bite Error Rate (BER)

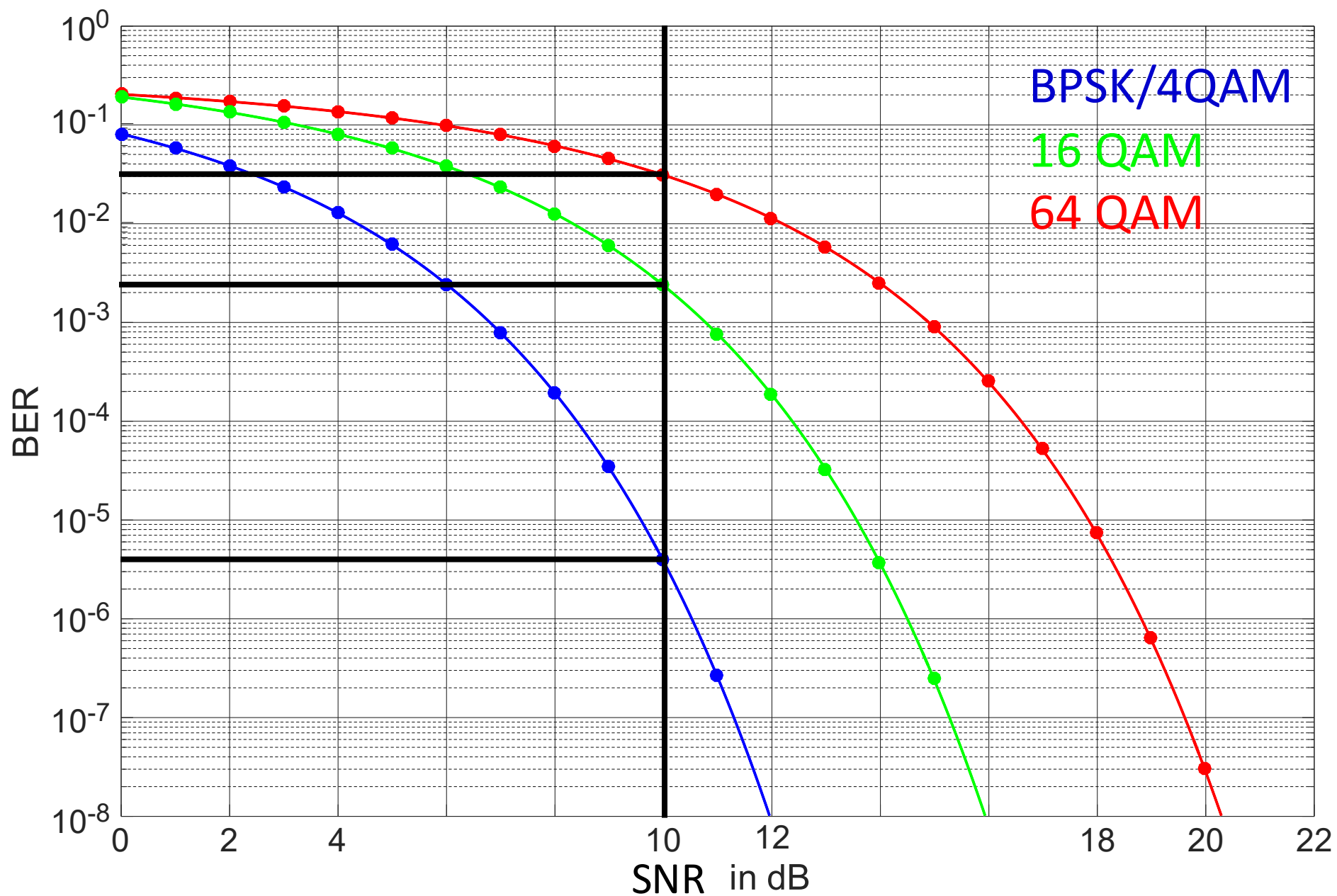
Bit Error Rate (BER)



Bit Error Rate (BER)

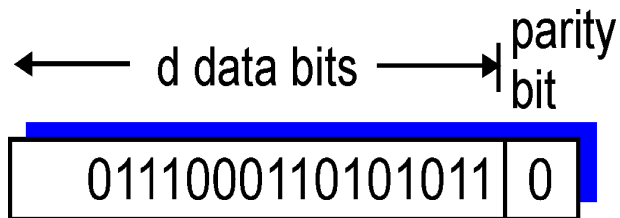


Bit Error Rate (BER)

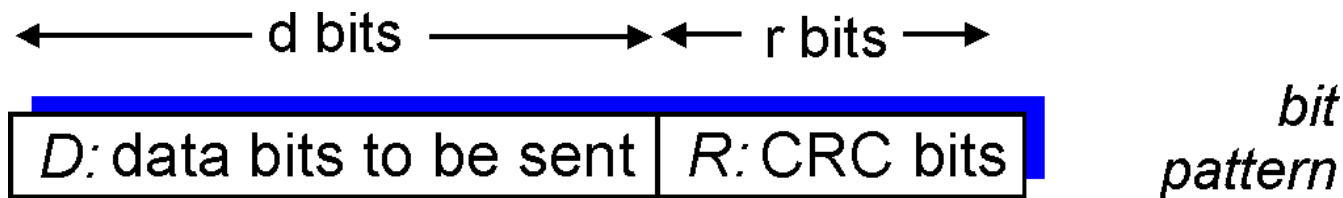


Error Detection

- Add Redundant bits in order to detect errors at receiver
- Checksums → Detect Errors
 - Parity Check



- CRC: Cyclic Redundancy Check

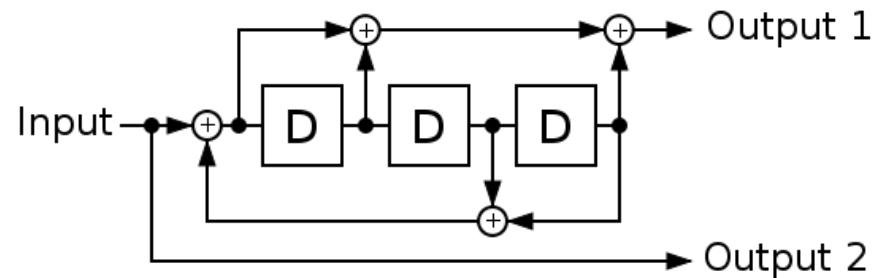


- If packet (frame, sequence of bits) fails the checksum test
 - Packet is dropped
 - Packet loss.

Coding: Forward Error Correction (FEC)

- Add Redundant bits in order to correct errors at receiver
- FEC: Forward Error Correction

- Repetition Code
- Convolutional codes
- Reed Solomon codes
- Turbo codes
- Polar codes
- LDPC codes



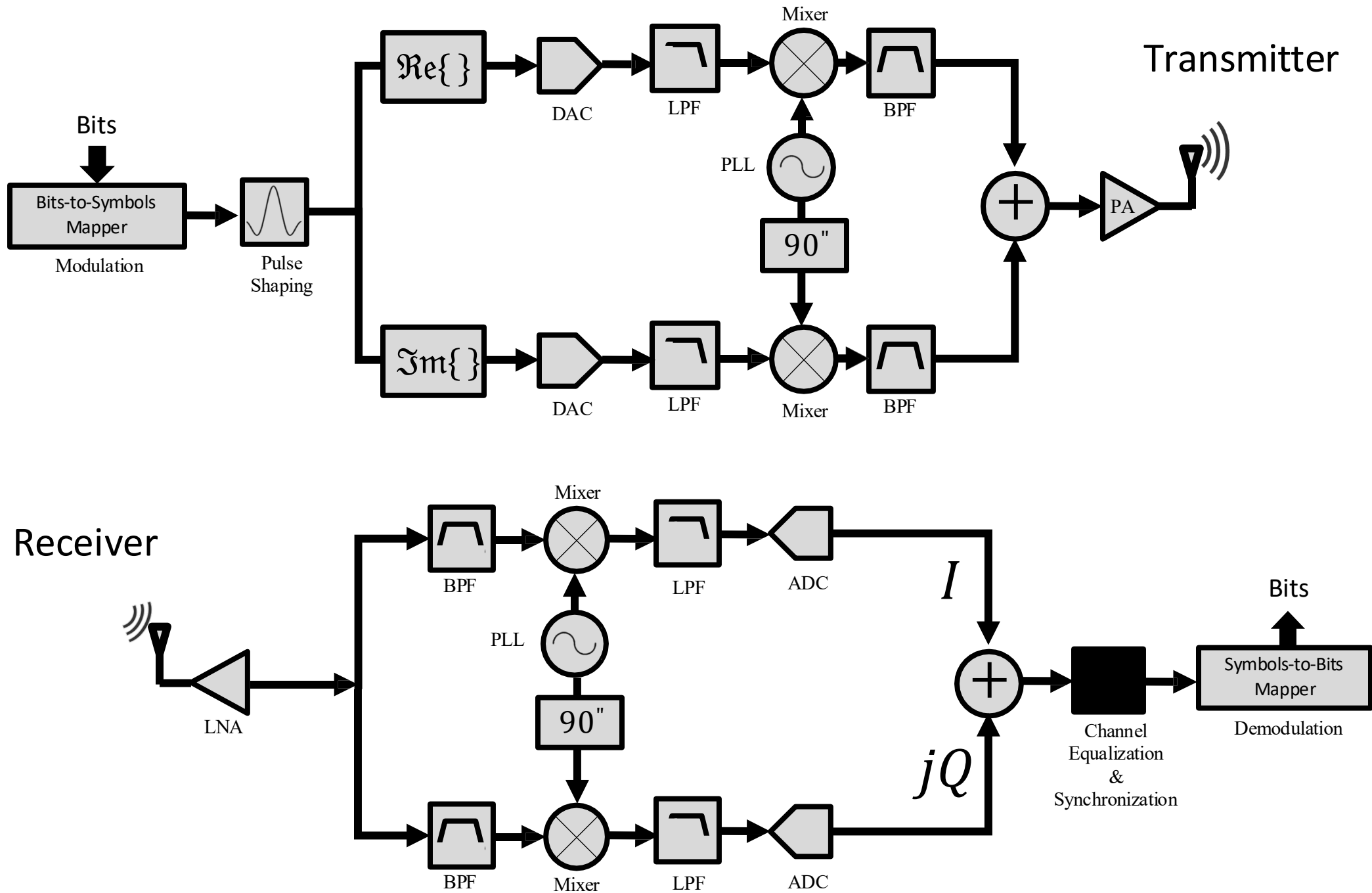
- Code Rate = $\frac{\text{\# of data bits (uncoded)}}{\text{\# of transmitted bits (coded)}}$

1 - 3 bits
1/2

- E.g. coding rate 2/3, for every 2 data bits, I transmit 3 bits.

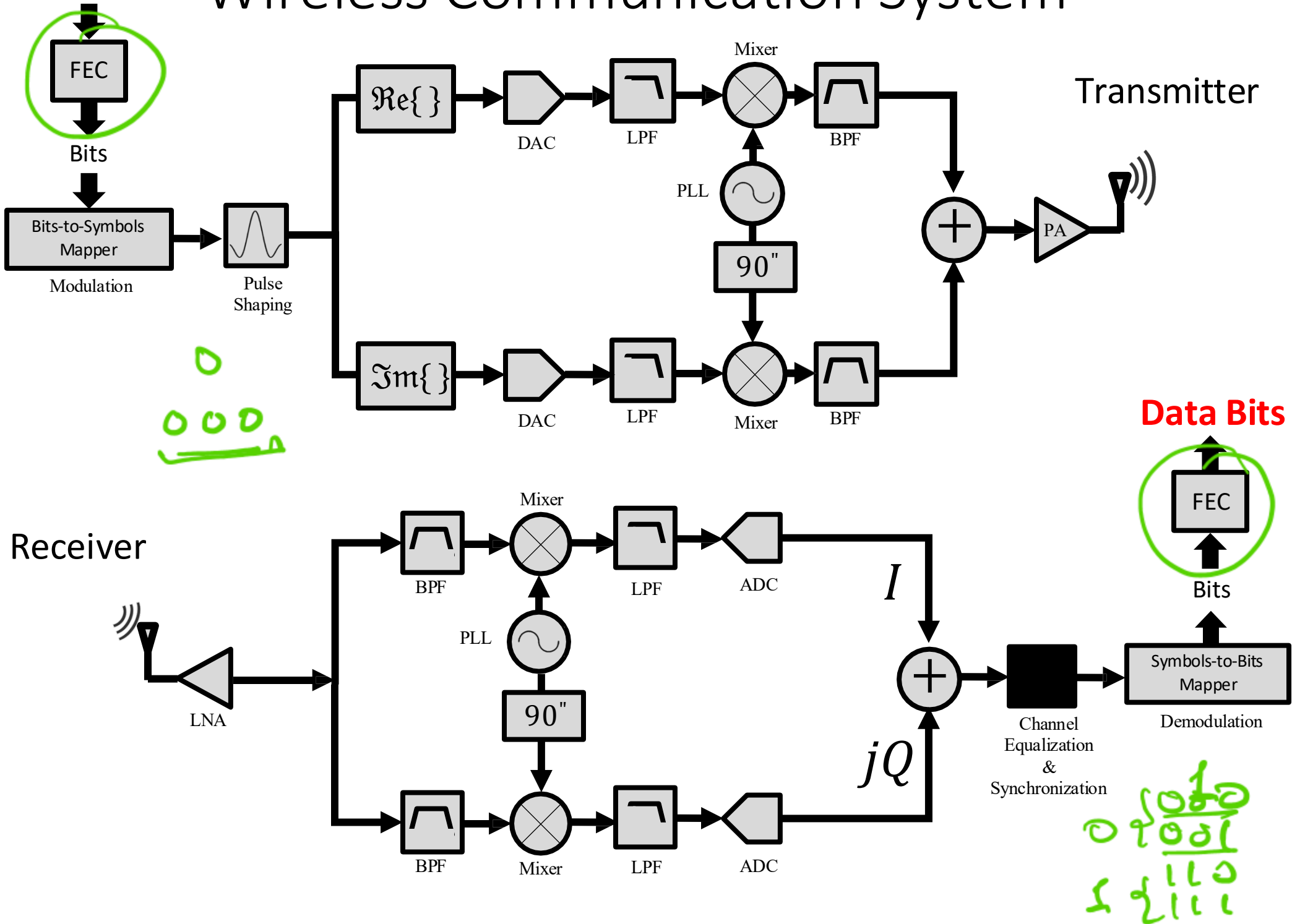
≈ 1

Wireless Communication System



Data Bits

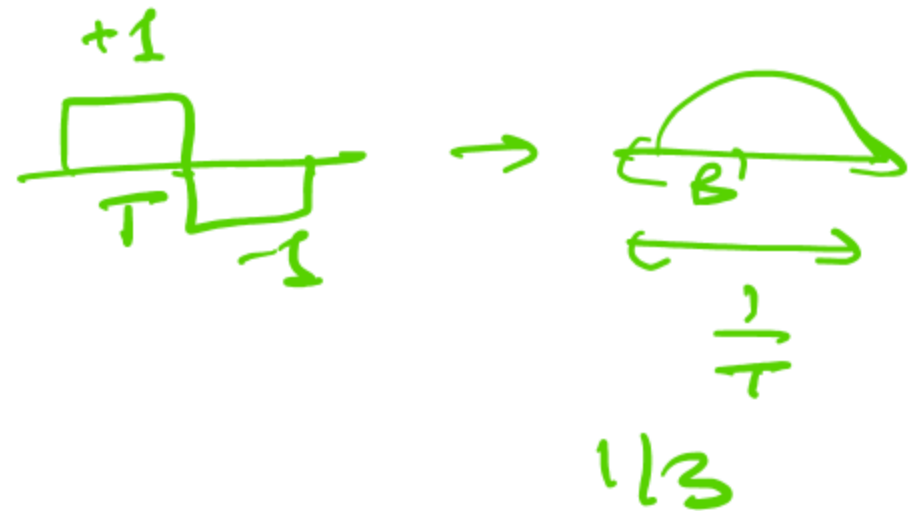
Wireless Communication System



Wireless Data Rates

- **Data Rate**

- Bandwidth: Samples/sec
- Modulation: Bits/sample
- Coding Rate: Data Bits/Coded Bits



$$\boxed{\text{Data Rate} = \text{Bandwidth} \times \text{Bits/sample} \times \text{Code Rate}}$$

sample/sec

useful bits/sample

- **Capacity**

- Maximum Achievable Data Rate
- Shannon Capacity Theorem:

$$\text{Capacity} = \text{Bandwidth} \times \log_2(1 + \text{SNR})$$